

Atoms and Nuclei

Question1

The average energy released per fission for the nucleus of ${}_{92}^{235}\text{U}$ is 190 MeV . When all the atoms of 47 g pure ${}_{92}^{235}\text{U}$ undergo fission process, the energy released is $\alpha \times 10^{23}\text{MeV}$. The value of α is ____ .

(Avogadro Number = 6×10^{23} per mole)

JEE Main 2026 (Online) 23rd January Evening Shift

Answer: 228

Solution:

The number of moles (n) is given by the formula :

$$n = \frac{\text{Given Mass}}{\text{Molar Mass}}$$

Given :

- Mass of Uranium = 47 g
- Molar Mass of $\text{U}_{92}^{235} = 235 \text{ g/mol}$

$$n = \frac{47}{235} = \frac{1}{5} = 0.2 \text{ moles}$$

Each mole contains a number of atoms equal to Avogadro's Number (N_A).

$$N = n \times N_A$$

$$\Rightarrow N = 0.2 \times (6 \times 10^{23})$$

Since each atom (nucleus) releases 190 MeV upon fission :

$$E = N \times \text{Energy per fission}$$

$$\Rightarrow E = (1.2 \times 10^{23}) \times 190 \text{ MeV}$$

$$\Rightarrow E = 228 \times 10^{23} \text{ MeV}$$

$$\Rightarrow \alpha \times 10^{23} \text{ MeV} = 228 \times 10^{23} \text{ MeV}$$

$$\Rightarrow \alpha = 228$$

Therefore, the correct answer is **228**.

Question2

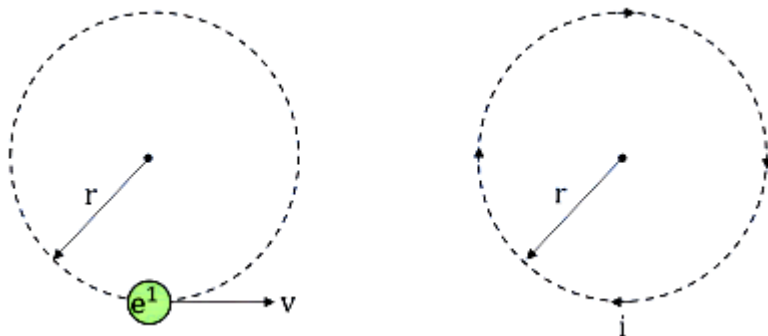
Using Bohr's model, calculate the ratio of the magnetic fields generated due to the motion of the electrons in the 2nd and 4th orbits of hydrogen atom _____.

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Answer: 32

Solution:

An electron moving in a circular orbit of radius r with speed v constitutes an electric current i .



The current is given by the charge per unit time:

$$i = \frac{e}{T} = \frac{e}{(2\pi r/v)} = \frac{ev}{2\pi r}$$

The magnetic field B at the centre of a circular current loop is:

$$B = \frac{\mu_0 i}{2r} = \frac{\mu_0 ev}{4\pi r^2}$$

In Bohr's model for a hydrogen atom:

The velocity of the electron $v \propto \frac{1}{n}$

The radius of the orbit $r \propto n^2$

Substituting these proportionalities into the magnetic field formula:

$$B \propto \frac{v}{r^2} \propto \frac{\left(\frac{1}{n}\right)}{(n^2)^2}$$

$$B \propto \frac{1}{n^5}$$

We need the ratio of the magnetic field in the 2nd orbit (B_2) to the 4th orbit (B_4):

$$\frac{B_2}{B_4} = \left(\frac{n_4}{n_2}\right)^5$$

Putting the value of principle quantum numbers $n_2 = 2$ and $n_4 = 4$:

$$\frac{B_2}{B_4} = \left(\frac{4}{2}\right)^5$$

$$\frac{B_2}{B_4} = (2)^5 = 32$$

Therefore, the ratio of the magnetic fields is 32:1. Thus, the correct answer is 32.

Question3

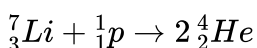
The energy released when $\frac{7}{17.13}$ kg of ${}^7_3\text{Li}$ is converted into ${}^4_2\text{He}$ by proton bombardment is $\alpha \times 10^{32}$ eV. The value of α is ____ . (Nearest integer) (Mass of ${}^7_3\text{Li} = 7.0183\text{u}$, mass of ${}^4_2\text{He} = 4.004\text{u}$, mass of proton = 1.008u and $1\text{u} = 931\text{MeV}/c^2$ and Avogadro number = 6.0×10^{23})

JEE Main 2026 (Online) 6th April Morning Shift

Answer: 6

Solution:

The given reaction is the proton bombardment of lithium:



Mass of reactants:

$$m_{\text{reactants}} = m({}^7_3\text{Li}) + m({}^1_1\text{p}) = 7.0183 + 1.008 = 8.0263 \text{ u}$$

Mass of products:

$$m_{\text{products}} = 2 \times m({}_2^4\text{He}) = 2 \times 4.004 = 8.008 \text{ u}$$

Mass defect:

$$\Delta m = m_{\text{reactants}} - m_{\text{products}} = 8.0263 - 8.008 = 0.0183 \text{ u}$$

$$E_{\text{per reaction}} = \Delta m \times 931 \text{ MeV}$$

$$E_{\text{per reaction}} = 0.0183 \times 931 = 17.04 \text{ MeV}$$

Convert this to eV:

$$E_{\text{per reaction}} = 17.04 \times 10^6 \text{ eV} = 1.704 \times 10^7 \text{ eV}$$

Given mass of Li:

$$m_{\text{given}} = \frac{7}{17.13} \text{ kg}$$

Convert to grams:

$$m_{\text{given}} = \frac{7}{17.13} \times 1000 = 408.6 \text{ g (approximately)}$$

Molar mass of Li-7 = 7 g/mol.

Number of moles:

$$n = \frac{408.6}{7} = 58.37 \text{ mol}$$

Number of atoms:

$$N = n \times N_A = 58.37 \times 6.0 \times 10^{23} = 3.50 \times 10^{25}$$

Each Li atom undergoes one reaction, so

$$E_{\text{total}} = N \times E_{\text{per reaction}}$$

$$E_{\text{total}} = 3.50 \times 10^{25} \times 1.704 \times 10^7$$

$$E_{\text{total}} = 5.96 \times 10^{32} \text{ eV}$$

The question states:

$$E_{\text{total}} = \alpha \times 10^{32} \text{ eV}$$

$$\Rightarrow \alpha = 5.96 \approx 6$$

Final Answer

$$\boxed{\alpha = 6}$$

Question4

If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is _____ m. (Atomic number of gold = 79 and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ in SI units)

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Options:

A.

$$2.95 \times 10^{-16}$$

B.

$$3.85 \times 10^{-14}$$

C.

$$2.95 \times 10^{-14}$$

D.

$$3.85 \times 10^{-16}$$

Answer: C

Solution:

Let an α -particle fired directly at the centre of a target nucleus, gold foil.

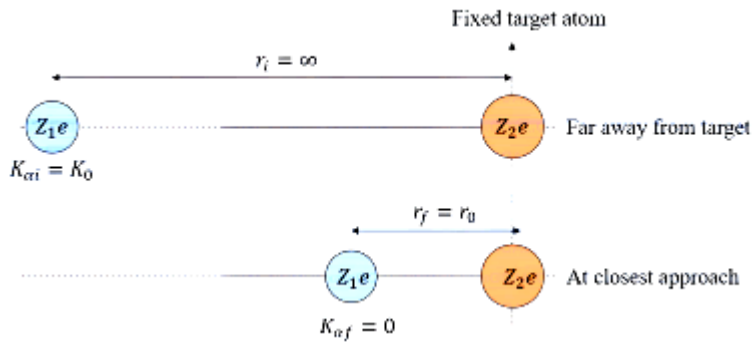
The α -particle has maximum kinetic energy (K) and practically zero electrostatic potential energy at very far away from target material.

The initial Kinetic Energy (K_0) of the alpha particle is :

$$K_0 = 7.7\text{MeV}$$

As it gets closer, the positive charge of the α -particle and the positive charge of the target nucleus repel each other. The α -particle slows down, converting its kinetic energy into electrostatic potential energy.

At the closest approach (r_0) it momentarily stops before reversing direction. At this exact point, all its initial kinetic energy has been converted into potential energy.



The electrostatic potential energy (U) between two point charges q_1 and q_2 separated by a distance r_0 is :

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_0}$$

By the Law of Conservation of Energy :

$$\text{Initial Kinetic Energy (} K_0 \text{)} = \text{Final Potential Energy (} U \text{)}$$

The electrostatic potential energy between two point charges (q_1 and q_2) separated by a distance r is :

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

An α -particle is a Helium nucleus, so it has 2 protons. So, the charge of α -particle (q_1) is $q_1 = +2e$. The atomic number of the target is $Z = 79$, so it has 79 protons. So, the charge of target nucleus (q_2) is $q_2 = +79e$.

Substituting into our energy conservation equation :

$$K_0 = \frac{1}{4\pi\epsilon_0} \frac{(2e)(79e)}{r_0}$$

$$\Rightarrow r_0 = \frac{1}{4\pi\epsilon_0} \frac{158e^2}{K_0}$$

The kinetic energy of alpha particle is,

$$K_0 = 7.7\text{MeV} = 7.7 \times 10^6 \text{eV}$$

Since $1\text{eV} = 1.6 \times 10^{-19} \text{J}$, we get :

$$K_0 = (7.7 \times 10^6) \times (1.6 \times 10^{-19}) \text{J}$$

$$\Rightarrow K_0 = 12.32 \times 10^{-13} \text{J}$$

Putting in the closest approach formula,

$$r_0 = \frac{(9 \times 10^9) \times 158 \times (1.6 \times 10^{-19})^2}{12.32 \times 10^{-13}}$$

$$\Rightarrow r_0 = \frac{(9 \times 158 \times 2.56) \times 10^{-29}}{12.32 \times 10^{-13}}$$

$$\Rightarrow r_0 = 295 \times 10^{-16} \text{m}$$

$$\Rightarrow r_0 = 2.95 \times 10^{-14} \text{m}$$

Therefore, the distance of closest approach is $r_0 = 2.95 \times 10^{-14} \text{m}$.

Hence, the correct option is (C).

Question5

The energy of an electron in an orbit of the Bohr's atom is $-0.04E_0$ eV where E_0 is the ground state energy. If L is the angular momentum of the electron in this orbit and h is the Planck's constant, then

$\frac{2\pi L}{h}$ is _____ :

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Options:

A.

6

B.

2

C.

5

D.

4

Answer: C

Solution:

In the Bohr model of the atom, the energy of an electron in the n^{th} orbit (E_n) is related to the ground state energy (E_0) by the formula :

$$E_n = -\frac{E_0}{n^2}$$

It is given that the energy in the current orbit is $-0.04E_0$ eV.

$$-0.04E_0 = -\frac{E_0}{n^2}$$

$$\Rightarrow 0.04 = \frac{1}{n^2}$$

$$\Rightarrow n^2 = \frac{100}{4} = 25$$

$$\Rightarrow n = \sqrt{25} = 5$$

So, the electron is in the 5th orbit of the Bohr's atom.

Bohr's second postulate states that the angular momentum of an electron is quantized and is an integral multiple of $h/2\pi$:

$$L = \frac{nh}{2\pi}$$

Where n is the principal quantum number (orbit number).

$$\frac{2\pi L}{h} = n = 5$$

Therefore, the value of $\frac{2\pi L}{h}$ is 5 .

Hence, the correct option is (C).

Question6

7.9MeV α -particle scatters from a target material of atomic number 79 . From the given data the estimated diameter of nuclei of the target material is (approximately) _____ m.

$$\left[\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2 \text{ and electron charge } = 1.6 \times 10^{-19} \text{C} \right]$$

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Options:

A.

$$2.88 \times 10^{-14}$$

B.

$$5.76 \times 10^{-14}$$

C.

$$1.44 \times 10^{-13}$$

D.

$$1.69 \times 10^{-12}$$

Answer: B

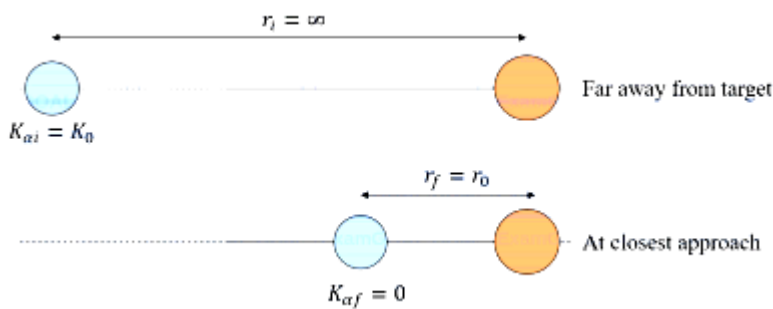
Solution:

Let an α -particle fired directly at the center of a target nucleus.

The α -particle has maximum kinetic energy (K) and practically zero electrostatic potential energy at very far away from target material.

As it gets closer, the positive charge of the α -particle and the positive charge of the target nucleus repel each other. The α -particle slows down, converting its kinetic energy into electrostatic potential energy.

At the closest approach (r_0) it momentarily stops before reversing direction. At this exact point, all its initial kinetic energy has been converted into potential energy.



By the Law of Conservation of Energy:

Initial Kinetic Energy (K_0) = Final Potential Energy (U)

The electrostatic potential energy between two point charges (q_1 and q_2) separated by a distance r is :

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

An α -particle is a Helium nucleus, so it has 2 protons. So, the charge of α -particle (q_1) is $q_1 = +2e$

The atomic number of the target is $Z = 79$, so it has 79 protons, So, the charge of target nucleus (q_2) is $q_2 = +79e$

Substituting into our energy conservation equation :

$$K_0 = \frac{1}{4\pi\epsilon_0} \frac{(2e)(79e)}{r_0}$$

$$\Rightarrow r_0 = \frac{1}{4\pi\epsilon_0} \frac{158e^2}{K_0}$$

The kinetic energy of alpha particle is,

$$K_0 = 7.9\text{MeV} = 7.9 \times 10^6 \text{eV}$$

Since $1\text{eV} = 1.6 \times 10^{-19} \text{J}$, we get:

$$K_0 = (7.9 \times 10^6) \times (1.6 \times 10^{-19}) \text{ J}$$

$$\Rightarrow K_0 = 12.64 \times 10^{-13} \text{ J}$$

Putting in the closest approach formula,

$$r_0 = \frac{(9 \times 10^9) \times 158 \times (1.6 \times 10^{-19})^2}{12.64 \times 10^{-13}}$$

$$\Rightarrow r_0 = \frac{(9 \times 158 \times 2.56) \times 10^{-29}}{12.64 \times 10^{-13}}$$

$$\Rightarrow r_0 = 288 \times 10^{-16} \text{ m}$$

$$\Rightarrow r_0 = 2.88 \times 10^{-14} \text{ m}$$

Therefore, the distance of closest approach is $r_0 = 2.88 \times 10^{-14} \text{ m}$.

The distance from the centre of the nucleus to this stopping point is the distance of closest approach. Because the α -particle cannot physically penetrate the nucleus using just the electrostatic force in this classical model, r_0 gives us an upper estimate for the radius of the target nucleus.

$$\text{Diameter} = 2 \times \text{Radius}$$

$$\text{Diameter} = 2 \times r_0$$

$$\text{Diameter} = 2 \times (2.88 \times 10^{-14} \text{ m})$$

$$\text{Diameter} = 5.76 \times 10^{-14} \text{ m}$$

Therefore, the diameter of nuclei of the target material is approximately $5.76 \times 10^{-14} \text{ m}$.

Hence, the correct option is (B).

Question 7

The minimum frequency of photon required to break a particle of mass 15.348 amu into 4α particles is ____ kHz .

[mass of He nucleus =

4.002 amu, 1 amu = $1.66 \times 10^{-27} \text{ kg}$, $h = 6.6 \times 10^{-34} \text{ J.s}$ and

$c = 3 \times 10^8 \text{ m/s}$]

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Options:

A.

$$14.94 \times 10^{20}$$

B.

$$9 \times 10^{19}$$

C.

$$9 \times 10^{20}$$

D.

$$14.94 \times 10^{19}$$

Answer: D

Solution:

Einstein's equation tells us that mass and energy are interchangeable.

$$E = mc^2$$

When a nuclear reaction occurs, the total mass of the products is often slightly different from the initial mass of the reactants.

- If the products are lighter, the missing mass is released as energy.
- If the products are heavier, energy must be supplied to create that extra mass.

It is given that a particle breaks into 4α particles.

Initial mass of the single particle. $M_{\text{initial}} = 15.348\text{amu}$

Final mass of 4α particles $= M_{\text{final}} = 4 \times \text{mass of He nucleus}$

$$M_{\text{final}} = 4 \times 4.002\text{amu} = 16.008\text{amu}$$

The mass defect (Δm) during this nuclear reaction is,

$$\Delta m = M_{\text{final}} - M_{\text{initial}}$$

$$\Rightarrow \Delta m = 16.008\text{amu} - 15.348\text{amu}$$

$$\Rightarrow \Delta m = 0.66\text{amu}$$

It is given that $1\text{amu} = 1.66 \times 10^{-27} \text{ kg}$.

$$\Delta m = 0.66 \times (1.66 \times 10^{-27})\text{kg}$$

$$\Rightarrow \Delta m = 1.0956 \times 10^{-27} \text{ kg}$$

Now using Einstein's equation,

$$E = \Delta m \cdot c^2$$

$$\Rightarrow E = (1.0956 \times 10^{-27}) \times (3 \times 10^8)^2$$

$$\Rightarrow E = (1.0956 \times 10^{-27}) \times (9 \times 10^{16})$$

$$\Rightarrow E = 9.8604 \times 10^{-11} \text{ Joules}$$

This energy is provided by a single photon. The energy of a photon is directly proportional to its frequency (ν), given by Planck's equation:

$$E = h\nu$$

Where h is Planck's constant ($6.6 \times 10^{-34} \text{ J} \cdot \text{s}$).

$$\nu = \frac{E}{h}$$

$$\Rightarrow \nu = \frac{9.8604 \times 10^{-11}}{6.6 \times 10^{-34}}$$

$$\Rightarrow \nu = 1.494 \times 10^{23} \text{ Hz}$$

$$\Rightarrow \nu_{\text{kHz}} = 1.494 \times 10^{23} \times 10^{-3} \text{ kHz}$$

$$\Rightarrow \nu_{\text{kHz}} = 14.94 \times 10^{19} \text{ kHz}$$

Therefore, the minimum frequency of photon required is $14.94 \times 10^{19} \text{ kHz}$.

Hence, the correct option is (D).

Question8

The smallest wavelength of Lyman series is 91 nm . The difference between the largest wavelengths of Paschen and Balmer series is nearly _____ nm.

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Options:

A.

1784

B.

1875

C.

1217

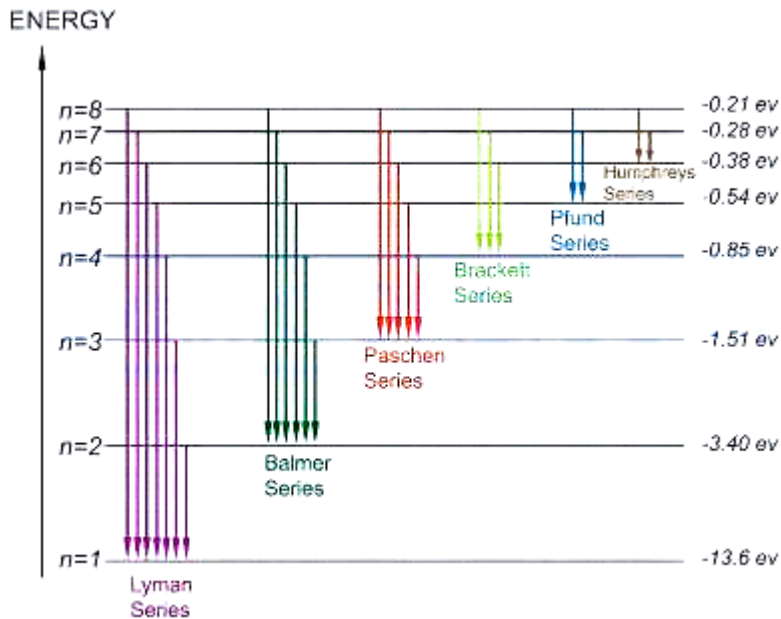
D.

1550

Answer: C

Solution:

The energy of n^{th} energy state of hydrogen atom is $E_n = -\frac{13.6}{n^2}$



The wavelengths of the spectral lines of hydrogen-like atoms are governed by the Rydberg formula

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Where :

- λ is the wavelength of the emitted photon.
- R is the Rydberg constant.
- n_1 is the principal quantum number of the lower energy level.
- n_2 is the principal quantum number of the higher energy level ($n_2 > n_1$).

The smallest wavelength corresponds to the highest energy transition (from $n_2 = \infty$ to n_1).

The largest wavelength corresponds to the lowest energy transition (from the immediate next energy level, $n_2 = n_1 + 1$, down to n_1).

For the Lyman series, the electron transitions fall to the ground state, so $n_1 = 1$. The smallest wavelength ($\lambda_{L(\min)}$) occurs when $n_2 = \infty$.

It is given that $\lambda_{L,\min} = 91 \text{ nm}$.

Using the Rydberg formula :

$$\frac{1}{\lambda_{L,\min}} = R \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right)$$

$$\Rightarrow \frac{1}{91} = R(1 - 0) \Rightarrow R = \frac{1}{91} \text{ nm}^{-1}$$

For the Balmer series, the lower energy level is $n_1 = 2$.

The largest wavelength ($\lambda_{B,\max}$) corresponds to the smallest energy jump, which is from $n_2 = 3$ to $n_1 = 2$.

Applying the formula :

$$\frac{1}{\lambda_{B,\max}} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R \left(\frac{5}{36} \right)$$

$$\Rightarrow \frac{1}{\lambda_{B,\max}} = \frac{1}{91} \times \frac{5}{36} \text{ nm}^{-1}$$

$$\Rightarrow \lambda_{B,\max} = \frac{91 \times 36}{5} \text{ nm} = 655.2 \text{ nm}$$

For the Paschen series, the lower energy level is $n_1 = 3$.

The largest wavelength ($\lambda_{P,\max}$) corresponds to the smallest energy jump, which is from $n_2 = 4$ to $n_1 = 3$.

Applying the formula :

$$\frac{1}{\lambda_{P,\max}} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = R \left(\frac{1}{9} - \frac{1}{16} \right)$$

$$\Rightarrow \frac{1}{\lambda_{P,\max}} = \frac{1}{91} \times \frac{7}{144} \text{ nm}^{-1}$$

$$\Rightarrow \lambda_{P,\max} = \frac{91 \times 144}{7} \text{ nm} = 1871 \text{ nm}$$

So, the difference between these two largest wavelengths :

$$\Delta\lambda = \lambda_{P,\max} - \lambda_{B,\max}$$

$$\Rightarrow \Delta\lambda = 1871 \text{ nm} - 655.2 \text{ nm}$$

$$\Rightarrow \Delta\lambda = 1215.8 \text{ nm} \approx 1216 \text{ nm}$$

Therefore, the difference between the largest wavelengths of Paschen and Balmer series is nearly 1216 nm.

Hence, the correct option is (C).

Question9

In hydrogen atom spectrum, ($R \rightarrow$ Rydberg's constant)

A. the maximum wavelength of the radiation of Lyman series is $\frac{4}{3R}$

B. the Balmer series lies in the visible region of the spectrum

C. the minimum wavelength of the radiation of Paschen series is $\frac{9}{R}$

D. the minimum wavelength of Lyman series is $\frac{5}{4R}$

Choose the correct answer from the options given below :

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Options:

A.

A, B Only

B.

B, D Only

C.

A, B and D Only

D.

A, B and C Only

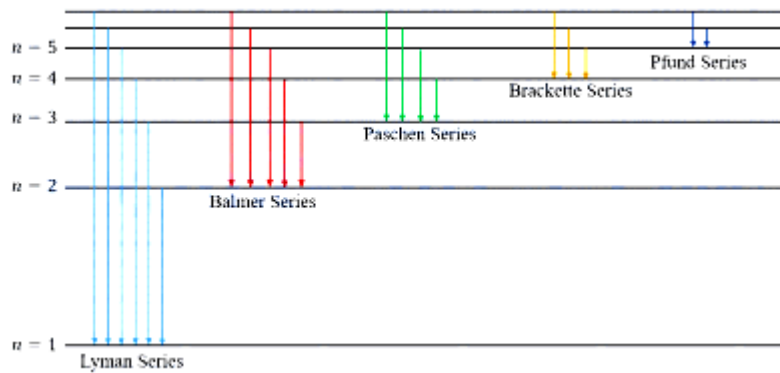
Answer: D

Solution:

Using the Rydberg formula :

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Where, λ is the wavelength of radiated photon when electron makes a transition from initial energy state n_2 to final energy state n_1 , and R is the Rydberg's constant.



For statement A

The given series is Lyman ($n_1 = 1$).

Maximum wavelength implies minimum energy. The transition corresponds to the smallest energy gap, which is between the first excited state ($n_2 = 2$) and the ground state ($n_1 = 1$).

$$\frac{1}{\lambda_{\max}} = R \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\Rightarrow \frac{1}{\lambda_{\max}} = R \left(1 - \frac{1}{4} \right) = \frac{3R}{4} \Rightarrow \lambda_{\max} = \frac{4}{3R}$$

Hence, the statement A is true.

For statement B

The given series is Balmer ($n_1 = 2$).

The Balmer series corresponds to electron transitions from higher shells ($n = 3, 4, 5 \dots$) to the second shell ($n = 2$). The wavelengths produced (e.g., 656 nm, 486 nm) fall within the range of approximately 400 nm to 700 nm, which is the visible spectrum

So, Balmer series lines are indeed visible. Hence, the statement B is true.

For Statement C

The given series is Paschen ($n_1 = 3$).

Minimum wavelength implies maximum energy. The transition corresponds to the largest energy gap, which is from infinity ($n_2 = \infty$) to the level $n_1 = 3$.

$$\frac{1}{\lambda_{\min}} = R \left(\frac{1}{3^2} - \frac{1}{\infty^2} \right)$$

$$\Rightarrow \frac{1}{\lambda_{\min}} = R \left(\frac{1}{9} \right) = \frac{R}{9} \Rightarrow \lambda_{\min} = \frac{9}{R}$$

Hence, the statement C is also true.

For statement D

The given series is Lyman ($n_1 = 1$).

Minimum wavelength means ($n_2 = \infty \rightarrow n_1 = 1$).

$$\frac{1}{\lambda_{\min}} = R \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right)$$

$$\Rightarrow \frac{1}{\lambda_{\min}} = R(1 - 0) = R$$

$$\Rightarrow \lambda_{\min} = 1R$$

Hence, the statement D is false.

- Statement A is Correct.
- Statement B is Correct.
- Statement C is Correct.
- Statement D is Incorrect.

Therefore, the correct option is (D).

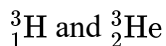
Question10

Which of the following pair of nuclei are isobars of the element?

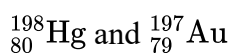
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Options:

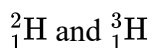
A.



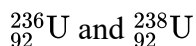
B.



C.



D.



Answer: A

Solution:

Atoms are represented by the notation X_Z^A , where :

- 1. A is the Mass Number (total number of protons and neutrons).

- 2. Z is the Atomic Number (number of protons).

Isobars are atoms of different chemical elements that have the same mass number (A) but different atomic numbers (Z).

Option A : H_1^3 and He_2^3

- Mass numbers: 3 and 3 (Same)
- Atomic numbers: 1 and 2 (Different)

Since they have the same mass number ($A = 3$) but represent different elements ($Z = 1$ vs $Z = 2$), they are Isobars. Hence, option (A) is correct.

Option B : Hg_{80}^{198} and Au_{79}^{197}

- Mass numbers: 198 and 197 (Different)
- Atomic numbers: 80 and 79 (Different)

So, Hg_{80}^{198} and Au_{79}^{197} are neither isobars nor isotopes. Hence, option (B) is incorrect.

Option C : H_1^2 and H_1^3

- Mass numbers: 2 and 3 (Different)
- Atomic numbers: 1 and 1 (Same)

So, H_1^2 and H_1^3 are Isotopes of Hydrogen (Deuterium and Tritium). Hence, option (C) is incorrect.

Option D : U_{92}^{236} and U_{92}^{238}

- Mass numbers: 236 and 238 (Different)
- Atomic numbers: 92 and 92 (Same)

So, U_{92}^{236} and U_{92}^{238} are Isotopes (same element, different mass) not isobars. Hence, option (D) is incorrect.

Question11

Given below are two statements :

Statement I : For all elements, greater the mass of the nucleus, greater is the binding energy per nucleon.

Statement II : For all elements, nuclei with less binding energy per nucleon transforms to nuclei with greater binding energy per nucleon.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Statement I is false but Statement II is true

B.

Both Statement I and Statement II are true

C.

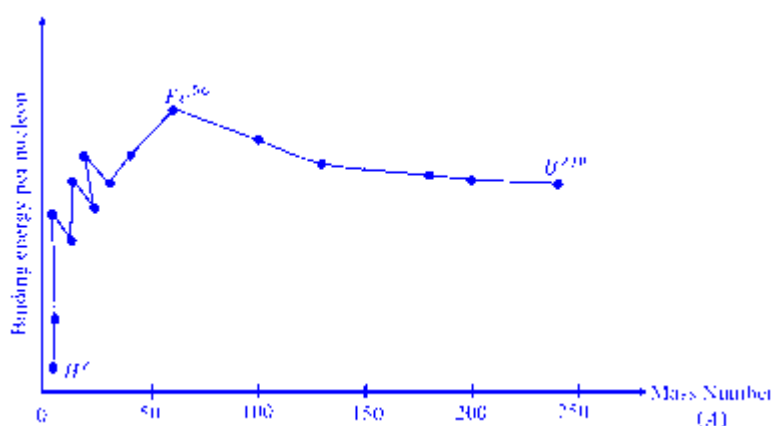
Both Statement I and Statement II are false

D.

Statement I is true but Statement II is false

Answer: A

Solution:



For light nuclei, the binding energy per nucleon ($\frac{BE}{A}$) increases rapidly as mass increases. The curve reaches a maximum around Mass Number $A \approx 56$ (Iron, Fe). This is the most stable region. For heavy nuclei the $\frac{BE}{A}$ actually decreases as mass increases.

So, the binding energy per nucleon does not constantly increase with mass. It increases initially, peaks, and then decreases.

Hence, statement I is false.

Nature always tends towards the state of lowest potential energy (highest stability). Higher Binding Energy per Nucleon means the nucleus is more tightly bound and therefore more stable.

So, in nuclear fission a heavy nucleus (low $\frac{BE}{A}$) splits into two lighter nuclei (higher $\frac{BE}{A}$). And in nuclear fusion the light nuclei (low BE/A) combine to form a heavier nucleus (higher BE/A).

In both fundamental nuclear processes (fission and fusion), the system evolves from a state of lower binding energy per nucleon to a state of higher binding energy per nucleon. This is the driving force behind nuclear energy release.

Hence, statement II is true.

Therefore, statement I is false, statement II is true. Hence, the correct option is (A).

Question12

Two electrons are moving in orbits of two hydrogen like atoms with speeds 3×10^5 m/s and 2.5×10^5 m/s respectively. If the radii of these orbits are nearly same then the possible order of energy states are ____ respectively.

JEE Main 2026 (Online) 24th January Morning Shift

Options:

A.

8 and 10

B.

10 and 12

C.

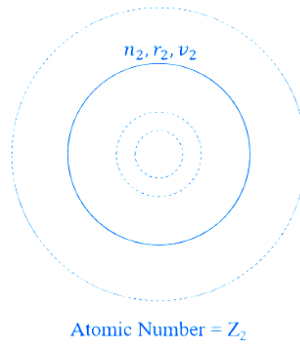
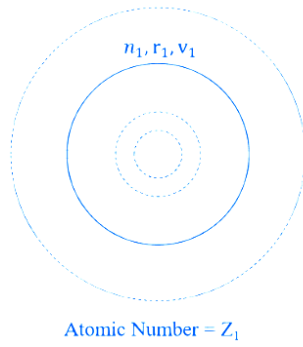
9 and 8

D.

6 and 5

Answer: D

Solution:



The velocity of an electron in the n^{th} orbit is proportional to the atomic number Z and inversely proportional to the principal quantum number n .

$$v \propto \frac{Z}{n}$$

$$\Rightarrow v = k_v \cdot \frac{Z}{n} \dots (i)$$

Where k_v is constant for the velocity.

The radius of the n^{th} orbit is proportional to the square of the principal quantum number n and inversely proportional to the atomic number Z .

$$r \propto \frac{n^2}{Z}$$

$$\Rightarrow r = k_r \cdot \frac{n^2}{Z} \dots (ii)$$

Let r_1 be the radius for the first atom (with state n_1, Z_1) and r_2 for the second (with state n_2, Z_2).

$$r_1 = r_2$$

$$\Rightarrow k_r \cdot \frac{n_1^2}{Z_1} = k_r \cdot \frac{n_2^2}{Z_2}$$

$$\Rightarrow \frac{n_1^2}{Z_1} = \frac{n_2^2}{Z_2}$$

$$\frac{Z_1}{Z_2} = \frac{n_1^2}{n_2^2} = \left(\frac{n_1}{n_2} \right)^2 \dots (iii)$$

Let v_1 and v_2 be the speeds.

$$v_1 = k_v \frac{Z_1}{n_1} \dots (iv)$$

$$v_2 = k_v \frac{Z_2}{n_2} \dots (v)$$

$$\therefore \frac{v_1}{v_2} = \frac{k_v \left(\frac{Z_1}{n_1} \right)}{k_v \left(\frac{Z_2}{n_2} \right)}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{Z_1}{n_1} \times \frac{n_2}{Z_2} \dots (vi)$$

$$\Rightarrow \frac{v_1}{v_2} = \left[\left(\frac{n_1}{n_2} \right)^2 \right] \cdot \left(\frac{n_2}{n_1} \right) = \frac{n_1}{n_2}$$

It is given that the speed of first electron is $v_1 = 3 \times 10^5$ m/s and that of electron in second hydrogen like atom is $v_2 = 2.5 \times 10^5$ m/s

$$\frac{n_1}{n_2} = \frac{3 \times 10^5}{2.5 \times 10^5}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{3}{2.5}$$

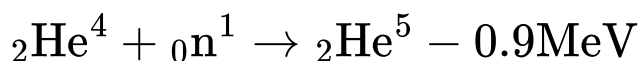
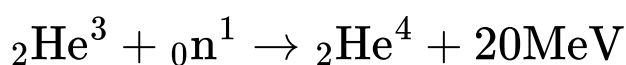
$$\Rightarrow \frac{n_1}{n_2} = \frac{6}{5}$$

So, the smallest possible energy states in two atoms are 6 and 5.

Hence, the correct option is (D).

Question13

The binding energy for the following nuclear reactions are expressed in MeV .



If X_3, X_4, X_5 denote the stability of ${}_2\text{He}^3, {}_2\text{He}^4$ and ${}_2\text{He}^5$, respectively, then the correct order is :

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Options:

A.

$$X_4 > X_5 > X_3$$

B.

$$X_4 < X_5 < X_3$$

C.

$$X_4 > X_5 < X_3$$

D.

$$X_4 = X_5 = X_3$$

Answer: A

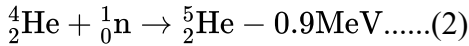
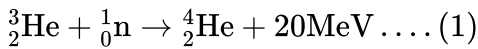
Solution:

A nucleus is more stable if it has a higher binding energy i.e., is in a lower energy state.

In exothermic reaction energy is released, $\Delta Q > 0$. The products are more stable than the reactants.

In endothermic reaction energy is absorbed, $\Delta Q < 0$. The products are less stable than the reactants.

The given two nuclear reactions are,



In ${}^3_2\text{He} + {}^1_0\text{n} \rightarrow {}^4_2\text{He} + 20\text{MeV}$ the released energy is 20MeV ($Q = +20\text{MeV}$).

Since energy is released, the product ${}^4_2\text{He}$ is significantly more stable than the reactant ${}^3_2\text{He}$.

Therefore, $X_4 > X_3$.

In ${}^4_2\text{He} + {}^1_0\text{n} \rightarrow {}^5_2\text{He} - 0.9\text{MeV}$ the absorbed energy is 0.9MeV ($Q = -0.9\text{MeV}$).

Since energy must be supplied to this reaction, the product ${}^5_2\text{He}$ is less stable than the reactant ${}^4_2\text{He}$.

Therefore, $X_4 > X_5$.

From reaction 1, the jump in stability from X_3 to X_4 is very large (20 MeV).

From reaction 2, the drop in stability from X_4 to X_5 is very small (0.9 MeV).

Because X_5 is only slightly less stable than X_4 , but X_3 is significantly less stable than X_4 , we can conclude that X_5 is still more stable than X_3 .

Therefore, the order of stability is $X_4 > X_5 > X_3$.

Hence, the correct option is (A).

Question14

An atom ${}^8_3\text{X}$ is bombarded by shower of fundamental particles and in 10 s this atom absorbed 10 electrons, 10 protons and 9 neutrons. The percentage growth in the surface area of the nucleons is recorded by :

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

150%

B.

900%

C.

125%

D.

225%

Answer: C

Solution:

The mass number of initial nucleus is $A_1 = 8$

The radius of a nucleus is proportional to $A^{1/3}$ ($R = R_0 A^{1/3}$).

So, the nuclear radius of initial nucleus is $R_1 = R_0(8)^{1/3} = 2R_0$

Hence, initial surface area

$$S_1 = 4\pi R_1^2$$

$$\Rightarrow S_1 = 4\pi(2R_0)^2 = 16\pi R_0^2$$

The atom absorbs 10 protons and 9 neutrons.

Total nucleons in final state of nucleus is 10 protons + 9 neutrons = 19 nucleons.

So, new mass number is $A_2 = A_1 + 19 = 8 + 19 = 27$

The final radius,

$$R_2 = R_0(27)^{1/3} = 3R_0$$

So, final surface area is

$$S_2 = 4\pi(3R_0)^2 = 36\pi R_0^2$$

The question has asked about the percentage growth recorded, that means final surface area is what percentage

So, the percentage growth is,

So, the percentage growth is,

$$\frac{\Delta S}{S_1} \times 100\% = \frac{36\pi R_0^2 - 16\pi R_0^2}{16\pi R_0^2} \times 100\% = 125\%$$

Question15

A nucleus has mass number α and radius R_α . Another nucleus has mass number β and radius R_β .

If $\beta = 8\alpha$ then R_α/R_β is :

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

1

B.

2

C.

8

D.

0.5

Answer: D

Solution:

The nucleus is assumed as a sphere where the volume is proportional to the total number of nucleons (mass number, A).

Because the density of nuclear matter is approximately constant, the radius R of a nucleus is related to its mass number A by the following formula :

$$R = R_0 A^{1/3}$$

Where R is the nuclear radius.

R_0 is a constant (approximately 1.2×10^{-15} m or 1.2 fm).

A is the mass number.

For the first nucleus : $R_\alpha = R_0(\alpha)^{1/3}$

For the second nucleus : $R_\beta = R_0(\beta)^{1/3}$

Taking the ratio $\frac{R_\alpha}{R_\beta}$:

$$\frac{R_\alpha}{R_\beta} = \frac{R_0\alpha^{1/3}}{R_0\beta^{1/3}} = \left(\frac{\alpha}{\beta}\right)^{1/3}$$

It is given that $\beta = 8\alpha$.

$$\frac{R_\alpha}{R_\beta} = \left(\frac{\alpha}{8\alpha}\right)^{1/3} = \frac{1}{2} = 0.5$$

Therefore, the ratio of the radius of the first nucleus to the second is 0.5 . Hence, the correct option is (D).

Question16

Angular momentum of an electron in a hydrogen atom is $\frac{3h}{\pi}$, then the energy of the electron is _____ eV.

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

-1.51

B.

-0.85

C.

-0.38

D.

-0.28

Answer: C

Solution:

According to Bohr's second postulate, the angular momentum (L) of an electron in a stationary orbit is quantized and given by :

$$L = \frac{nh}{2\pi}$$

The angular momentum is $L = \frac{3h}{\pi}$.

$$\Rightarrow \frac{nh}{2\pi} = \frac{3h}{\pi}$$

$$\Rightarrow \frac{n}{2} = 3$$

$$\Rightarrow n = 6$$

In the Bohr model, the total energy (E_n) of an electron in the n^{th} orbit of a hydrogen atom is the sum of its kinetic and potential energies. The energy in electron volts (eV) is derived as :

$$E_n = -13.6n^2 \text{ eV}$$

For $n = 6$:

$$E_6 = -\frac{13.6}{6^2}$$

$$E_6 = -\frac{13.6}{36}$$

$$E_6 \approx -0.3777 \dots \text{ eV}$$

Therefore, the energy of the electron is -0.38 eV . Therefore, the correct option is (C).

Question17

The binding energy per nucleon of ${}^{209}_{83}\text{Bi}$ is _____ MeV.

[Take $m({}^{209}_{83}\text{Bi}) = 208.980388 \text{ u}$, $m_p = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$, $1 \text{ u} = 931 \text{ MeV}/c^2$]

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

7.48

B.

7.84

C.

8.79

D.

6.94

Answer: B

Solution:

The mass defect is the difference between the sum of the masses of the individual nucleons (protons and neutrons) and the actual mass of the nucleus.

For Bi_{83}^{209} :

Number of protons (Z) = 83

Number of neutrons (N) = $A - Z = 209 - 83 = 126$

$$\Delta m = [Z m_p + N m_n] - M_{\text{nucleus}}$$

$$\Delta m = [83 \times 1.007825 + 126 \times 1.008665] - 208.980388$$

$$\Delta m = [83.649475 + 127.09179] - 208.980388$$

$$\Delta m = 210.741265 - 208.980388 = 1.760877 \text{ u}$$

Using Einstein's mass-energy equivalence, the binding energy is:

$$\text{B.E.} = \Delta m \times 931 \text{ MeV/u}$$

$$\text{B.E.} = 1.760877 \times 931 \approx 1639.376 \text{ MeV}$$

The binding energy per nucleon is the total binding energy divided by the mass number ($A = 209$).

$$\frac{\text{B.E.}}{A} = \frac{1639.376}{209} \approx 7.8439 \text{ MeV}$$

Therefore, the binding energy per nucleon of Bi_{83}^{209} is 7.8439 MeV. Hence, the correct option is (B).

Question18

Two nuclei of mass number 3 combine with another nucleus of mass number 4 to yield a nucleus of mass number 10. If the binding energy per nucleon for the mass numbers 3,4 and 10 are 5.6MeV, 7.4MeV and 6.1 MeV , respectively, then in the process, $\Delta Mc^2 = \underline{\hspace{1cm}} \text{ MeV}$.

JEE Main 2026 (Online) 4th April Morning Shift

Options:

A.

6.9

B.

7.9

C.

2.2

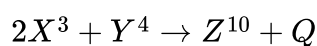
D.

4.3

Answer: C

Solution:

In nuclear reactions, the energy released or absorbed (ΔMc^2 or Q-value) is the difference between the total binding energy of the product nuclei and the total binding energy of the reactant nuclei.



The reactants consist of two nuclei of mass number $A = 3$ and one nucleus of mass number $A = 4$.

Binding energy (BE) is calculated as, $BE = (\text{Binding Energy per nucleon}) \times (\text{Mass Number})$.

For one nucleus of $A = 3$: $BE_3 = 3 \times 5.6\text{MeV} = 16.8\text{MeV}$

For two such nuclei: $2 \times BE_3 = 33.6\text{MeV}$

For one nucleus of $A = 4$: $BE_4 = 4 \times 7.4\text{MeV} = 29.6\text{MeV}$

Total BE of reactants = $33.6 + 29.6 = 63.2\text{MeV}$

The product is one nucleus of mass number $A = 10$.

For one nucleus of $A = 10$: $BE_{10} = 10 \times 6.1\text{MeV} = 61.0\text{MeV}$

The energy released or the change in mass-energy is :

$$\Delta Mc^2 = BE_{\text{products}} - BE_{\text{reactants}}$$

$$\Rightarrow \Delta Mc^2 = 61.0\text{MeV} - 63.2\text{MeV} = -2.2\text{MeV}$$

The negative sign indicates that the process is endothermic, meaning energy must be supplied for the reaction to occur because the product is less stable than the reactants.

Therefore, the magnitude of the energy change in the process is 2.2 MeV .

Thus, the correct option is (C).

Question19

In Rutherford's alpha-particle scattering experiment, only a few alpha particles rebound back because

- A. The size of gold nucleus is very small as compared to the size of gold atom.**
- B. Alpha particle and gold nucleus have equal charge.**
- C. The impact parameter is minimum for a few alpha particles.**
- D. A few alpha particles have very high kinetic energy.**
- E. Only a few alpha particles undergo head-on collision with the nuclei.**

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 4th April Evening Shift

Options:

A.

A, B Only

B.

B, E Only

C.

C, D Only

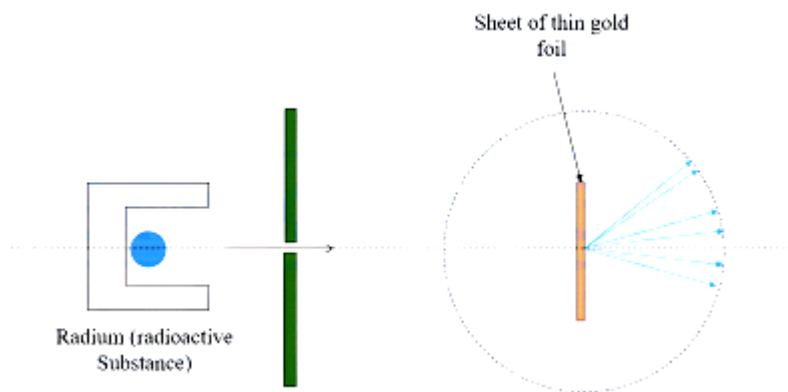
D.

A, C, E Only

Answer: D

Solution:

In the experiment, alpha particles (positively charged helium nuclei) are fired at a thin gold foil. Rutherford observed that most particles passed straight through, but a tiny fraction (about 1 in 8000) reflected back at angles greater than 90° .

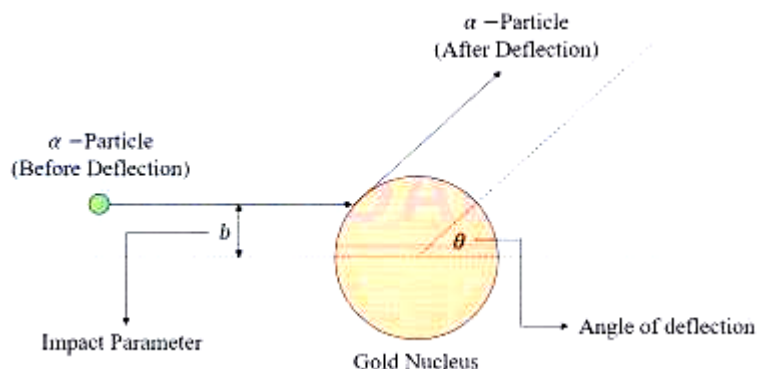


Statement A: Rutherford concluded that the positive charge and most of the mass of the atom are concentrated in an extremely small region called the nucleus. Since the nucleus is tiny compared to the total volume of the atom, the probability of an alpha particle hitting it or coming close enough to be repelled backward is very low.

Hence, statement A is true.

Statement B: Alpha particles have a charge of $+2e$, while a gold nucleus has a charge of $+79e$. They do not have equal charges. Hence, statement B is false.

Statement C: The impact parameter (b) is the perpendicular distance of the initial velocity vector of the alpha particle from the centre of the nucleus.



A rebound or large-angle scattering occurs when the impact parameter is very small (close to zero), meaning the particle is heading almost directly toward the nucleus. Only a few particles have such a small impact parameter. Hence, the statement C is true.

Statement D: The kinetic energy of the incident alpha particles is generally uniform in the beam; high kinetic energy would actually make it easier to penetrate closer to the nucleus rather than causing the few rebounds mentioned. Hence, the statement D is false.

Statement E: A head-on collision occurs when the particle moves directly toward the centre of the nucleus ($b = 0$). Because the nucleus is so small, only a very small number of particles are aligned for such a collision. Hence, the statement E is true.

Therefore, statements A, C, and E are the correct reasons for the observed phenomenon.

Thus, the correct option is (D).

Question20

In the hydrogen atom, the electron makes a transition from the higher orbit (i) to a lower orbit (f). The ratio of the radius of the orbits is given by $r_i : r_f = 16 : 4$. The wavelength of photon emitted due to this transition is _____ nm.

(Given Rydberg constant = $1.0973 \times 10^7/\text{m}$)

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

121

B.

242

C.

486

D.

974

Answer: C

Solution:

The radius of an electron in the n^{th} orbit of a hydrogen atom is given by the formula:

$$r_n = \left(\frac{\epsilon_0 h^2}{\pi m e^2} \right) n^2 \Rightarrow r_n \propto n^2$$

The ratio of radii of i^{th} orbit to f^{th} orbit is $r_i : r_f = 16 : 4$,

$$\frac{n_i^2}{n_f^2} = \frac{16}{4} = 4$$

$$\Rightarrow \frac{n_i}{n_f} = 2 \Rightarrow n_i = 2n_f$$

For a transition to a lower orbit in a hydrogen atom, the simplest integers satisfying this are $n_f = 2$ and $n_i = 4$.

The wavelength of the emitted photon is given by the Rydberg formula:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Substituting the values $n_f = 2$ and $n_i = 4$:

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R \left(\frac{1}{4} - \frac{1}{16} \right)$$

$$\Rightarrow \frac{1}{\lambda} = R \left(\frac{4-1}{16} \right) = \frac{3R}{16}$$

$$\Rightarrow \lambda = \frac{16}{3R}$$

Putting the value of Rydberg's constant $R = 1.0973 \times 10^7 \text{ m}^{-1}$

$$\lambda = \frac{16}{3 \times 1.0973 \times 10^7} \approx 4.86 \times 10^{-7} \text{ m} = 486 \times 10^{-9} \text{ m}$$

$$\Rightarrow \lambda = 486 \text{ nm}$$

Therefore, the wavelength of photon emitted due to this transition is 486 nm .

Hence, the correct option is (C).

Question21

Assuming the experimental mass of ${}^{12}_6\text{C}$ as $12u$, the mass defect of ${}^{12}_6\text{C}$ atom is _____ MeV/c^2 .

(Mass of proton = $1.00727u$. mass of neutron = $1.00866u$, $1u = 931.5\text{MeV}/c^2$ and c is the speed of the light in vacuum).

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

127.5

B.

89.03

C.

272.0

D.

92.0

Answer: B

Solution:

Mass defect (Δm) is the difference between the sum of the masses of the individual nucleons (protons and neutrons) and the actual experimental mass of the nucleus.

For the atom C_6^{12} , atomic number is $Z = 6$ (Number of protons)

Mass number is $A = 12$

Number of neutrons $N = A - Z = 12 - 6 = 6$

So, mass of 6 protons $= 6 \times 1.00727u = 6.04362u$

And the mass of 6 neutrons $= 6 \times 1.00866u = 6.05196u$

So, total theoretical mass $= 6.04362 + 6.05196 = 12.09558u$

It is given that the experimental mass of $C_6^{12} = 12u$:

Hence, the mass defect is

$$\Delta m = (\text{Total theoretical mass}) - (\text{Experimental mass})$$

$$\Rightarrow \Delta m = 12.09558u - 12u = 0.09558u$$

The energy equivalent of 1 u is given as $931.5\text{MeV}/c^2$.

$$\Delta m \text{ (in MeV}/c^2) = 0.09558 \times 931.5$$

$$\Rightarrow \Delta m \approx 89.03277 \approx 89.03\text{MeV}/c^2$$

Therefore, the mass defect of the C_6^{12} atom is approximately $89.03\text{MeV}/c^2$.

Thus, the correct option is (B).

Question22

The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β . The possible values of α and β are:-

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

27 and 20

B.

3 and 16

C.

5 and 36

D.

20 and 27

Answer: D

Solution:

According to the de Broglie relation, the momentum (p) of a photon is inversely proportional to its wavelength (λ):

$$p = \frac{h}{\lambda}$$

Therefore, the ratio of momenta is the inverse ratio of their wavelengths:

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1}$$

The Balmer series occurs when an electron transitions from a higher energy level (n_2) to the second energy level ($n_1 = 2$). The wavelength is given by Rydberg formula:

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

For 1st line of Balmer transition from $n = 3$ to $n = 2$.

$$\frac{1}{\lambda_1} = R \left(\frac{1}{4} - \frac{1}{9} \right) = R \left(\frac{9-4}{36} \right) = \frac{5R}{36} \Rightarrow \lambda_1 = \frac{36}{5R}$$

For 2nd line of Balmer transition from $n = 4$ to $n = 2$.

$$\frac{1}{\lambda_2} = R \left(\frac{1}{4} - \frac{1}{16} \right) = R \left(\frac{4-1}{16} \right) = \frac{3R}{16} \Rightarrow \lambda_2 = \frac{16}{3R}$$

So, the ratio (α/β) ,

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{16/3R}{36/5R} = \frac{16}{3} \times \frac{5}{36}$$

$$\Rightarrow \frac{p_1}{p_2} = \frac{4 \times 5}{3 \times 9} = \frac{20}{27} = \frac{\alpha}{\beta}$$

So, $\alpha = 20$ and $\beta = 27$.

Therefore, the correct option is (D).

Question23

The energy released if hydrogen atoms are combined to form ${}^4_2\text{He}$ is _____ MeV .

(Take binding energies per nucleon of ${}^2_1\text{H}$ and ${}^4_2\text{He}$ as 1.1 MeV and 7.2 MeV , respectively)

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

6.1

B.

24.4

C.

26.6

D.

5

Answer: B

Solution:

The binding energy (B.E.) of a nucleus is the energy required to disassemble it into its constituent nucleons.

$$\text{Total B.E.} = (\text{Binding Energy per nucleon}) \times (\text{Number of nucleons } A)$$

$$\text{Energy Released (Q)} = (\text{Total B.E. of Products}) - (\text{Total B.E. of Reactants})$$

$$\text{The given reaction is } H_1^2 + H_1^2 \rightarrow He_2^4$$

The reactants are two nuclei of H_1^2 .

$$\text{B.E. per nucleon of } H_1^2 = 1.1\text{MeV}$$

$$\text{Number of nucleons (A) for one } H_1^2 = 2$$

$$\text{So, total B.E. of reactants} = 2 \times (1.1 \times 2) = 4.4\text{MeV}$$

The product is one nucleus of He_2^4 .

$$\text{B.E. per nucleon of } He_2^4 = 7.2\text{MeV}$$

$$\text{Number of nucleons (A) for } He_2^4 = 4$$

$$\text{So, total B.E. of product} = 7.2 \times 4 = 28.8\text{MeV}$$

Hence, energy released during reaction is,

$$Q = 28.8\text{MeV} - 4.4\text{MeV}$$

$$\Rightarrow Q = 24.4\text{MeV}$$

Therefore, the energy released if hydrogen atoms are combined to form He_2^4 is 24.4 MeV .

Thus, the correct option is (D).

Question24

Two radioactive substances A and B of mass numbers 200 and 212 respectively, shows spontaneous α -decay with same Q value of 1 MeV . The ratio of energies of α -rays produced by A and B is ____ .

JEE Main 2026 (Online) 8th April Evening Shift

Options:

A.

$$\frac{2548}{2650}$$

B.

$$\frac{2706}{2646}$$

C.

$$\frac{2597}{2600}$$

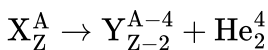
D.

$$\frac{2862}{2499}$$

Answer: C

Solution:

In α -decay, a parent nucleus (X) of mass number A decays into a daughter nucleus (Y) and an α -particle (Helium nucleus, mass number 4).



Using conservation of momentum, the momentum of the α -particle (p_α) must be equal and opposite to the momentum of the daughter nucleus (p_D).

And using energy balance (Q-value) the total kinetic energy released (Q) is shared between the α -particle and the daughter nucleus.

$$Q = K_\alpha + K_D$$

Kinetic energy can be expressed in terms of momentum (p) and mass (m) as $K = \frac{p^2}{2m}$.

Since $p_\alpha = p_D$, we have :

$$K_D = \frac{p^2}{2m_D} \text{ and } K_\alpha = \frac{p^2}{2m_\alpha}$$

$$\text{Therefore, } \frac{K_D}{K_\alpha} = \frac{m_\alpha}{m_D}.$$

Substituting $K_D = K_\alpha \left(\frac{m_\alpha}{m_D} \right)$ into the Q-value equation:

$$Q = K_\alpha + K_\alpha \left(\frac{m_\alpha}{m_D} \right) = K_\alpha \left(1 + \frac{m_\alpha}{m_D} \right) = K_\alpha \left(\frac{m_D + m_\alpha}{m_D} \right)$$

Since $m_D + m_\alpha \approx A$ (mass number of parent) and $m_D \approx A - 4$:

$$K_\alpha = Q \left(\frac{A-4}{A} \right)$$

The Q-value is the same (1 MeV) for both substances.

For Substance A ($A_A = 200$) :

$$K_{\alpha A} = Q \left(\frac{200-4}{200} \right) = Q \left(\frac{196}{200} \right)$$

For Substance B ($A_B = 212$) :

$$K_{\alpha B} = Q \left(\frac{212-4}{212} \right) = Q \left(\frac{208}{212} \right)$$

So, the ratio is;

$$\frac{K_{\alpha A}}{K_{\alpha B}} = \frac{Q \left(\frac{196}{200} \right)}{Q \left(\frac{208}{212} \right)} = \frac{196}{200} \times \frac{212}{208}$$

$$\Rightarrow \frac{K_{\alpha A}}{K_{\alpha B}} = \frac{49}{50} \times \frac{53}{52} = \frac{49 \times 53}{50 \times 52} = \frac{2597}{2600}$$

Therefore, the ratio of energies of α -rays produced by A and B is $\frac{2597}{2600}$.

Hence, the correct option is (C).

Question 25

An electron in the hydrogen atom initially in the fourth excited state makes a transition to n^{th} energy state by emitting a photon of energy 2.86 eV . The integer value of n will be _____.

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Answer: 2

Solution:

To find the integer value of n for which an electron transitions from the fourth excited state in a hydrogen atom, thus emitting a photon with an energy of 2.86 eV, we can follow these steps:

We use the formula for the energy difference associated with electron transitions in a hydrogen atom:

$$E = 13.6 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

However, in this question, it seems there's a typo in the original explanation, as both indices are given as n_1 . To correct it, we should use this formula:

$$E = 13.6 \left(\frac{1}{n^2} - \frac{1}{5^2} \right)$$

where $n_1 = 5$ (the fifth energy level or fourth excited state) and $n_2 = n$ (the state to which the electron transitions).

Given the photon's energy is 2.86 eV, set up the equation:

$$2.86 = 13.6 \left(\frac{1}{n^2} - \frac{1}{25} \right)$$

Solve for $\frac{1}{n^2}$:

$$\frac{1}{n^2} = \frac{2.86}{13.6} + \frac{1}{25}$$

Calculate the value:

$$\frac{1}{n^2} = 0.21 + 0.04 = 0.25$$

Find n^2 :

$$n^2 = \frac{1}{0.25} = 4$$

Consequently:

$$n = \sqrt{4} = 2$$

Thus, the electron transitions to the $n = 2$ energy state.

Question26

A radioactive nucleus n_2 has 3 times the decay constant as compared to the decay constant of another radioactive nucleus n_1 . If initial number of both nuclei are the same, what is the ratio of number of nuclei of n_2 to the number of nuclei of n_1 , after one half-life of n_1 ?

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Options:

A. $1/4$

B. $1/8$

C. 4

D. 8

Answer: A

Solution:

Let's work through the problem step by step.

The decay of a radioactive nucleus is given by the formula:

$$N(t) = N_0 e^{-\lambda t},$$

where:

$N(t)$ is the number of nuclei remaining at time t ,

N_0 is the initial number of nuclei,

λ is the decay constant.

For nucleus n_1 with decay constant λ_1 , its half-life $t_{1/2}$ is:

$$t_{1/2} = \frac{\ln 2}{\lambda_1}.$$

After one half-life of n_1 , the number of n_1 nuclei remaining is:

$$N_1 = N_0 e^{-\lambda_1 t_{1/2}} = N_0 e^{-\lambda_1 \frac{\ln 2}{\lambda_1}} = N_0 e^{-\ln 2} = \frac{N_0}{2}.$$

Nucleus n_2 has a decay constant that is 3 times that of n_1 :

$$\lambda_2 = 3\lambda_1.$$

For nucleus n_2 , the number of nuclei remaining after one half-life of n_1 (which is $t = \frac{\ln 2}{\lambda_1}$) is:

$$N_2 = N_0 e^{-\lambda_2 t} = N_0 e^{-3\lambda_1 \frac{\ln 2}{\lambda_1}} = N_0 e^{-3 \ln 2} = N_0 \cdot \frac{1}{2^3} = \frac{N_0}{8}.$$

Now, we find the ratio of n_2 nuclei to n_1 nuclei after this time:

$$\frac{N_2}{N_1} = \frac{\frac{N_0}{8}}{\frac{N_0}{2}} = \frac{1/8}{1/2} = \frac{1}{4}.$$

Therefore, the correct ratio is $\frac{1}{4}$, which corresponds to Option A.

Question 27

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The binding energy per nucleon is found to be practically independent of the atomic number A , for nuclei with mass numbers between 30 and 170 .

Reason (R) : Nuclear force is long range. In the light of the above statements, choose the correct answer from the options given below :

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Options:

- A. (A) is true but (R) is false
- B. Both (A) and (R) are true and (R) is the correct explanation of (A)
- C. (A) is false but (R) is true
- D. Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

Answer: A

Solution:

Let's analyze both statements:

Assertion (A): "The binding energy per nucleon is found to be practically independent of the atomic number A, for nuclei with mass numbers between 30 and 170."

In fact, for nuclei in this mass range, the binding energy per nucleon is nearly constant (around 7–8 MeV per nucleon). This is because the attractive nuclear force saturates, leading to a relatively uniform binding energy per nucleon in medium to heavy nuclei.

Therefore, Assertion (A) is true.

Reason (R): "Nuclear force is long range."

The nuclear force (or strong force) that holds the nucleons (protons and neutrons) together is actually very short-ranged. Its effective range is approximately 1 to 3 femtometers (fm), after which the force rapidly decreases.

Hence, Reason (R) is false.

Since the assertion is correct and the reason is incorrect, the correct answer is:

Option A: (A) is true but (R) is false.

Question28

During the transition of electron from state A to state C of a Bohr atom, the wavelength of emitted radiation is $2000\overset{o}{\text{\AA}}$ and it becomes $6000\overset{o}{\text{\AA}}$ when the electron jumps from state B to state C. Then the wavelength of the radiation emitted during the transition of electrons from state A to state B is

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Options:

A. $4000\overset{o}{A}$

B. $6000\overset{o}{A}$

C. $2000\overset{o}{A}$

D. $3000\overset{o}{A}$

Answer: D

Solution:

$$\frac{1}{\lambda_{AC}} = \frac{E_A - E_C}{hc}, \quad \frac{1}{\lambda_{BC}} = \frac{E_B - E_C}{hc}$$

Since the electron transitions from state A to state C in two steps (A to B and then B to C), the energy difference for the transition from A to B is given by

$$E_A - E_B = (E_A - E_C) - (E_B - E_C)$$

Expressing these energy differences in terms of wavelengths,

$$\frac{hc}{\lambda_{AB}} = \frac{hc}{\lambda_{AC}} - \frac{hc}{\lambda_{BC}}$$

Cancelling out hc , we obtain

$$\frac{1}{\lambda_{AB}} = \frac{1}{\lambda_{AC}} - \frac{1}{\lambda_{BC}}$$

Substitute the given values:

$$\frac{1}{\lambda_{AB}} = \frac{1}{2000} - \frac{1}{6000}$$

Finding a common denominator:

$$\frac{1}{\lambda_{AB}} = \frac{3-1}{6000} = \frac{2}{6000} = \frac{1}{3000}$$

Thus, the wavelength for the A to B transition is

$$\lambda_{AB} = 3000 \overset{o}{A}.$$

This matches Option D.

Question29

The energy E and momentum p of a moving body of mass m are related by some equation. Given that c represents the speed of light, identify the correct equation

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Options:

A. $E^2 = pc^2 + m^2c^4$

B. $E^2 = pc^2 + m^2c^2$

C. $E^2 = p^2c^2 + m^2c^4$

D. $E^2 = p^2c^2 + m^2c^2$

Answer: C

Solution:

$$[E] = M^1 L^2 T^{-2}$$

$$[Pc] = M^1 L^1 T^{-1} \cdot L^1 T^{-1} = M^1 L^2 T^{-2}$$

$$[mc^2] = M^1 L^2 T^{-2}$$

$$E^2 = M^1 L^2 T^{-2}$$

$$E^2 = P^2 c^2 + m^2 c^4$$

Question30

Choose the correct nuclear process from the below options [p : proton, n : neutron, e^- : electron, e^+ : positron, ν : neutrino, $\bar{\nu}$: antineutrino]

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Options:

A. $n \rightarrow p + e^+ + \nu$

B. $n \rightarrow p + e^- + \bar{\nu}$

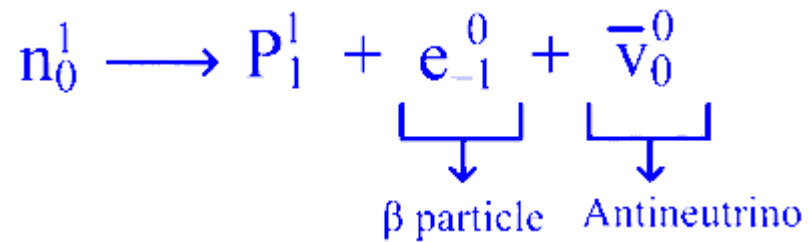
C. $n \rightarrow p + e^+ + \bar{\nu}$

D. $n \rightarrow p + e^- + \nu$

Answer: B

Solution:

We know, theoretical equation for β^- decay,



Hence, option 2 is correct.

Question31

The frequency of revolution of the electron in Bohr's orbit varies with n , the principal quantum number as:

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Options:

A.

$$\frac{1}{n^4}$$

B.

$$\frac{1}{n^2}$$

C.

$$\frac{1}{n^3}$$

D.

$$\frac{1}{n}$$

Answer: C

Solution:

Frequency of revolution $\propto \frac{1}{n^3}$

Question32

The number of spectral lines emitted by atomic hydrogen that is in the 4th energy level, is

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Options:

A.

3

B.

6

C.

1

D.

0

Answer: B

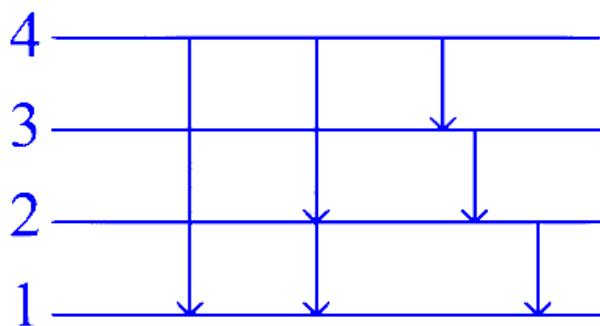
Solution:

When an e transition from n^{th} to ground state then $\sum (n - 1)$ spectral lines are produced.

Given, $n = 4$

So, $\sum (4 - 1) = \sum 3 = 3 + 2 + 1 = 6$

Hence, 6 spectral lines are emitted by atomic Hydrogen that is in the 4^{th} energy level



Question33

Considering Bohr's atomic model for hydrogen atom :

- (A) the energy of H atom in ground state is same as energy of He^+ ion in its first excited state.
- (B) the energy of H atom in ground state is same as that for Li^{++} ion in its second excited state.
- (C) the energy of H atom in its ground state is same as that of He^+ ion for its ground state.
- (D) the energy of He^+ ion in its first excited state is same as that for Li^{++} ion in its ground state.

Choose the correct answer from the options given below :

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Options:

- A. (A), (B) only
- B. (A), (D) only
- C. (A), (C) only
- D. (B), (D) only

Answer: A

Solution:

In Bohr's atomic model, the energy of an electron in an atom is given by the formula:

$$E \propto \frac{Z^2}{n^2}$$

where Z is the atomic number and n is the principal quantum number of the electron's orbit. Let's consider the implications:

For hydrogen (H), $Z = 1$.

For helium ion (He^+), $Z = 2$.

For lithium ion (Li^{++}), $Z = 3$.

Now, identifying the relevant states:

Ground State: This corresponds to $n = 1$.

First Excited State: This corresponds to $n = 2$.

Second Excited State: This corresponds to $n = 3$.

Let's evaluate the energy comparisons:

Hydrogen Atom in Ground State: $E \propto \frac{1^2}{1^2} = 1$.

Helium Ion (He^+) in the First Excited State:

$$E \propto \frac{2^2}{2^2} = 1$$

This matches the energy of hydrogen in its ground state.

Lithium Ion (Li^{++}) in the Second Excited State:

$$E \propto \frac{3^2}{3^2} = 1$$

This also matches the energy of hydrogen in its ground state.

From this analysis, we can conclude:

The energy of a hydrogen atom in its ground state is equal to the energy of a He^+ ion in its first excited state.

The energy of a hydrogen atom in its ground state also equals the energy of a Li^{++} ion in its second excited state.

Hence, statements (A) and (B) are correct.

Question34

Assuming the validity of Bohr's atomic model for hydrogen like ions the radius of Li^{++} ion in its ground state is given by $\frac{1}{X}a_0$, where $X = \underline{\hspace{2cm}}$ (Where a_0 is the first Bohr's radius.)

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Options:

- A. 2
- B. 9
- C. 1
- D. 3

Answer: D

Solution:

The radius of a hydrogen-like ion according to Bohr's atomic model is given by the formula:

$$r = r_0 \frac{n^2}{Z}$$

Where:

r_0 is the first Bohr radius (a_0)

n represents the principal quantum number

Z is the atomic number of the ion

For the ion Li^{++} , the atomic number $Z = 3$ and we are considering the ground state, so $n = 1$.

Plugging these values into the formula, we get:

$$r = r_0 \frac{1^2}{3} = \frac{r_0}{3}$$

This shows that the radius of Li^{++} in its ground state is $\frac{1}{3}a_0$, meaning $X = 3$.

Question35

Energy released when two deuterons (${}_1\text{H}^2$) fuse to form a helium nucleus (${}_2\text{He}^4$) is : (Given : Binding energy per nucleon of ${}_1\text{H}^2 = 1.1\text{MeV}$ and binding energy per nucleon of ${}_2\text{He}^4 = 7.0\text{MeV}$)

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Options:

A. 26.8 MeV

B. 8.1 MeV

C. 23.6 MeV

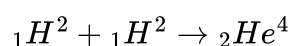
D. 5.9 MeV

Answer: C

Solution:

To calculate the energy released when two deuterons (${}_1\text{H}^2$) fuse to form a helium nucleus (${}_2\text{He}^4$), consider the following:

The reaction is represented as:



Given the binding energy per nucleon:

For ${}_1\text{H}^2$: 1.1 MeV

For ${}_2\text{He}^4$: 7.0 MeV

Calculating Binding Energy

Binding Energy for Reactants:

Each deuteron (${}_1\text{H}^2$) has a binding energy of 1.1 MeV per nucleon.

Since there are two nucleons in a deuteron, the total binding energy for one deuteron is $1.1 \times 2 = 2.2$ MeV.

Therefore, for two deuterons: $2.2 \times 2 = 4.4$ MeV.

Binding Energy for Product:

For ${}_2\text{He}^4$: Each of the four nucleons has a binding energy of 7.0 MeV.

Total binding energy for the helium nucleus: $7.0 \times 4 = 28.0$ MeV.

Energy Released (Q)

The energy released, Q , is the difference between the binding energy of the products and the reactants:

$$Q = \text{BE}_{\text{product}} - \text{BE}_{\text{reactants}} = 28.0 \text{ MeV} - 4.4 \text{ MeV} = 23.6 \text{ MeV}$$

Thus, the energy released when two deuterons fuse to form a helium nucleus is **23.6 MeV**.

Question36

Match the LIST-I with LIST-II

List - I		List - II	
A.	${}_0^1\text{n} + {}_{92}^{235}\text{U} \rightarrow {}_{54}^{140}\text{Xe} + {}_{38}^{94}\text{Sr} + 2{}_0^1\text{n}$	I.	Chemical reaction
B.	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$	II.	Fusion with +ve Q value
C.	${}_1^2\text{H} + {}_1^2\text{H} \rightarrow {}_2^3\text{He} + {}_0^1\text{n}$	III.	Fission
D.	${}_1^1\text{H} + {}_1^3\text{H} \rightarrow {}_1^2\text{H} + {}_1^2\text{H}$	IV.	Fusion with -ve Q value

Choose the correct answer from the options given below:

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Options:

A. A-II, B-I, C-III, D-IV

B. A-III, B-I, C-II, D-IV

C. A-III, B-I, C-IV, D-II

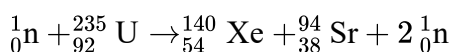
D. A-II, B-I, C-IV, D-III

Answer: B

Solution:

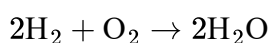
Here's the correct matching:

A.



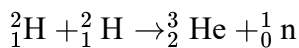
— This is nuclear fission $\Rightarrow$ III.

B.



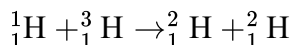
— This is a chemical reaction $\Rightarrow$ I.

C.



— This D–D fusion releases about 3.3 MeV $\Rightarrow$ fusion with +ve Q $\Rightarrow$ II.

D.



— The mass of products exceeds reactants $\Rightarrow$ requires energy $\Rightarrow$ fusion with –ve Q $\Rightarrow$ IV.

So the answer is

Option B: A–III, B–I, C–II, D–IV.

Question 37

Given below are two statements: one is labelled as Assertion **A** and the other is labelled as Reason **R**

Assertion A : The Bohr model is applicable to hydrogen and hydrogen-like atoms only.

Reason R : The formulation of Bohr model does not include repulsive force between electrons.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A. **A** is true but **R** is false

B. Both **A** and **R** are true and **R** is the correct explanation of **A**

C. **A** is false but **R** is true

D. Both **A** and **R** are true but **R** is NOT the correct explanation of **A**

Answer: B

Solution:

Option B

Both statements are true, and the absence of electron–electron repulsion in Bohr’s formulation is exactly why it works only for one-electron systems (hydrogen and hydrogen-like ions).

- Assertion A is true: Bohr’s model applies only to atoms with a single electron.
 - Reason R is true: Bohr treated only the Coulomb attraction between nucleus and electron, omitting any repulsive force between multiple electrons.
 - Since ignoring e–e repulsion prevents extension to multi–electron atoms, R correctly explains A.
-

Question38

Considering the Bohr model of hydrogen like atoms, the ratio of the radius of 5th orbit of the electron in Li²⁺ and He⁺ is

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Options:

A. $\frac{3}{2}$

B. $\frac{2}{3}$

C. $\frac{4}{9}$

D. $\frac{9}{4}$

Answer: B

Solution:

In the Bohr model for a hydrogen-like atom, the radius of the n^{th} orbit is given by

$$r_n = \frac{n^2 a_0}{Z}$$

where:

n is the principal quantum number,

a_0 is the Bohr radius, and

Z is the atomic number (the effective nuclear charge).

For the 5th orbit ($n = 5$):

For Li^{2+} (where $Z = 3$):

$$r_5(\text{Li}^{2+}) = \frac{5^2 a_0}{3} = \frac{25a_0}{3}$$

For He^+ (where $Z = 2$):

$$r_5(\text{He}^+) = \frac{5^2 a_0}{2} = \frac{25a_0}{2}$$

Now, the ratio of the radius of the 5th orbit of Li^{2+} to that of He^+ is:

$$\frac{r_5(\text{Li}^{2+})}{r_5(\text{He}^+)} = \frac{\frac{25a_0}{3}}{\frac{25a_0}{2}} = \frac{25a_0}{3} \times \frac{2}{25a_0} = \frac{2}{3}$$

Thus, the ratio is $\frac{2}{3}$.

The correct answer is Option B.

Question39

Given below are two statements :

Statement (I) : The dimensions of Planck's constant and angular momentum are same.

Statement (II) : In Bohr's model electron revolve around the nucleus only in those orbits for which angular momentum is integral multiple of Planck's constant.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

- A. Both Statement I and Statement II are correct
- B. Statement I is correct but Statement II is incorrect
- C. Both Statement I and Statement II are incorrect
- D. Statement I is incorrect but Statement II is correct

Answer: B

Solution:

$$E = hf$$

$$ML^2 T^{-2} = [h] \times [T^{-1}]$$

$$[h] = [ML^2 T^{-1}]$$

$$L = [MVR] = [ML^2 T^{-1}]$$

$$L = \frac{nh}{2\pi}$$

L is integral multiple of $\frac{h}{2\pi}$

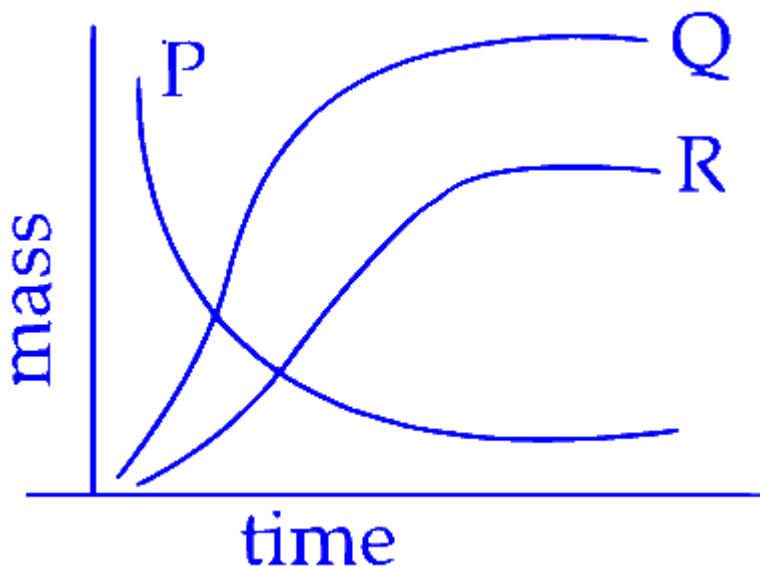
Question40

A radioactive material P first decays into Q and then Q decays to non-radioactive material R . Which of the following figure represents time dependent mass of P , Q and R ?

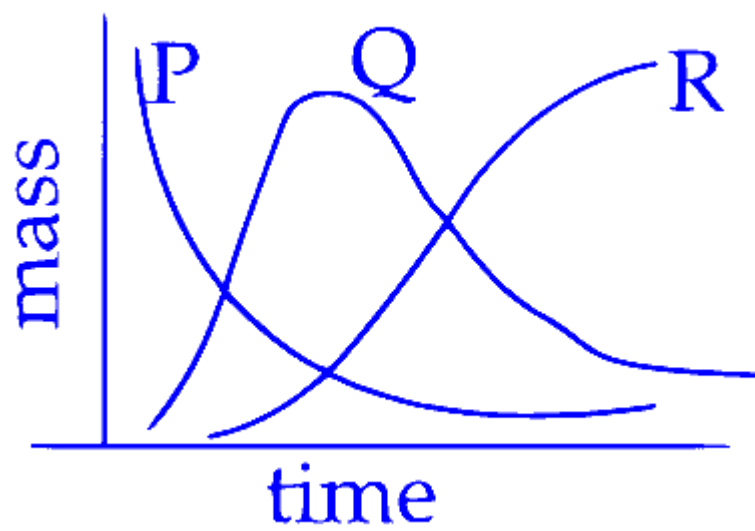
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Options:

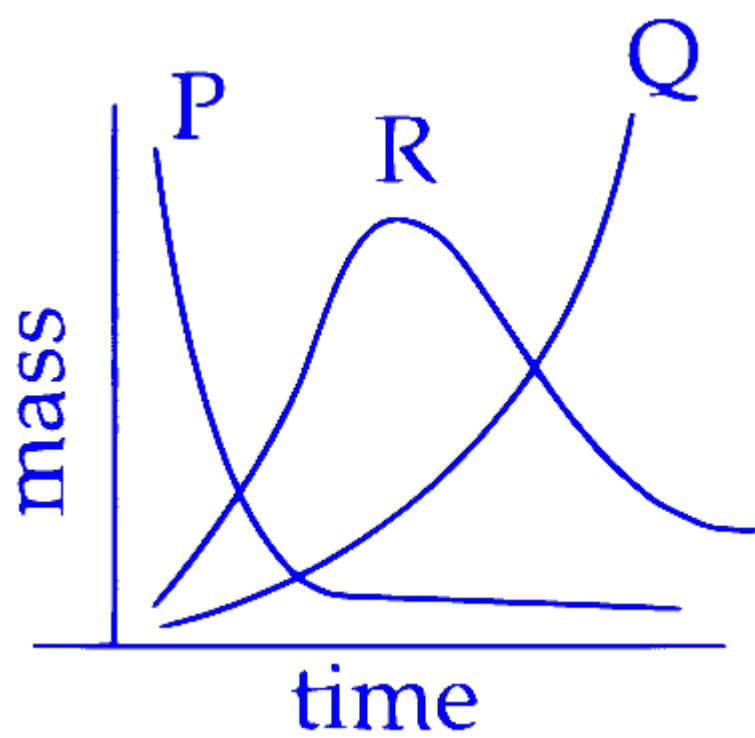
A.



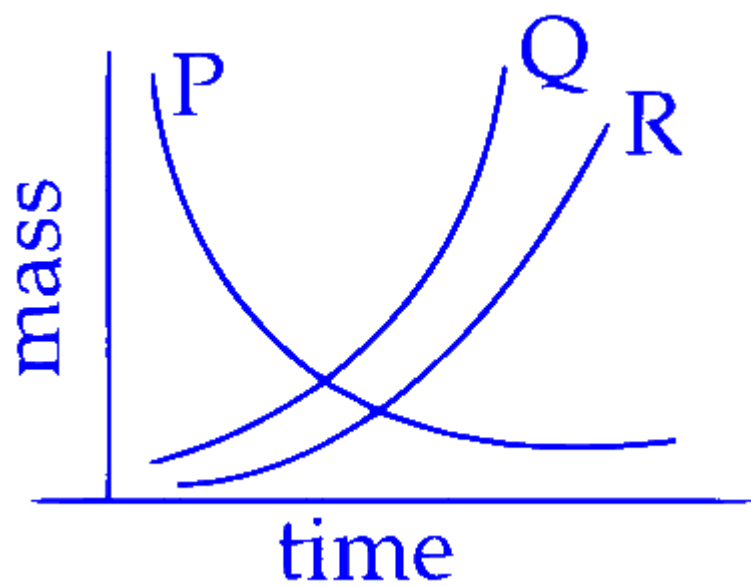
B.



C.



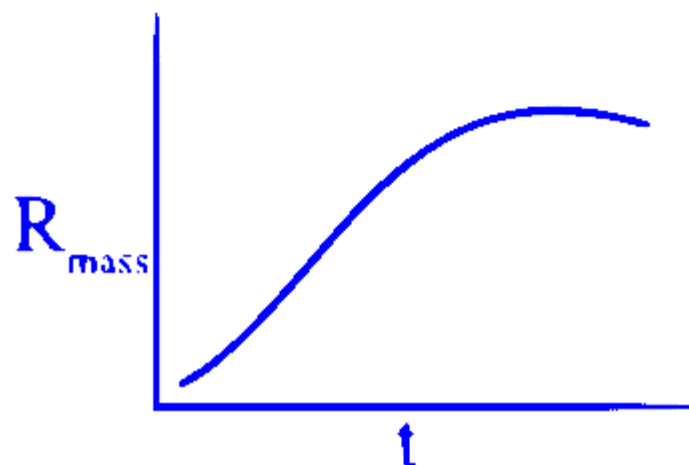
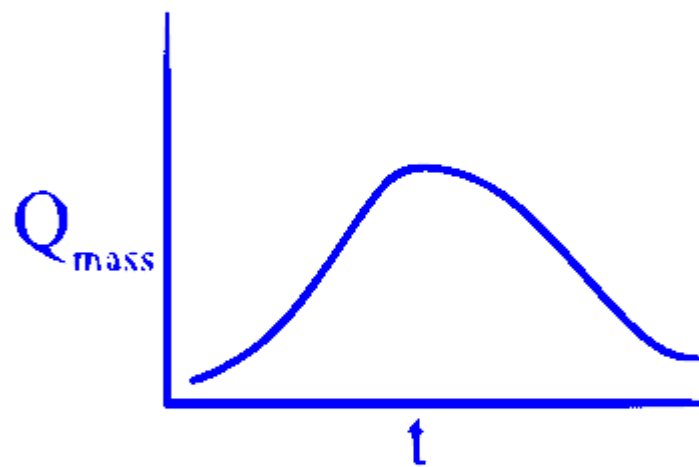
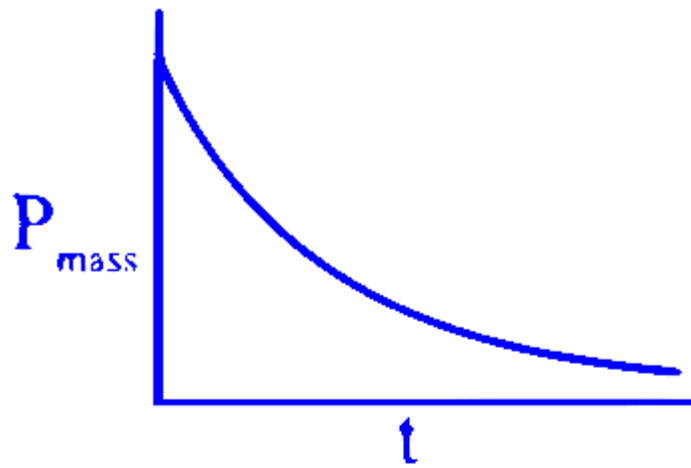
D.



Answer: B

Solution:





Question41

For a hydrogen atom, the ratio of the largest wavelength of Lyman series to that of the Balmer series is

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Options:

A. 5 : 27

B. 27 : 5

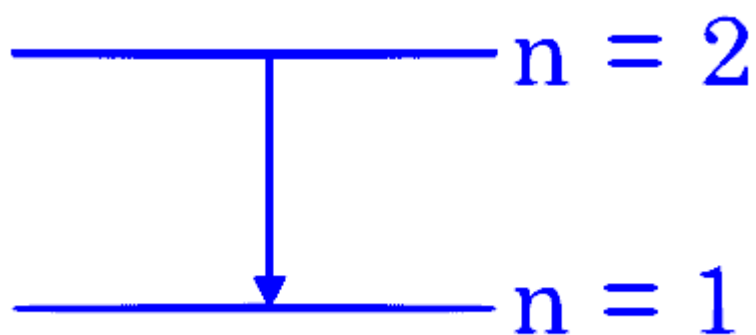
C. 3 : 4

D. 5 : 36

Answer: A

Solution:

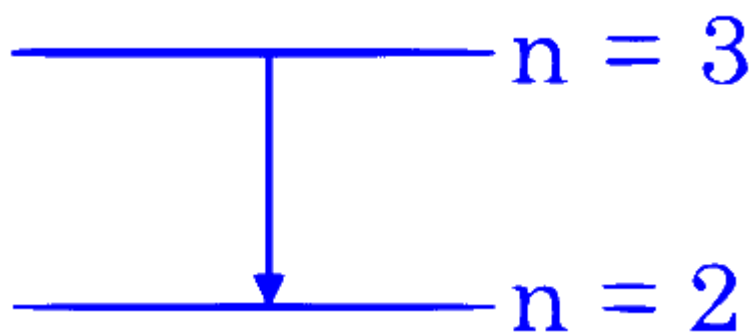
Lyman



$$\frac{1}{\lambda_1} = R \left[\frac{1}{1} - \frac{1}{4} \right] = \frac{3R}{4}$$

$$\lambda_1 = \frac{4}{3R} \quad \dots (1)$$

and Balmer



$$\frac{1}{\lambda_2} = R \left[\frac{1}{4} - \frac{1}{9} \right] = \frac{5R}{36}$$

$$\lambda_2 = \frac{36}{5R}$$

$$\text{Then, } \frac{\lambda_1}{\lambda_2} = \frac{5}{27}$$

Question42

In a hydrogen like ion, the energy difference between the 2nd excitation energy state and ground is 108.8 eV . The atomic number of the ion is:

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Options:

A. 1

B. 4

C. 3

D. 2

Answer: C

Solution:

To determine the atomic number (Z) of a hydrogen-like ion, given that the energy difference between its second excitation state and the ground state is 108.8 eV, we use the following formula for the energy difference between two energy levels in a hydrogen-like atom:

$$\Delta E = 13.6 \text{ eV} \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Here, for the second excitation, $n_2 = 3$ (since excitation means moving to the second higher energy level beyond the ground state, which means $n_1 = 1$ for the ground state to $n_2 = 3$). Plug these into the energy difference formula:

$$\Delta E = 13.6 Z^2 \left[\frac{1}{1^2} - \frac{1}{3^2} \right]$$

Simplifying the term in brackets:

$$\Delta E = 13.6 Z^2 \left[1 - \frac{1}{9} \right] = 13.6 Z^2 \times \frac{8}{9}$$

Given that $\Delta E = 108.8 \text{ eV}$, set the equations equal:

$$13.6 \frac{8Z^2}{9} = 108.8$$

Solving for Z , first simplify:

$$13.6 \times \frac{8}{9} \times Z^2 = 108.8$$

$$\frac{108.8 \times 9}{13.6 \times 8} = Z^2$$

$$Z^2 = 9$$

Thus, the atomic number Z is:

$$Z = 3$$

The atomic number of the ion is 3.

Question 43

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The density of the copper (${}^{64}_{29}\text{Cu}$) nucleus is greater than that of the carbon (${}^{12}_6\text{C}$) nucleus.

Reason (R): The nucleus of mass number A has a radius proportional to $A^{1/3}$.

In the light of the above statements, choose the most appropriate answer from the options given below:

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

(A) is correct but (R) is not correct

B.

Both (A) and (R) are correct but (R) is **not** the correct explanation of (A)

C.

(A) is not correct but (R) is correct

D.

Both (A) and (R) are correct and (R) is the correct explanation of (A)

Answer: C

Solution:

To evaluate the assertion and reason provided, we need to consider the formula for the density of a nucleus. The density ρ can be expressed as follows:

$$\rho = \frac{M}{V} = \frac{m_n \times A}{\frac{4}{3}\pi R^3}$$

Here, m_n represents the nucleon mass, A is the mass number, and R is the radius of the nucleus. According to the formula for the radius:

$$R = R_0 A^{1/3}$$

This indicates that the radius R is proportional to $A^{1/3}$. By substituting the expression for R in terms of A back into the formula for density, we get:

$$\rho = \frac{m_n \times A}{\frac{4}{3}\pi (A^{1/3} R_0)^3}$$

Simplifying the equation shows that the A terms cancel out:

$$\rho = \frac{m_n}{\frac{4}{3}\pi R_0^3}$$

This reveals that the nuclear density ρ is approximately constant and independent of the mass number A . Therefore, the densities of copper (${}^{64}_{29}\text{Cu}$) and carbon (${}^{12}_6\text{C}$) nuclei are effectively the same because the density formula results in a similar value regardless of the specific mass number.

Question44

For a nucleus of mass number A and radius R, the mass density of nucleus can be represented as

JEE Main 2025 (Online) 8th April Evening Shift

Options:

A.

$$A^{\frac{2}{3}}$$

B.

Independent of A

C.

$$A^3$$

D.

$$A^{\frac{1}{3}}$$

Answer: B

Solution:

The radius R of a nucleus is given by the empirical relation:

$$R = r_0 A^{\frac{1}{3}},$$

where r_0 is a constant and A is the mass number.

The volume V of the nucleus is approximately that of a sphere:

$$V = \frac{4}{3}\pi R^3.$$

Substituting the expression for R into the volume formula:

$$V = \frac{4}{3}\pi \left(r_0 A^{\frac{1}{3}}\right)^3 = \frac{4}{3}\pi r_0^3 A.$$

The mass density ρ is defined as the mass (which is proportional to A) divided by the volume:

$$\rho = \frac{A}{V} = \frac{A}{\frac{4}{3}\pi r_0^3 A} = \frac{1}{\frac{4}{3}\pi r_0^3}.$$

Notice that A cancels out in the numerator and denominator, so the density is:

$$\rho = \text{constant},$$

meaning it is independent of the mass number A .

Thus, the correct option is:

Option B: Independent of A.

Question 45

The radius of third stationary orbit of electron for Bohr's atom is R. The radius of fourth stationary orbit will be:

[27-Jan-2024 Shift 1]

Options:

A. $\frac{4}{3}R$

B. $\frac{16}{9}R$

C. $\frac{3}{4}R$

D. $\frac{9}{16}R$

Answer: B

Solution:

$$r \propto \frac{n^2}{Z}$$

$$\frac{r_4}{r_3} = \frac{4^2}{3^2}$$

$$r_4 = \frac{16}{9}R$$

Question46

If Rydberg's constant is R , the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where $\alpha = \underline{\hspace{2cm}}$

[27-Jan-2024 Shift 2]

Answer: 144

Solution:

Longest wavelength corresponds to transition between $n = 3$ and $n = 4$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = RZ^2 \left(\frac{1}{9} - \frac{1}{16} \right)$$

$$= \frac{7RZ^2}{9 \times 16}$$

$$\Rightarrow \lambda = \frac{144}{7R} \text{ for } Z = 1$$

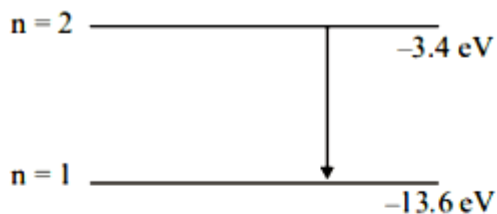
$$\therefore \alpha = 144$$

Question47

When a hydrogen atom going from $n = 2$ to $n = 1$ emits a photon, its recoil speed is $\frac{x}{5} \text{ m/s}$. Where $x = \underline{\hspace{1cm}}$ (Use : mass of hydrogen atom $= 1.6 \times 10^{-27} \text{ kg}$)
[29-Jan-2024 Shift 1]

Answer: 17

Solution:



$$\Delta E = 10.2 \text{ eV}$$

$$\text{Recoil speed } (v) = \frac{\Delta E}{mc}$$

$$= \frac{10.2 \text{ eV}}{1.6 \times 10^{-27} \times 3 \times 10^8}$$

$$= \frac{10.2 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-27} \times 3 \times 10^8}$$

$$v = 3.4 \text{ m/s} = \frac{17}{5} \text{ m/s}$$

Therefore, $x = 17$

Question48

Given below are two statements:

Statement I: Fluorine has most negative electron gain enthalpy in its group.

Statement II: Oxygen has least negative electron gain enthalpy in its group.

In the light of the above statements, choose the most appropriate from the options given below.

[29-Jan-2024 Shift 2]

Options:

- A. Both Statement I and Statement II are true
- B. Statement I is true but Statement II is false
- C. Both Statement I and Statement II are false
- D. Statement I is false but Statement II is true

Answer: D

Solution:

Statement-1 is false because chlorine has most negative electron gain enthalpy in its group.

Question49

Hydrogen atom is bombarded with electrons accelerated through a potential different of V , which causes excitation of hydrogen atoms. If the experiment is being formed at $T = 0\text{K}$. The minimum potential difference needed to observe any Balmer series lines in the emission spectra will be $\frac{\alpha}{10}V$, where $\alpha = \underline{\hspace{2cm}}$

[29-Jan-2024 Shift 2]

Answer: 121

Solution:

For minimum potential difference electron has to make transition from $n=3$ to $n=2$ state but first electron has to reach to $n = 3$ state from ground state. So, energy of bombarding electron should be equal to energy difference of $n=3$ and $n=1$ state.

$$\Delta E = 13.6 \left[1 - \frac{1}{3^2} \right] e = eV$$

$$\frac{13.6 \times 8}{9} = V$$

$$V = 12.09V \approx 12.1V$$

So, $\alpha = 121$

Question50

The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is :
[30-Jan-2024 Shift 1]

Options:

A. 4

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. 1

Answer: C

Solution:

$$\frac{1}{2} |PE| = KE \text{ for each value of } n \text{ (orbit)}$$

$$\therefore \frac{KE}{|PE|} = \frac{1}{2}$$

Question51

The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is :
[30-Jan-2024 Shift 1]

Options:

A. 4

B. $\frac{1}{4}$

C. $\frac{1}{2}$

D. 1

Answer: C

Question52

A electron of hydrogen atom on an excited state is having energy $E_n = -0.85 \text{ eV}$. The maximum number of allowed transitions to lower energy level is
[30-Jan-2024 Shift 1]

Answer: 6

Solution:

$$E_n = -\frac{13.6}{n^2} = -0.85$$

$$\Rightarrow n = 4$$

No of transition

$$= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

Question53

An electron revolving in n^{th} Bohr orbit has magnetic moment μ_n . If $\mu_n \propto n^x$, the value of x is:

[30-Jan-2024 Shift 2]

Options:

A. 2

B. 1

C. 3

D. 0

Answer: B

Solution:

$$\text{Magnetic moment} = i\pi r^2$$

$$\mu = \frac{evr}{2}$$

$$\mu \propto \left(\frac{1}{n}\right)n^2$$

$$\mu \propto n$$

$$x = 1$$

Question54

The mass defect in a particular reaction is 0.4g. The amount of energy liberated is $n \times 10^7$ kWh, where $n = \underline{\hspace{2cm}}$ (speed of light $= 3 \times 10^8 \text{ m/s}$)
[31-Jan-2024 Shift 1]

Answer: 1

Solution:

$$\begin{aligned} E &= \Delta mc^2 \\ &= 0.4 \times 10^{-3} \times (3 \times 10^8)^2 \\ &= 3600 \times 10^7 \text{ kWs} \\ &= \frac{3600 \times 10^7}{3600} \text{ kWh} = 1 \times 10^7 \text{ kWh} \end{aligned}$$

Question55

In a nuclear fission process, a high mass nuclide ($A \approx 236$) with binding energy 7.6 MeV/Nucleon dissociated into middle mass nuclides ($A \approx 118$), having binding energy of 8.6 MeV/Nucleon. The energy released in the process would be $\underline{\hspace{2cm}}$ MeV.
[27-Jan-2024 Shift 1]

Answer: 236

Solution:

$$\begin{aligned}
 Q &= BE_{\text{Product}} - BE_{\text{Reactant}} \\
 &= 2(118)(8.6) - 236(7.6) \\
 &= 236 \times 1 = 236 \text{ MeV}
 \end{aligned}$$

Question56

The atomic mass of ${}_6\text{C}^{12}$ is 12.000000u and that of ${}_6\text{C}^{13}$ is 13.003354u. The required energy to remove a neutron from ${}_6\text{C}^{13}$, if mass of neutron is 1.008665u, will be :
[27-Jan-2024 Shift 2]

Options:

- A. 62.5 MeV
- B. 6.25 MeV
- C. 4.95 MeV
- D. 49.5 MeV

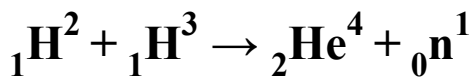
Answer: C

Solution:

$$\begin{aligned}
 {}_6\text{C}^{13} + \text{Energy} &\rightarrow {}_6\text{C}^{12} + {}_0\text{n}^1 \\
 \Delta m &= (12.000000 + 1.008665) - 13.003354 = -0.00531\text{u} \\
 \therefore \text{Energy required} &= 0.00531 \times 931.5 \text{ MeV} = 4.95 \text{ MeV}
 \end{aligned}$$

Question57

The explosive in a Hydrogen bomb is a mixture of ${}_1\text{H}^2$, ${}_1\text{H}^3$ and ${}_3\text{Li}^6$ in some condensed form. The chain reaction is given by
 ${}_3\text{Li}^6 + {}_0\text{n}^1 \rightarrow {}_2\text{He}^4 + {}_1\text{H}^3$



During the explosion the energy released is approximately

[Given : $M(\text{Li}) = 6.01690 \text{ amu}$. $M({}_1\text{H}^2) = 2.01471 \text{ amu}$.

$M({}_2\text{He}^4) = 4.00388 \text{ amu}$, and $1 \text{ amu} = 931.5 \text{ MeV}$]

[29-Jan-2024 Shift 1]

Options:

A. 28.12 MeV

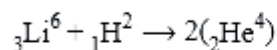
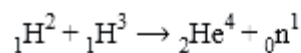
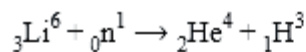
B. 12.64 MeV

C. 16.48 MeV

D. 22.22 MeV

Answer: D

Solution:



Energy released in process

$$Q = \Delta mc^2$$

$$Q = [M(\text{Li}) + M({}_1\text{H}^2) - 2 \times M({}_2\text{He}^4)] \times 931.5 \text{ MeV}$$

$$Q = [6.01690 + 2.01471 - 2 \times 4.00388] \times 931.5 \text{ MeV}$$

$$Q = 22.216 \text{ MeV}$$

$$Q = 22.22 \text{ MeV}$$

Question58

In a nuclear fission reaction of an isotope of mass M , three similar daughter nuclei of same mass are formed. The speed of a daughter nuclei in terms of mass defect ΔM will be :
[30-Jan-2024 Shift 2]

Options:

A. $\sqrt{\frac{2c \Delta M}{M}}$

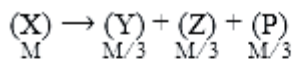
B. $\frac{\Delta Mc^2}{3}$

C. $c \sqrt{\frac{2 \Delta M}{M}}$

D. $c \sqrt{\frac{3 \Delta M}{M}}$

Answer: C

Solution:



$$\Delta Mc^2 = \frac{1}{2} \frac{M}{3} V^2 + \frac{1}{2} \frac{M}{3} V^2 + \frac{1}{2} \frac{M}{3} V^2$$

$$V = c \sqrt{\frac{2 \Delta M}{M}}$$

Question59

The mass number of nucleus having radius equal to half of the radius of nucleus with mass number 192 is:
[31-Jan-2024 Shift 2]

Options:

A. 24

B. 32

C. 40

D. 20

Answer: A

Solution:

$$R_1 = \frac{R_2}{2}$$

$$R_0(A_1)^{1/3} = \frac{R_0}{2}(A_2)^{1/3}$$

$$A_1 = \frac{1}{8}A_2$$

$$A_1 = \frac{192}{8} = 24$$

Question60

A nucleus has mass number A_1 and volume V_1 . Another nucleus has mass number A_2 and volume V_2 . If relation between mass number is

$A_2 = 4A_1$, then $\frac{V_2}{V_1} = \underline{\hspace{2cm}}$

[31-Jan-2024 Shift 2]

Answer: 4

Solution:

For a nucleus

Volume:

$$V = \frac{4}{3}\pi R^3$$

$$R = R_0(A)^{1/3}$$

$$V = \frac{4}{3}\pi R_0^3 A$$

$$\Rightarrow \frac{V_2}{V_1} = \frac{A_2}{A_1} = 4$$

Question61

The minimum energy required by a hydrogen atom in ground state to emit radiation in Balmer series is nearly :

[1-Feb-2024 Shift 1]

Options:

A. 1.5 eV

B. 13.6 eV

C. 1.9 eV

D. 12.1 eV

Answer: D

Solution:

Transition from $n = 1$ to $n = 3$
 $\Delta E = 12.1\text{eV}$

Question62

A particular hydrogen - like ion emits the radiation of frequency 3×10^{15} Hz when it makes transition from $n = 2$ to $n = 1$. The frequency of radiation emitted in transition from $n = 3$ to $n = 1$ is

$\frac{x}{9} \times 10^{15}$ Hz, when $x =$ _____

[1-Feb-2024 Shift 2]

Answer: 32

Solution:

$$E = -13.6z^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$E = C \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$h\nu = C \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$\frac{\nu_1}{\nu_2} = \frac{\left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]_{2 \rightarrow 1}}{\left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]_{3 \rightarrow 1}}$$

$$= \frac{\left[\frac{1}{1} - \frac{1}{4} \right]}{\left[\frac{1}{1} - \frac{1}{9} \right]} = \frac{3/4}{8/9}$$

$$= \frac{3}{4} \times \frac{9}{8}$$

$$\frac{\nu_1}{\nu_2} = \frac{27}{32}$$

$$\nu_2 = \frac{32}{27} \nu_1 = \frac{32}{27} \times 3 \times 10^{15} \text{ Hz} = \frac{32}{9} \times 10^{15} \text{ Hz}$$

Question63

The radius of a nucleus of mass number 64 is 4.8 fermi. Then the mass number of another nucleus having radius of 4 fermi is $\frac{1000}{x}$, where x is _____

[1-Feb-2024 Shift 1]

Answer: 27

Solution:

$$R = R_0 A^{1/3}$$

$$R^3 \propto A$$

$$\left(\frac{4.8}{4} \right)^3 = \frac{64}{A}$$

$$= \frac{64}{A} = (1.2)^3$$

$$\frac{64}{A} = 1.44 \times 1.2$$

$$A = \frac{64}{1.44 \times 1.2} = \frac{1000}{x}$$

$$x = \frac{144 \times 12}{64} = 27$$

Question 64

From the statements given below :

- (A) The angular momentum of an electron in n^{th} orbit is an integral multiple of h .**
- (B) Nuclear forces do not obey inverse square law.**
- (C) Nuclear forces are spin dependent.**
- (D) Nuclear forces are central and charge independent.**
- (E) Stability of nucleus is inversely proportional to the value of packing fraction.**

Choose the correct answer from the options given below :

[1-Feb-2024 Shift 2]

Options:

A. (A), (B), (C), (D) only

B. (A), (C), (D), (E) only

C. (A), (B), (C), (E) only

D. (B), (C), (D), (E) only

Answer: C

Solution:

Part of theory

Question65

If the binding energy of ground state electron in a hydrogen atom is 13.6 eV, then, the energy required to remove the electron from the second excited state of Li^{2+} will be: $x \times 10^{-1}$ eV. The value of x is

[31-Jan-2023 Shift 2]

Answer: 136

Solution:

$$E_{\text{H}} = 13.6$$

$$E_{\text{Li}^{2+}} = 13.6 \frac{Z^2}{n^2} = 13.6 \times \frac{9}{9} = 13.6 \text{ eV}$$
$$= 136 \times 10^{-1} \text{ eV}$$

Question66

Consider the following radioactive decay process

${}_{84}^{218}\text{A} \xrightarrow{\alpha} \text{A}_1 \xrightarrow{\beta^-} \text{A}_2 \xrightarrow{\gamma} \text{A}_3 \xrightarrow{\alpha} \text{A}_4 \xrightarrow{\beta^+} \text{A}_5 \xrightarrow{\gamma} \text{A}_6$ The mass number and the atomic number A_6 are given by :

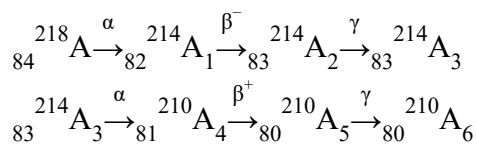
[24-Jan-2023 Shift 1]

Options:

- A. 210 and 82
- B. 210 and 84
- C. 210 and 80
- D. 211 and 80

Answer: C

Solution:



Question67

The energy released per fission of nucleus of ${}^{240}\text{X}$ is 200 MeV. The energy released if all the atoms in 120g of pure ${}^{240}\text{X}$ undergo fission is $\underline{\hspace{1cm}} \times 10^{25}$ MeV.

(. Given $N_A = 6 \times 10^{23}$)

[24-Jan-2023 Shift 2]

Answer: 0

Solution:

$$\text{No. of mole} = \frac{120}{240} = \frac{1}{2}$$

$$\text{No. of molecules} = \frac{1}{2} \times N_A$$

$$\begin{aligned} \text{Energy released} &= \frac{1}{2} \times 6 \times 10^{23} \times 200 \\ &= 6 \times 10^{25} \text{ MeV} \end{aligned}$$

Question 68

The ratio of the density of oxygen nucleus (${}^8_{16}\text{O}$) and helium nucleus (${}^2_4\text{He}$) is

[25-Jan-2023 Shift 1]

Options:

A. 4 : 1

B. 8 : 1

C. 1 : 1

D. 2 : 1

Answer: C

Solution:

Nuclear density is independent of mass number

$$\text{As nuclear density} = \frac{Au}{\frac{4}{3}\pi R^3}$$

$$\text{Also, } R = R_0 A^{\frac{1}{3}}$$

$$\text{And } R^3 = R_0^3 A$$

$$\Rightarrow \text{Nuclear density} = \frac{Au}{\frac{4}{3}\pi R_0^3 A}$$

$$\text{Nuclear density} = \frac{3u}{4\pi R_0^3}$$

$\Rightarrow$ Nuclear density is independent of A

Question69

If a radioactive element having half-life of 30 min is undergoing beta decay, the fraction of radioactive element remains undecayed after 90 min. will be :

[29-Jan-2023 Shift 1]

Options:

A. $\frac{1}{8}$

B. $\frac{1}{16}$

C. $\frac{1}{4}$

D. $\frac{1}{2}$

Answer: A

Solution:

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/t_{1/2}} = \left(\frac{1}{2}\right)^{\frac{90}{30}}$$
$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Question70

A radioactive element ${}^{242}_{92}\text{X}$ emits two α -particles, one electron and two positrons. The product nucleus is represented by ${}^{234}_p\text{Y}$. The value of P is _____

[29-Jan-2023 Shift 1]

Answer: 87

Solution:

$$P = 92 - 2 - 2 + 1 - 1 - 1$$

$$P = 92 - 5$$

$$P = 87$$

Question 71

Substance A has atomic mass number 16 and half life of 1 day.

Another substance B has atomic mass number 32 and half life of $\frac{1}{2}$ day. If both A and B simultaneously start undergo radio activity at the same time with initial mass 320g each, how many total atoms of A and B combined would be left after 2 days.

[29-Jan-2023 Shift 2]

Options:

A. 3.38×10^{24}

B. 6.76×10^{24}

C. 6.76×10^{23}

D. 1.69×10^{24}

Answer: A

Solution:

$$(N_0)_A = \frac{320}{16} = 20 \text{ moles}$$

$$(N_0)_B = \frac{320}{32} = 10 \text{ moles}$$

$$N_A = \frac{(N_0)_A}{(2)^{2/1}} = \frac{20}{4} = 5$$

$$N_B = \frac{(N_0)_B}{(2)^{2/.5}} = \frac{10}{2^4} = 0.625$$

$$\text{Total } N = 5.625$$

$$\begin{aligned}\text{No. of atoms} &= 5.625 \times 6.023 \times 10^{23} \\ &= 3.38 \times 10^{24}\end{aligned}$$

Question72

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The nuclear density of nuclides

${}_5^{10}\text{B}$, ${}_3^6\text{Li}$, ${}_{26}^{56}\text{Fe}$, ${}_{10}^{20}\text{Ne}$ and ${}_{83}^{209}\text{Bi}$ can be arranged as

$$\rho_{\text{Bi}}^{\text{N}} > \rho_{\text{Fe}}^{\text{N}} > \rho_{\text{Ne}}^{\text{N}} > \rho_{\text{B}}^{\text{N}} > \rho_{\text{Li}}^{\text{N}}.$$

Reason R : The radius R of nucleus is related to its mass number A as $R = R_0 A^{1/3}$, where R_0 is a constant.

In the light of the above statement, choose the correct answer from the options given below :

[30-Jan-2023 Shift 2]

Options:

- A. Both A and R are true and R is the correct explanation of A
- B. A is false but R is true
- C. A is true but R is false
- D. Both A and R are true but R is NOT the correct explanation of A

Answer: B

Solution:

Nuclear density is independent of A .

Question73

A radioactive nucleus decays by two different process. The half life of the first process is 5 minutes and that of the second process is 30 s.

The effective half-life of the nucleus is calculated to be $\frac{\alpha}{11}$ s. The value of α is _____.
[30-Jan-2023 Shift 2]

Answer: 300

Solution:

$$\begin{aligned}\frac{dN_1}{dt} &= -\lambda_1 N & \frac{dN_2}{dt} &= -\lambda_2 N \\ \frac{dN}{dt} &= -(\lambda_1 + \lambda_2)N \\ \Rightarrow \lambda_{eq} &= \lambda_1 + \lambda_2 \\ \Rightarrow \frac{1}{t_{1/2}} &= \frac{1}{300} + \frac{1}{30} = \frac{11}{300} \\ \Rightarrow t_{1/2} &= \frac{300}{11}\end{aligned}$$

Question74

A free neutron decays into a proton but a free proton does not decay into neutron. This is because
[31-Jan-2023 Shift 1]

Options:

- A. neutron is an uncharged particle
- B. proton is a charged particle
- C. neutron is a composite particle made of a proton and an electron
- D. neutron has larger rest mass than proton

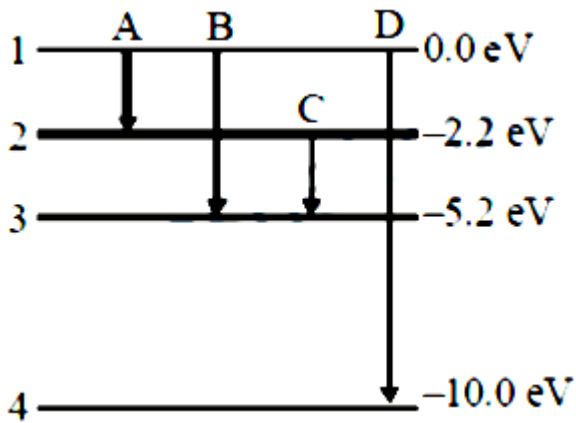
Answer: D

Solution:

As neutron has more rest mass than proton it will require energy to decay proton into neutron.
Option 4.

Question 75

The energy levels of an atom is shown in figure. Which one of these transitions will result in the emission of a photon of wavelength 124.1 nm ? Given ($h = 6.62 \times 10^{-34}$ Js)



[25-Jan-2023 Shift 2]

Options:

A. B

B. A

C. C

D. D

Answer: D

Solution:

$$\lambda = \frac{hc}{\Delta E}$$

$$\Delta E_A = 2.2 \text{ eV}$$

$$\Delta E_B = 5.2 \text{ eV}$$

$$\Delta E_C = 3 \text{ eV}$$

$$\Delta E_D = 10 \text{ eV}$$

$$\lambda_A = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{2.2 \times 1.6 \times 10^{-19}}$$

$$= \frac{12.41 \times 10^{-7}}{2.2} \text{m}$$

$$= \frac{1241}{2.2} \text{nm} = 564 \text{ nm}$$

$$\lambda_B = \frac{1241}{5.2} \text{nm} = 238.65 \text{ nm}$$

$$\lambda_C = \frac{1241}{3} \text{nm} = 413.66 \text{ nm}$$

$$\lambda_D = \frac{1241}{10} = 124.1 \text{ nm}$$

Question 76

Speed of an electron in Bohr's 7th orbit for Hydrogen atom is $3.6 \times 10^6 \text{ m/s}$. The corresponding speed of the electron in 3rd orbit, in m/s is :

[30-Jan-2023 Shift 1]

Options:

A. (1.8×10^6)

B. (7.5×10^6)

C. (3.6×10^6)

D. (8.4×10^6)

Answer: D

Solution:

$$V_n \propto \frac{Z}{n}$$

$$Z = 1, \therefore V_n \propto \frac{1}{n}$$

$$\therefore \frac{V_3}{V_7} = \frac{7}{3}$$

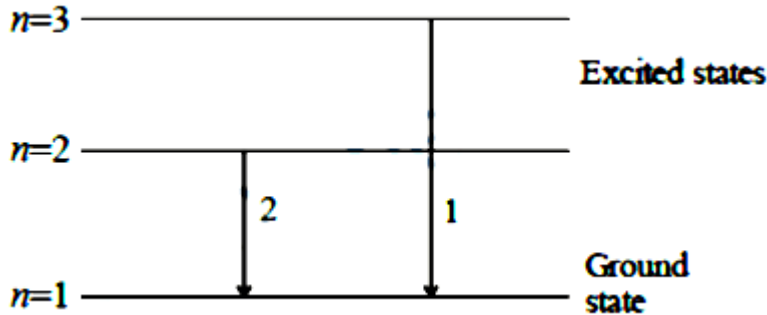
$$\therefore V_3 = \frac{7}{3} V_7$$

$$= \frac{7}{3} \times 3.6 \times 10^6 \text{ m/s}$$

$$= 8.4 \times 10^6 \text{ m/s}$$

Question77

For hydrogen atom, λ_1 and λ_2 are the wavelengths corresponding to the transitions 1 and 2 respectively as shown in figure. The ratio of λ_1 and λ_2 is $\frac{x}{32}$. The value of x is _____.



[31-Jan-2023 Shift 1]

Answer: 27

Solution:

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda_1} = RZ^2 \left[\frac{1}{1^2} - \frac{1}{3^2} \right] = \frac{8}{9}RZ^2 \dots (1)$$

$$\frac{1}{\lambda_2} = RZ^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = \frac{3}{4}RZ^2 \dots (2)$$

$$1/2 \Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{8}{9} \times \frac{4}{3} = \frac{32}{27}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{27}{32}$$

Ans. 27

Question78

The radius of electron's second stationary orbit in Bohr's atom is R. The radius of 3 rd orbit will be

[31-Jan-2023 Shift 2]

Options:

A. $2.25R$

B. $3R$

C. $\frac{R}{3}$

D. $9R$

Answer: B

Solution:

$$r \propto \frac{n^2}{Z}$$

$$\frac{r_{2\text{nd}}}{r_{3\text{rd}}} = \left(\frac{n_2}{n_3} \right)^2$$

$$\Rightarrow \frac{R}{r_{3\text{rd}}} = \left(\frac{2}{3} \right)^2$$

$$\Rightarrow r_{3\text{rd}} = \frac{9}{4}R$$

$$= 2.25R$$

Question 79

Assume that protons and neutrons have equal masses. Mass of a nucleon is 1.6×10^{-27} kg and radius of nucleus is $1.5 \times 10^{-15} A^{1/3}$ m. The approximate ratio of the nuclear density and water density is $n \times 10^{13}$. The value of n is _____
[24-Jan-2023 Shift 1]

Answer: 11

Solution:

$$\text{density of nuclei} = \frac{\text{mass of nuclei}}{\text{volume of nuclei}}$$

$$\rho = \frac{1.6 \times 10^{-27} \text{ A}}{\frac{4}{3}\pi(1.5 \times 10^{-15})^3 \text{ A}}$$

$$= \frac{1.6 \times 10^{-27}}{14.14 \times 10^{-45}} = 0.113 \times 10^{18}$$

$$\rho_w = 10^3$$

$$\frac{\rho}{\rho_w} = 11.31 \times 10^{13}$$

Question80

A photon is emitted in transition from $n = 4$ to $n = 1$ level in hydrogen atom. The corresponding wavelength for this transition is (given, $h = 4 \times 10^{-15} \text{ eVs}$) :
[24-Jan-2023 Shift 2]

Options:

A. 94.1 nm

B. 941 nm

C. 97.4 nm

D. 99.3 nm

Answer: A

Solution:

$$\frac{hc}{\lambda} = \left[1 - \frac{1}{16} \right] (13.6 \text{ eV})$$

$$\text{So, } \lambda = 94.1 \text{ nm}$$

Question81

The wavelength of the radiation emitted is λ_0 when an electron jumps from the second excited state to the first excited state of hydrogen atom. If the electron jumps from the third excited state to the second orbit of the hydrogen atom, the wavelength of the radiation emitted will be $20/x\lambda_0$. The value of x is _____.

[25-Jan-2023 Shift 1]

Answer: 27

Solution:

_____ n=4
 _____ n=3
 _____ n=2
 _____ n=1

Second excited state $\rightarrow$ first excited state

$n = 3 \rightarrow n = 2$

$$\frac{hc}{\lambda_0} = 13.6 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \dots (i)$$

Third excited state $\rightarrow$ second orbit

$n = 4 \rightarrow n = 2$

$$\frac{hc}{(20\lambda_0/x)} = 13.6 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) \dots (ii)$$

(ii) $\div$ (i)

$$\frac{x}{20} = \frac{\frac{1}{2^2} - \frac{1}{4^2}}{\frac{1}{2^2} - \frac{1}{3^2}}$$

$x = 27$

Question82

The mass of proton, neutron and helium nucleus are respectively 1.0073u, 1.0087u and 4.0015u. The binding energy of helium nucleus is:

[1-Feb-2023 Shift 1]

Options:

A. 14.2 MeV

B. 28.4 MeV

C. 56.8 MeV

D. 7.1 MeV

Answer: B

Solution:

$$\begin{aligned}\text{B.E of Helium} &= (2m_p + 2m_n - m_{\text{He}})c^2 \\ &= 28.4 \text{ MeV}\end{aligned}$$

Question83

A light of energy 12.75 eV is incident on a hydrogen atom in its ground state. The atom absorbs the radiation and reaches to one of its excited states. The angular momentum of the atom in the excited state is $\frac{x}{\pi} \times 10^{-17}$ eVs. The value of x is _____ (use

$$h = 4.14 \times 10^{-15} \text{ eVs}, c = 3 \times 10^8 \text{ ms}^{-1})$$

[1-Feb-2023 Shift 1]

Answer: 828

Solution:

In the ground state energy = -13.6 eV

So energy

$$\frac{-13.6 \text{ eV}}{n^2} = -13.6 + 12.75$$

$$\frac{-13.6 \text{ eV}}{n^2} = -0.85$$

$$\begin{aligned}n &= \sqrt{16} \\ n &= 4\end{aligned}$$

$$\begin{aligned}\text{Angular momentum} &= \frac{nh}{2\pi} = \frac{4h}{2\pi} = \frac{2h}{\pi} \\ \text{Angular momentum} &= \frac{2}{\pi} \times 4.14 \times 10^{-15} \\ &= \frac{828 \times 10^{-17}}{\pi} \text{ eVs}\end{aligned}$$

Question84

An electron of a hydrogen like atom, having $Z = 4$, jumps from 4th energy state to 2nd energy state, The energy released in this process, will be:

(Given $R_{ch} = 13.6 \text{ eV}$)

Where R = Rydberg constant

c = Speed of light in vacuum

h = Planck's constant

[1-Feb-2023 Shift 2]

Options:

A. 13.6 eV

B. 10.5 eV

C. 3.4 eV

D. 40.8 eV

Answer: D

Solution:

$$\begin{aligned}\Delta E &= 13.6Z^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] \text{ eV} \\ &= 13.6 \times (4)^2 \left(\frac{1}{4} - \frac{1}{16} \right) \text{ eV} \\ &= 13.6[4 - 1] \text{ eV} \\ &= 13.6 \times 3 = 40.8 \text{ eV}\end{aligned}$$

Question85

Nucleus a having $Z = 17$ and equal number of protons and neutrons has 1.2 MeV binding energy per nucleon.

Another nucleus B of $Z = 12$ has total 26 nucleons and 1.8 MeV binding energy per nucleons.

**The difference of binding energy of B and A will be _____ MeV.
[1-Feb-2023 Shift 2]**

Answer: 6

Solution:

For A mass number = 34

Total binding energy = $1.2 \times 34 = 40.8 \text{ MeV}$

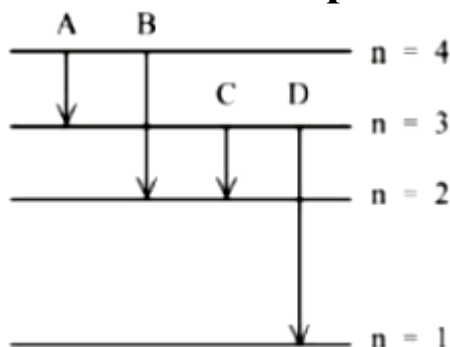
For B mass number = 26

total binding energy = $1.8 \times 26 \text{ MeV}$
= 46.8 MeV

Difference of BE = 6 MeV

Question86

The energy levels of an hydrogen atom are shown below. The transition corresponding to emission of shortest wavelength is



[6-Apr-2023 shift 1]

Options:

A. A

B. D

C. C

D. B

Answer: B

Solution:

$$E = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{E}$$

For $\lambda_{\min}$, E must be maximum. And E is maximum for D.

Question87

The radius of fifth orbit of the Li^{++} is _____ $\times 10^{-12}\text{m}$ Take : radius of hydrogen atom = $0.51\text{\AA}$
[6-Apr-2023 shift 1]

Answer: 425

Solution:

$$r_n = \frac{0.51n^2}{z}\text{\AA}$$

For Li^{++} , $z = 3$.

$$\text{So } r_5 = 0.51 \times \frac{25}{3} \times 10^{-10}\text{m} = 17 \times 25 \times 10^{-12}\text{m} = 425 \times 10^{-12}\text{m}$$

Question88

The ratio of wavelength of spectral lines H_α and H_β in the Balmer series is $\frac{x}{20}$. The value of x is
[8-Apr-2023 shift 2]

Answer: 27

Question89

The angular momentum for the electron in Bohr's orbit is L. If the electron is assumed to revolve in second orbit of hydrogen atom, then the change in angular momentum will be
[10-Apr-2023 shift 1]

Options:

A. $\frac{L}{2}$

B. zero

C. L

D. 2L

Answer: C

Solution:

$$\text{Angular momentum} = \frac{nh}{2\pi}$$

$$n = 1, L_1 = \frac{h}{2\pi} = L$$

$$n = 2, L_2 = \frac{2h}{2\pi} = 2L$$

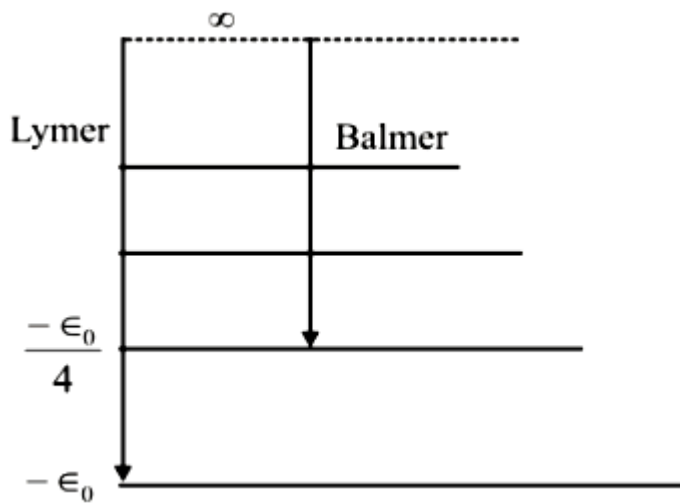
$$\Delta L = 2L - L = L$$

Question90

If $917\text{\AA}$ be the lowest wavelength of Lyman series then the lowest wavelength of Balmer series will be _____ $\text{\AA}$
 [10-Apr-2023 shift 2]

Answer: 3668

Solution:



Lowest wavelength of by may sense will be obtained for trasion $n = \infty \rightarrow n = 1$
 and for balmer series, Lyman Series $n = \infty \rightarrow n = 2$

for Lyman, $E_0 = \frac{hC}{917\text{\AA}}$

for balmer, $\frac{E_0}{4} = \frac{hC}{\lambda(\text{\AA})}$

using this

$\lambda = 917 \times 4 = 3668$

Question91

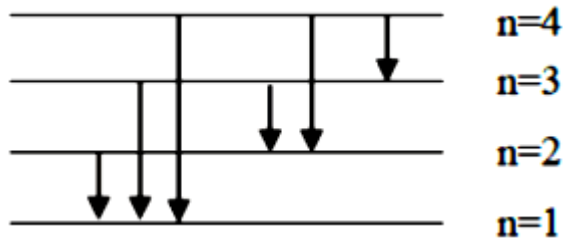
A monochromatic light is incident on a hydrogen sample in ground state. Hydrogen atoms absorb a fraction of light and subsequently emit radiation of six different wavelengths. The frequency of incident light is $x \times 10^{15}$ Hz. The value of x is _____.

(Given $h = 4.25 \times 10^{-15}$ eVs)

[11-Apr-2023 shift 1]

Answer: 3

Solution:



Total emission lines = 6 (given)

So electron absorbed energy and jump from $n = 1$ to $n = 4$

$$\Delta E = 13.6 \left[\frac{1}{1^2} - \frac{1}{4^2} \right] \text{ eV}$$

$$= 13.6 \left[1 - \frac{1}{16} \right] \text{ eV}$$

$$\Delta E = hf$$

$$12.75 = 4.5 \times 10^{-15},$$

$$f = \frac{12.75}{4.25} \times 10^{15} = 3 \times 10^{15} \text{ Hz}$$

$$x = 3 \text{ Ans.}$$

Question92

The energy of He^+ ion in its first excited state is, (The ground state energy for the Hydrogen atom is -13.6 eV):

[11-Apr-2023 shift 2]

Options:

A. -13.6 eV

B. -54.4 eV

C. -27.2 eV

D. -3.4 eV

Answer: D

Solution:

$$E = - \left(\frac{13.6z^2}{n^2} \right) \text{ eV}$$

First excited state

$$n = 2$$

$$E = \frac{-13.6 \times z^2}{2^2} = -13.6 \text{ eV}$$

Question93

A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. The number of spectral lines emitted will be : [12-Apr-2023 shift 1]

Options:

A. 2

B. 3

C. 4

D. 1

Answer: B

Solution:

$$12.5 = -13.6 \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right] \quad n_i = 1$$

$$12.5 = -13.6 \left[\frac{1}{n_f^2} - 1 \right]$$

$$12.5 = 13.6 - \frac{13.6}{n_f^2}$$

$$-1.1 = - \frac{13.6}{n_f^2}$$

$$n_f = 3.5$$

$n_f = 3$ so transition is 3 to 1

$$\text{No. of lines} = {}^nC_2 \quad \left({}^nC_r = \frac{{}^nC_n}{x|n-r| \mid xr} \right)$$

$$= \frac{3}{(3-2)(2)}$$

$$\frac{3 \times 2 \times 1}{1 \times 2} = 3 \text{ lines}$$

Question94

The radius of 2nd orbit of He⁺ of Bohr's model is r_1 and that of fourth orbit of Be³⁺ is represented as r_2 . Now the ratio $\frac{r_2}{r_1}$ is $x : 1$. The value of x is _____
[13-Apr-2023 shift 1]

Answer: 2

Solution:

$$r \propto \frac{n^2}{Z}$$

$$\frac{r_2}{r_1} = \left(\frac{n_2}{n_1} \right)^2 \times \frac{Z_1}{Z_2}$$

$$\frac{r_2}{r_1} = \left(\frac{4}{2} \right)^2 \times \frac{2}{4}$$

$$\frac{r_2}{r_1} = 2$$

$$x = 2$$

Question95

An atom absorbs a photon of wavelength 500 nm and emits another photon of wavelength 600 nm. The net energy absorbed by the atom in this process is $n \times 10^{-4}$ eV. The value of n is [Assume the atom to be stationary during the absorption and emission process] (. Take $h = 6.6 \times 10^{-34}$ Js and $c = 3 \times 10^8$ m/s)
[13-Apr-2023 shift 2]

Answer: 4125

Solution:

$$E = E_1 - E_2 = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

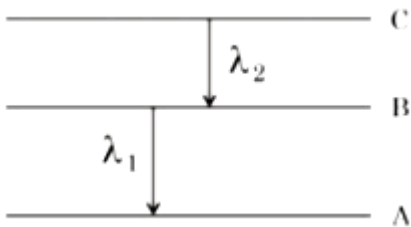
$$E = 6.6 \times 10^{-20} \text{ J}$$

$$E = 4.125 \times 10^{-1} \text{ eV}$$

$$E = 4125 \times 10^{-4} \text{ eV}$$

Question96

As per given figure A, B and C are the first, second and third excited energy levels of hydrogen atom respectively. If the ratio of the two wavelengths (. i.e. $\frac{\lambda_1}{\lambda_2}$) is $\frac{7}{4n}$, then the value of n will be _____.



[15-Apr-2023 shift 1]

Answer: 5

Solution:

For A, $n = 2$

B, $n = 3$

C, $n = 4$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda_2} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right)$$

$$\frac{1}{\lambda_2} = \frac{7R}{144} \dots (1)$$

$$\frac{1}{\lambda_1} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$\frac{1}{\lambda_1} = \frac{5R}{36} \dots (2)$$

(1) and (2)

$$\frac{\lambda_1}{\lambda_2} = \frac{7}{20} = \frac{7}{4 \times 5}$$

$$n = 5$$

Question97

For a nucleus ${}_Z^AX$ having mass number A and atomic number Z

A. The surface energy per nucleon (b_s) = $-a_1 A^{2/3}$.

B. The Coulomb contribution to the binding energy $b_c = -a_2 \frac{Z(Z-1)}{A^{4/3}}$

C. The volume energy $b_v = a_3 A$

D. Decrease in the binding energy is proportional to surface area.

E. While estimating the surface energy, it is assumed that each nucleon interacts with 12 nucleons. (a_1 , a_2 and a_3 are constants)

Choose the most appropriate answer from the options given below:
[8-Apr-2023 shift 1]

Options:

A. B, C only

B. A, B, C, D only

C. B, C, E only

D. C, D only

Answer: D

Solution:

$$E_B = a_v A - a_s A^{2/3} - a_A \frac{(A - 2Z)^2}{A^{1/3}} - a_c \frac{Z(Z - 1)}{A^{1/3}} + \delta(A, Z)$$

Most appropriate is option (4)

Question98

A nucleus with mass number 242 and binding energy per nucleon as 7.6 MeV breaks into two fragment each with mass number 121. If each fragment nucleus has binding energy per nucleon as 8.1 MeV, the total gain in binding energy is _____ MeV.
[8-Apr-2023 shift 1]

Answer: 121

Solution:

$$\begin{aligned} \text{Gain in binding energy} &= B \cdot E_f - B_i \\ &= 2(121 \times 8.1) - 242 \times 7.6 \\ &= 121 \text{ MeV} \end{aligned}$$

Question99

A radio active material is reduced to $1/8$ of its original amount in 3 days. If 8×10^{-3} kg of the material is left after 5 days the initial amount of the material is
[8-Apr-2023 shift 2]

Options:

A. 64g

B. 40g

C. 32g

D. 256g

Answer: D

Solution:

$$m = m_0 e^{-\lambda t}$$

$$\frac{m_0}{8} = m_0 e^{-\lambda t}$$

$$-\ln 8 = -\lambda t$$

$$= \lambda = \frac{\ln 8}{3} \text{ per day}$$

$$m = m_0 e^{-\lambda t}$$

$$8 = m_0 e^{-\frac{\ln 8}{3} \times 5}$$

$$\Rightarrow 8 = m_0 e^{-\frac{3 \ln 2}{3} \times 5}$$

$$8 = m_0 e^{\ln 2^{-5}}$$

$$= 8m_0 \left(\frac{1}{2^5} \right)$$

$$m_0 = 8 \times 2^5$$

$$= 8 \times 32$$

$$m_0 = 256 \text{ gm}^2$$

Question100

The decay constant for a radioactive nuclide is $1.5 \times 10^{-5} \text{ s}^{-1}$. Atomic weight of the substance is 60 gmole^{-1} , ($N_A = 6 \times 10^{23}$). The activity of $1.0 \mu\text{g}$ of the substance is _____ $\times 10^{10} \text{ Bq}$.
[10-Apr-2023 shift 1]

Answer: 15

Solution:

$$\text{No. of moles} = \frac{1 \times 10^{-6}}{60} = \frac{10^{-7}}{6}$$

$$\text{No. of atom} = n(N_A) = \frac{10^{-7}}{6} \times 6 \times 10^{23} = 10^{16}$$

$$\text{at } (t=0)A_0 = N_0 \lambda = 10^{16} \times 1.5 \times 10^{-5} = 15 \times 10^{10} \text{ Bq}$$

Question101

The half life of a radioactive substance is T. The time taken, for disintegrating $\frac{7}{8}$ th part of its original mass will be:

[10-Apr-2023 shift 2]

Options:

- A. T
- B. 2T
- C. 3T
- D. 8T

Answer: C

Solution:

If $\frac{7}{8}$ th is disintegrated it means only $\frac{1}{8}$ th part is radioactive active no. of nuclears after ' n ' half lives

$$\Rightarrow \frac{N_o}{2^n} = \frac{N_o}{8}$$

$$2^n = 8 = n = 3$$

So, the elapsed is 3 half lives = 3T

Question102

Two radioactive elements A and B initially have same number of atoms. The half life of A is same as the average life of B. If λ_A and λ_B are decay constants of A and B respectively, then choose the correct

relation from the given options.

[11-Apr-2023 shift 1]

Options:

A. $\lambda_A = 2\lambda_B$

B. $\lambda_A = \lambda_B$

C. $\lambda_A \ln 2 = \lambda_B$

D. $\lambda_A = \lambda_B \ln 2$

Answer: D

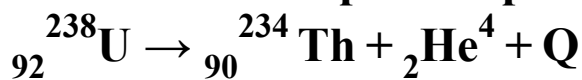
Solution:

$$\begin{array}{ccc} \text{A} & & \text{B} \\ t=0 & \text{N}_0 & \text{N}_0 \\ T_A = \tau_B & \rightarrow & \begin{array}{l} T \rightarrow \text{half life} \\ \tau \rightarrow \text{ang life} \\ \text{given in question} \end{array} \end{array}$$

Now $\frac{\ln(2)}{\lambda_A} = \frac{1}{\lambda_B} \Rightarrow \lambda_A = \lambda_B \cdot \ln(2)$

Question103

A common example of alpha decay is



Given :

$${}_{92}^{238}\text{U} = 238.05060\text{u},$$

$${}_{90}^{234}\text{Th} = 234.04360\text{u},$$

$${}_2^4\text{He} = 4.00260\text{u} \text{ and}$$

$$1\text{u} = 931.5 \frac{\text{MeV}}{c^2}$$

The energy released (Q) during the alpha decay of ${}^{238}\text{U}$ is _____
MeV

[12-Apr-2023 shift 1]

Answer: 4.25

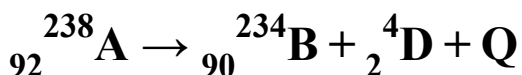
Solution:

$$\alpha = (m_u - M_{th} - M_{Ho})C^2$$

$$\alpha = 0.00456 \times 931.5$$

$$\alpha = 4.25 \text{ MeV}$$

Question104



In the given nuclear reaction, the approximate amount of energy released will be: [Given, mass of ${}_{92}^{238}\text{A} = 238.05079 \times 931.5 \text{ MeV}/c^2$,

$$\text{mass of } {}_{90}^{234}\text{B} = 234.04363 \times 931.5 \text{ MeV}/c^2 ,$$

$$\text{mass of } {}_2^4\text{D} = 4.00260 \times 931.5 \text{ MeV}/c^2]$$

[13-Apr-2023 shift 1]

Options:

A. 4.25 MeV

B. 5.9 MeV

C. 3.82 MeV

D. 2.12 MeV

Answer: A

Solution:

$$Q = \Delta m C^2$$

$$Q = (238.05079 - 234.04363 - 4.00260) \times 931.5 \text{ MeV}$$

$$Q = 0.00456 \times 931.5 \text{ MeV}$$

$$Q = 4.25 \text{ MeV}$$

Question105

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170 .

Reason R : Nuclear force is short ranged.

In the light of the above statements, choose the correct answer from the options given below

[13-Apr-2023 shift 2]

Options:

- A. Both A and R are true but R is NOT the correct explanation of A
- B. Both A and R are true and R is the correct explanation of A
- C. A is true but R is false
- D. A is false but R is true

Answer: B

Question106

The half-life of a radioactive nucleus is 5 years. The fraction of the original sample that would decay in 15 years is :

[15-Apr-2023 shift 1]

Options:

- A. $\frac{3}{4}$
- B. $\frac{1}{8}$

C. $\frac{7}{8}$

D. $\frac{1}{4}$

Answer: C

Solution:

$$N = N_0 \left(\frac{1}{2} \right)^{t/T_{1/2}}$$

$$N = N_0 \left(\frac{1}{2} \right)^{15/5}$$

$$N = \frac{N_0}{8}$$

$$\begin{aligned} \text{Decayed nuclei} &= N_0 - \frac{N_0}{8} \\ &= \frac{7N_0}{8} \end{aligned}$$

Question107

**Choose the correct option from the following options given below :
In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's
[24-Jun-2022-Shift-1]**

Options:

A. model electrons always experience a net-force.

An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-

B. uniform mass distribution in Thomson's model.

C. A classical atom based on Rutherford's model is doomed to collapse.

D. The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

Answer: C

Solution:

An atom based on classical theory of Rutherford's model should collapse as the electrons in continuous circular motion that is a continuously accelerated charge should emit EM waves and so should lose energy. These electrons losing energy should soon fall into heavy nucleus collapsing the whole atom.

Question108

**In Bohr's atomic model of hydrogen, let K, P and E are the kinetic energy, potential energy and total energy of the electron respectively. Choose the correct option when the electron undergoes transitions to a higher level :
[24-Jun-2022-Shift-2]**

Options:

- A. All K, P and E increase.
- B. K decreases, P and E increase.
- C. P decreases, K and E increase.
- D. K increases, P and E decrease.

Answer: B

Solution:

$$T.E. = \frac{-Z^2 me^4}{8(nh\epsilon_0)^2}$$

$$P.E. = \frac{-Z^2 me^4}{4(nh\epsilon_0)^2}$$

$$K.E. = \frac{Z^2 me^4}{8(nh\epsilon_0)^2}$$

As electron makes transition to higher level, total energy and potential energy increases (due to negative sign) while the kinetic energy reduces.

Question109

The ratio for the speed of the electron in the 3rd orbit of He⁺ to the speed of the electron in the 3rd orbit of hydrogen atom will be :
[25-Jun-2022-Shift-1]

Options:

A. 1 : 1

B. 1 : 2

C. 4 : 1

D. 2 : 1

Answer: D

Solution:

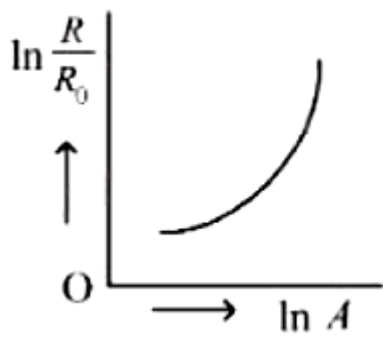
$$\begin{aligned}\text{We know that } v &\propto \frac{Z}{n} \\ \Rightarrow \text{Required ratio} &= \frac{\frac{2}{3}}{\frac{1}{3}} \\ &= 2 : 1\end{aligned}$$

Question110

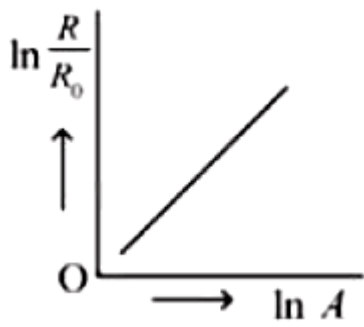
Which of the following figure represents the variation of $I_n \left(\frac{R}{R_0} \right)$ with $I_n A$ (if R = radius of a nucleus and A = its mass number)
[25-Jun-2022-Shift-2]

Options:

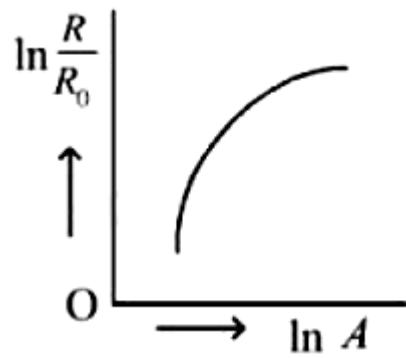
A.



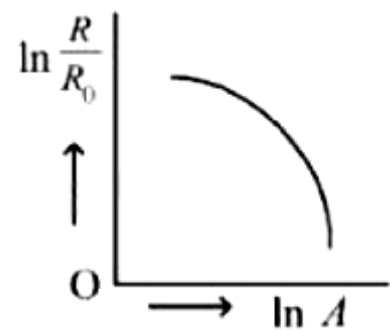
B.



C.



D.



Answer: B

Solution:

We know that

$$R = R_0 A^{1/3}$$

$$\Rightarrow \underbrace{\ln\left(\frac{R}{R_0}\right)}_y = \underbrace{\frac{1}{3}}_m \underbrace{\ln(A)}_x$$

$\Rightarrow$ Straight line

Question111

A hydrogen atom in its ground state absorbs 10.2 eV of energy. The angular momentum of electron of the hydrogen atom will increase by the value of :

(Given, Planck's constant = 6.6×10^{-34} Js).

[27-Jun-2022-Shift-1]

Options:

A. 2.10×10^{-34} Js

B. 1.05×10^{-34} Js

C. 3.15×10^{-34} Js

D. 4.2×10^{-34} Js

Answer: B

Solution:

$$-13.6 + 10.2 = \frac{-13.6}{n^2}$$

$$\Rightarrow \frac{13.6}{n^2} = 3.4$$

$$\Rightarrow n = 2$$

$$\Rightarrow \Delta L = 2 \times \frac{h}{2\lambda} - 1 \times \frac{h}{2\lambda}$$

$$= \frac{h}{2\lambda}$$

$$\Rightarrow \Delta L \simeq 1.05 \times 10^{-34} \text{ Js}$$

Question112

A beam of monochromatic light is used to excite the electron in Li^{++} from the first orbit to the third orbit. The wavelength of monochromatic light is found to be $x \times 10^{-10} \text{ m}$. The value of x is _____
[Given $hc = 1242 \text{ eV nm}$]
[27-Jun-2022-Shift-1]

Answer: 114

Solution:

$$\begin{aligned} E(\text{eV}) &= 13.6 \times 9 \left(1 - \frac{1}{9}\right) \\ &= 13.6 \times 8 \text{ eV} \\ \Rightarrow \lambda &= \frac{12420}{13.6 \times 8} \text{ \AA} \\ &= 114.15 \text{ \AA} \end{aligned}$$

Question113

$\sqrt{d_1}$ and $\sqrt{d_2}$ are the impact parameters corresponding to scattering angles 60° and 90° respectively, when an α particle is approaching a gold nucleus. For $d_1 = x d_2$, the value of x will be _____
[29-Jun-2022-Shift-1]

Answer: 3

Solution:

Impact parameter $\propto \cot \frac{\theta}{2}$

$$\Rightarrow \sqrt{\frac{d_1}{d_2}} = \frac{\sqrt{3}}{1}$$

$$\Rightarrow d_1 = 3d_2$$

$$\Rightarrow x = 3$$

Question114

Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits in two fragments ' B ' and ' C ' of mass numbers 105 and 115 . The binding energy of nucleons in ' B ' and ' C ' is 6.4 MeV per nucleon. The energy Q released per fission will be :

[24-Jun-2022-Shift-1]

Options:

A. 0.8 MeV

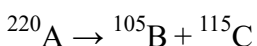
B. 275 MeV

C. 220 MeV

D. 176 MeV

Answer: D

Solution:



$$\Rightarrow Q = [105 \times 6.4 + 115 \times 6.4] - [220 \times 5.6] \text{ MeV}$$

$$\Rightarrow Q = 176 \text{ MeV}$$

Question115

A sample contains 10^{-2} kg each of two substances A and B with half lives 4s and 8 s respectively. The ratio of their atomic weights is 1 : 2. The ratio of the amounts of A and B after 16s is $\frac{x}{100}$. The value of x

is _____
[24-Jun-2022-Shift-2]

Answer: 25

Solution:

$$N_1 = \frac{\left(\frac{10^{-2}}{1}\right)}{2^4}$$

$$N_2 = \frac{\left(\frac{10^{-2}}{2}\right)}{2^2}$$

$$\Rightarrow \frac{N_1}{N_2} = \frac{1}{2}$$

∴ Mass ratio of A and B,

$$\frac{m_1}{m_2} = \frac{N_1}{N_2} \times \left(\frac{M_1}{M_2}\right)$$

$$= \frac{1}{2} \times \left(\frac{1}{2}\right)$$

$$= \frac{1}{4}$$

$$= \frac{25}{100}$$

$$\therefore x = 25$$

Question116

How many alpha and beta particles are emitted when Uranium ${}_{92}\text{U}^{238}$ decays to lead ${}_{82}\text{Pb}^{206}$?

[26-Jun-2022-Shift-1]

Options:

A. 3 alpha particles and 5 beta particles

B. 6 alpha particles and 4 beta particles

C. 4 alpha particles and 5 beta particles

D. 8 alpha particles and 6 beta particles

Answer: D

Solution:

$$\begin{aligned} {}_{92}\text{U}^{238} &\longrightarrow {}_{82}\text{Pb}^{206} + x({}_2\text{H}\text{e}^4) + {}_y(-1\beta^0) \\ 238 &= 206 + 4x + 0 \\ \Rightarrow 4x &= 32 \Rightarrow x = 8 \\ \text{also, } 92 &= 82 + 2x - y \\ y &= 82 + 16 - 92 = 6 \end{aligned}$$

Question117

A radioactive nucleus can decay by two different processes. Half-life for the first process is 3.0 hours while it is 4.5 hours for the second process. The effective half-life of the nucleus will be:
[26-Jun-2022-Shift-2]

Options:

A. 3.75 hours

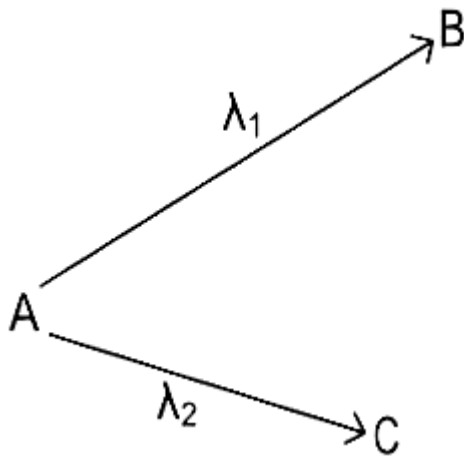
B. 0.56 hours

C. 0.26 hours

D. 1.80 hours

Answer: D

Solution:



$$\begin{aligned} \frac{dA}{dt} &= (-\lambda_1 A) + (-\lambda_2 A) \\ \Rightarrow \frac{dA}{dt} &= -(\lambda_1 + \lambda_2)A \\ \Rightarrow \lambda_{\text{eff}} &= \lambda_1 + \lambda_2 \\ \Rightarrow \frac{\ln 2}{(t_{1/2})_{\text{eff}}} &= \frac{\ln 2}{(t_{1/2})_1} + \frac{\ln 2}{(t_{1/2})_2} \\ \Rightarrow (t_{1/2})_{\text{eff}} &= \frac{4.5 \times 3}{7.5} \text{ hours} = 1.8 \text{ hours} \end{aligned}$$

Question 118

The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as :

[28-Jun-2022-Shift-1]

Options:

- A. $Q = K_p$
- B. $(K_p + Q) < 0$
- C. $Q < K_p$
- D. $(K_p + Q) > 0$

Answer: D

Solution:

$$K_p > 0$$

If Q is released $\Rightarrow Q > 0$

$$\Rightarrow K_p + Q > 0$$

Even the particle has to be given kinetic energy greater than magnitude of Q to maintain momentum conservation.

$$\Rightarrow K + Q > 0$$

Question119

Following statements related to radioactivity are given below :

(A) Radioactivity is a random and spontaneous process and is dependent on physical and chemical conditions.

(B) The number of un-decayed nuclei in the radioactive sample decays exponentially with time.

(C) Slope of the graph of $\log_e$ (no. of undecayed nuclei) Vs. time represents the reciprocal of mean life time (τ).

(D) Product of decay constant (λ) and half-life time ($T_{1/2}$) is not constant.

Choose the most appropriate answer from the options given below :
[28-Jun-2022-Shift-2]

Options:

A. (A) and (B) only

B. (B) and (D) only

C. (B) and (C) only

D. (C) and (D) only

Answer: C

Solution:

Radioactive decay is a random and spontaneous process it depends on unbalancing of nucleus.

$$N = N_0 e^{-\lambda t} \dots (B)$$

$$\ln N = -\lambda t + \ln N_0$$

So, slope = $-\lambda \dots (C)$

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

$$\text{So } t_{1/2} \times \lambda = \ln 2 = \text{Constant}$$

Question120

The activity of a radioactive material is 2.56×10^{-3} Ci. If the half life of the material is 5 days, after how many days the activity will become 2×10^{-5} Ci ?

[29-Jun-2022-Shift-1]

Options:

A. 30 days

B. 35 days

C. 40 days

D. 25 days

Answer: B

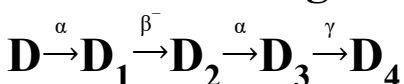
Solution:

$$\frac{A}{A_0} = \frac{N}{N_0}$$
$$\frac{2 \times 10^{-5}}{2.56 \times 10^{-3}} = \frac{N}{N_0}$$
$$\frac{N}{N_0} = \frac{1}{128} \Rightarrow N = \frac{N_0}{128}$$

After 7 half life activity comes down to given value $T = 7 \times 5$
= 35 days

Question121

In the following nuclear reaction,



Mass number of D is 182 and atomic number is 74 . Mass number and atomic number of D_4 respectively will be _____

[29-Jun-2022-Shift-2]

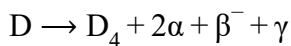
Options:

- A. 174 and 71
- B. 174 and 69
- C. 172 and 69
- D. 172 and 71

Answer: A

Solution:

Equivalent reaction can be written as



$$\Rightarrow \text{Mass number of } D_4 = \text{Mass number of } D - 2 \times 4$$

$$= 182 - 8 = 174$$

$$\Rightarrow \text{Atomic number of } D_4$$

$$= \text{Atomic number of } D - 2 \times 2 + 1$$

$$= 74 - 4 + 1 = 71$$

Question122

The half life of a radioactive substance is 5 years. After x years a given sample of the radioactive substance gets reduced to 6.25% of its initial value. The value of x is _____
[29-Jun-2022-Shift-2]

Answer: 20

Solution:

$$N = N_0 e^{-\lambda t}$$

$$\Rightarrow \frac{6.25}{100} = e^{-\lambda t}$$

$$\Rightarrow e^{-\lambda t} = \frac{1}{16} = \left(\frac{1}{2}\right)^4$$

$$\Rightarrow t = 4t_{1/2}$$

$$\Rightarrow t = 20 \text{ years}$$

Question123

The momentum of an electron revolving in n^{th} orbit is given by:
(Symbols have their usual meanings)

[25-Jul-2022-Shift-1]

Options:

A. $\frac{nh}{2\pi r}$

B. $\frac{nh}{2r}$

C. $\frac{nh}{2\pi}$

D. $\frac{2\pi r}{nh}$

Answer: A

Solution:

$$\because mvr = \frac{nh}{2\pi}$$
$$\Rightarrow mv = \frac{nh}{2\pi r}$$

Question124

The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by :

[25-Jul-2022-Shift-1]

Options:

A. $\vec{\mu}_L = \frac{eL}{2m}$

B. $\vec{\mu}_L = -\frac{eL}{2m}$

C. $\vec{\mu}_l = -\frac{eL}{m}$

D. $\vec{\mu}_l = \frac{2eL}{m}$

Answer: B

Solution:

$$\because \vec{\mu} = \frac{q}{2m}$$

$$\Rightarrow \vec{\mu} = \frac{-eL}{2m}$$

Question125

Hydrogen atom from excited state comes to the ground state by emitting a photon of wavelength λ . The value of principal quantum number ' n ' of the excited state will be : (R : Rydberg constant)
[25-Jul-2022-Shift-2]

Options:

A. $\sqrt{\frac{\lambda R}{\lambda - 1}}$

B. $\sqrt{\frac{\lambda R}{\lambda R - 1}}$

C. $\sqrt{\frac{\lambda}{\lambda R - 1}}$

D. $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$

Answer: B

Solution:

$$\begin{aligned}
 \because \frac{1}{\lambda} &= R \left(\frac{1}{1^2} - \frac{1}{n^2} \right) \\
 \Rightarrow \frac{1}{\lambda R} &= 1 - \frac{1}{n^2} \\
 \Rightarrow \frac{1}{n^2} &= 1 - \frac{1}{\lambda R} = \frac{\lambda R - 1}{\lambda R} \\
 \Rightarrow n &= \sqrt{\frac{\lambda R}{\lambda R - 1}}
 \end{aligned}$$

Question 126

$\frac{x}{x+4}$ is the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its
 (i) third permitted energy level to the second level and
 (ii) the highest permitted energy level to the second permitted level.
 The value of x will be
 [25-Jul-2022-Shift-2]

Answer: 5

Solution:

$$\begin{aligned}
 E_n &= -\frac{13.6}{n^2} \text{ eV} \\
 \frac{\frac{1}{2^2} - \frac{1}{3^2}}{\frac{1}{2^2}} &= \frac{x}{x+4} \\
 \Rightarrow \frac{9-4}{9 \times 4 \times \frac{1}{4}} &= \frac{x}{x+4} = \frac{5}{9} \\
 x &= 5
 \end{aligned}$$

Question127

In the hydrogen spectrum, λ be the wavelength of first transition line of Lyman series. The wavelength difference will be " $a \lambda$ " between the wavelength of 3rd transition line of Paschen series and that of 2nd transition line of Balmer series where $a =$ _____.
[26-Jul-2022-Shift-1]

Answer: 5

Solution:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\frac{1}{\lambda_3} = R_H \left(\frac{1}{3^2} - \frac{1}{6^2} \right)$$

$$\frac{1}{\lambda_2} = R_H \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

$$\therefore \lambda_3 - \lambda_2 = a\lambda$$

$$a = 5$$

Question128

A nucleus of mass M at rest splits into two parts having masses $\frac{M'}{3}$ and $\frac{2M'}{3}$ ($M' < M$). The ratio of de Broglie wavelength of two parts will be :
[26-Jul-2022-Shift-2]

Options:

A. 1 : 2

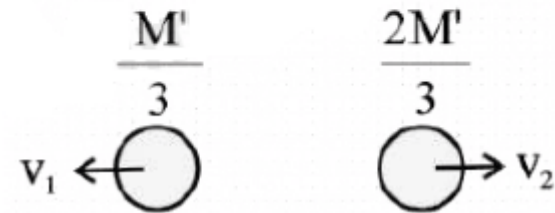
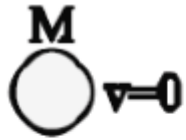
B. 2 : 1

C. 1 : 1

D. 2 : 3

Answer: C

Solution:



$$|\vec{P}_1| = |\vec{P}_2|$$

Here $\vec{P}$ is momentum

$$\text{So } \lambda = \frac{h}{P}$$

Hence both will have same de broglie wavelength.

Question129

Mass numbers of two nuclei are in the ratio of 4 : 3. Their nuclear densities will be in the ratio of
[26-Jul-2022-Shift-2]

Options:

A. 4 : 3

B. $\left(\frac{3}{4}\right)^{\frac{1}{3}}$

C. 1 : 1

D. $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

Answer: C

Solution:

$$\text{Radius of nucleus } R = R_0 A^{\frac{1}{3}}$$

$$\text{Density of nucleus} = \frac{\text{Mass of nucleus}}{\text{volume of nucleus}}$$

$$\rho = \frac{m \times A}{\frac{4}{3}\pi R^3} \text{ Where } m: \text{ mass of proton or neutron}$$

$$\rho = \frac{m \times A}{\frac{4}{3}\pi R_0^3 A}$$

$$\rho \propto A^0$$

Hence density of nucleus is independent of mass number

Question130

An electron (mass m) with an initial velocity $\vec{v} = v_0 \hat{i}$ ($v_0 > 0$) is moving in an electric field $\vec{E} = -E_0 \hat{i}$ ($E_0 > 0$) where E_0 is constant. If at $t = 0$ de Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de Broglie wavelength after time t is given by
[27-Jul-2022-Shift-1]

Options:

A. λ_0

B. $\lambda_0 \left(1 + \frac{eE_0 t}{mv_0} \right)$

C. $\lambda_0 t$

D. $\frac{\lambda_0}{\left(1 + \frac{eE_0 t}{mv_0} \right)}$

Answer: D

Solution:



$$E = -E_0 \hat{i}$$

$$\lambda_0 = \frac{h}{mv}$$

$$v = v_0 + \frac{eE_0 t}{m}$$

$$\lambda = \frac{h}{mv} = \frac{h}{m \left(v_0 + \frac{eE_0 t}{m} \right)}$$

$$\lambda' = \frac{h}{mv_0 \left(1 + \frac{eE_0 t}{mv_0} \right)}$$

$$\lambda' = \frac{\lambda_0}{\left(1 + \frac{eE_0 t}{mv_0} \right)}$$

Question131

Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

[29-Jul-2022-Shift-1]

Options:

A. 3 : 4

B. 4 : 3

C. 1 : 4

D. 4 : 1

Answer: A

Solution:

$$E_n = \frac{-13.6}{n^2} \text{ eV}$$

$$\Rightarrow \frac{E_2 - E_1}{E_\infty - E_1} = \frac{13.6 \left(1 - \frac{1}{4}\right)}{13.6} = \frac{3}{4}$$

Question 132

Read the following statements:

(A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass number.

(D) Density of the nucleus is directly proportional to the cube root of the mass number.

(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options.

[29-Jul-2022-Shift-2]

Options:

A. (A) and (D) only.

B. (A) and (E) only.

C. (B) and (E) only.

D. (A) and (C) only.

Answer: B

Solution:

We know,

$$\text{Radius of nucleus, } r = r_0 A^{\frac{1}{3}}$$

where, A = mass number

$$\therefore \text{Volume (v)} = \frac{4}{3} \pi r^3$$

$$= \frac{4}{3} \pi r_0^3 \cdot A$$

$$\therefore v \propto A$$

∴ Volume is directly proportional to mass number.

We know,

$$\begin{aligned}\text{density}(d) &= \frac{m}{v} \\&= \frac{zm_p + (A - z)m_N}{\frac{4}{3}\pi r_0^3 (A^{1/3})^3} \\&= \frac{zm_p + Am_p - zm_p}{\frac{4}{3}\pi r_0^3 A} [\because m_p \simeq m_N] \\&= \frac{Am_p}{\frac{4}{3}\pi r_0^3 A} \\&= \frac{m_p}{\frac{4}{3}\pi r_0^3}\end{aligned}$$

∴ Density of nucleus is independent of mass number.

Question133

The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is :

(Take $\log_{10} 1.88 = 0.274$)

[26-Jul-2022-Shift-1]

Options:

- A. 0.02min^{-1}
- B. 2.7min^{-1}
- C. 0.063min^{-1}
- D. 6.3min^{-1}

Answer: C

Solution:

$$A_0 = 4250$$

$$A = 2250 = A_0 e^{-\lambda t}$$

$$\Rightarrow \frac{2250}{4250} = e^{-\lambda t}$$

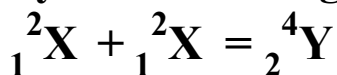
$$\Rightarrow \lambda(10) = \ln\left(\frac{4250}{2250}\right)$$

$$\lambda(10) = 0.636$$

$$\lambda = 0.063$$

Question134

Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below :



The binding energies per nucleon for ${}_1^2\text{X}$ and ${}_2^4\text{Y}$ are 1.1 MeV and 7.6 MeV respectively. The energy released in this process is

_____ MeV

[26-Jul-2022-Shift-2]

Answer: 26

Solution:

$$\begin{aligned} \text{Energy released in the given process} &= \text{Binding energy of product} - \text{Binding energy of reactants} \\ &= 7.6 \times 4 - (1.1 \times 2) \times 2 \\ &= 30.4 - 4.4 \\ &= 26 \text{ MeV} \end{aligned}$$

Question135

What is the half-life period of a radioactive material if its activity drops to $1/16^{\text{th}}$ of its initial value in 30 years?

[27-Jul-2022-Shift-1]

Options:

A. 9.5 years

- B. 8.5 years
- C. 7.5 years
- D. 10.5 years

Answer: C

Solution:

$$A = A_0 e^{-\lambda t}$$

$$\Rightarrow -\lambda t = \ln \left(\frac{A}{A_0} \right)$$

$$\Rightarrow -\frac{\ln 2}{t_{1/2}} \times 30 = \ln \left(\frac{1}{16} \right)$$

$$\Rightarrow -\frac{\ln 2}{t_{1/2}} \times 30 = -4 \ln 2$$

$$\Rightarrow t_{1/2} = \frac{30}{4} = 7.5 \text{ yrs}$$

Question136

The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after :
[27-Jul-2022-Shift-2]

Options:

- A. 7 days
- B. 15 days
- C. 25 days
- D. 35 days

Answer: D

Solution:

$$A_0 = 6.4 \times 10^{-4} \text{ Curie}$$

$$T_{1/2} = 5 \text{ days} = \frac{\ln 2}{\lambda}$$

$$A = A_0 e^{-\lambda t}$$

$$5 \times 10^{-6} = 6.4 \times 10^{-4} e^{-\lambda t}$$

$$\frac{5}{6.4} \times 10^{-2} = e^{-\lambda t}$$

$$7.8 \times 10^{-3} = e^{-\lambda t}$$

$$\log(7.8 \times 10^{-3}) = -\lambda t \ln e$$

$$\ln(7.8 \times 10^{-3}) = -\frac{\lambda \ln 2}{5} \cdot t$$

$$\therefore \frac{5 \times 4.853}{0.693} = t = 35 \text{ days}$$

Question 137

The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$ th of its original mass to disintegrate will be :

[28-Jul-2022-Shift-1]

Options:

A. 120 days

B. 130 days

C. 180 days

D. 20 days

Answer: C

Solution:

$$\therefore N = \frac{N_0}{2^{\frac{t}{T_{1/2}}}}$$

$$\Rightarrow 2^{\frac{t}{T_{1/2}}} = \frac{N_0}{N} = \frac{N_0}{\left(\frac{N_0}{8}\right)}$$

$$\Rightarrow 2^{\frac{t}{T_{1/2}}} = 8 = 2^3$$

$$\Rightarrow t = 3 \times T_{1/2} = 3 \times 60 \\ = 180 \text{ days}$$

Question138

A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours.
[28-Jul-2022-Shift-1]

Answer: 15

Solution:

$$A = A_0 \times 2^{-t/T} \\ \frac{A_0}{64} = A_0 \times 2^{-t/T} \\ \therefore t = 6T = 6 \times 2 \cdot 5 = 15 \text{ hours}$$

Question139

A radioactive sample decays $\frac{7}{8}$ times its original quantity in 15 minutes. The half-life of the sample is
[28-Jul-2022-Shift-2]

Options:

- A. 5 min
- B. 7.5 min
- C. 15 min
- D. 30 min

Answer: A

Solution:

$$N = \frac{N_0}{2^{\frac{t}{T_{1/2}}}}$$
$$\Rightarrow 2^{\frac{t}{T_{1/2}}} = \frac{N_0}{N} = \frac{N_0}{\left(\frac{N_0}{8}\right)} = 8$$

$$\Rightarrow \frac{t}{T_{1/2}} = 3$$

$$\Rightarrow T_{1/2} = \frac{15}{3} = 5 \text{ min}$$

Question 140

Two radioactive materials A and B have decay constants 25λ and 16λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of B to that of A will be "e" after a time $\frac{1}{a\lambda}$. The value of a is _____.

[29-Jul-2022-Shift-2]

Answer: 9

Solution:

$$N_A = N_0 e^{-25\lambda t}$$

$$N_B = N_0 e^{-16\lambda t}$$

$$\frac{N_B}{N_A} = e = e^{9\lambda t}$$

$$t = \frac{1}{9\lambda}$$

Question141

The recoil speed of a hydrogen atom after it emits a photon in going from $n = 5$ state to $n = 1$ state will be
[26 Feb 2021 Shift 2]

Options:

A. 4.17m/s

B. 2.19m/s

C. 3.25m/s

D. 4.34m/s

Answer: A

Solution:

As we know, mass of photon, $m = 1.6 \times 10^{-27}$ kg Using de-Broglie wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{mv} \dots (i)$$

where, h is Planck's constant $= 6.63 \times 10^{-34}$ J – s and p is momentum of photon.

By using Rydberg wavelength equation,

$$\frac{1}{\lambda} = R \left[\frac{1}{n_2^2} - \frac{1}{n_1^2} \right]$$

$$\frac{1}{\lambda} = 1.09 \times 10^7 \left[\frac{1}{1^2} - \frac{1}{25} \right]$$

$$\Rightarrow \frac{1}{\lambda} = 1.09 \times 10^7 \left[\frac{24}{25} \right]$$

$$\Rightarrow \lambda = \frac{25}{1.09 \times 10^7 \times 24}$$

Substituting values in Eq. (i), we get

$$\begin{aligned} v &= \frac{h}{m\lambda} = \frac{6.63 \times 10^{-34} \times 1.09 \times 10^7 \times 24}{1.6 \times 10^{-27} \times 25} \\ &= \frac{6.63 \times 1.09 \times 24}{1.6 \times 2.5} = 4.17 \text{ ms}^{-1} \end{aligned}$$

Question142

If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of

$\lambda_1 : \lambda_2$ is

[26 Feb 2021 Shift 1]

Options:

A. 1 : 9

B. 7 : 108

C. 7 : 135

D. 1 : 3

Answer: C

Solution:

By using Rydberg's formula, $\frac{1}{\lambda} = R \left[\frac{1}{n_2^2} - \frac{1}{n_1^2} \right]$

where, R is Rydberg constant.

For wavelength of third member of Lyman series $n_2 = 1$ and $n_1 = 4$.

$$\therefore \frac{1}{\lambda_1} = R \left[\frac{1}{1^2} - \frac{1}{4^2} \right] = R \left[\frac{1}{1} - \frac{1}{16} \right] \dots (i)$$

For wavelength of first member of Paschen series, $n_2 = 3$ and $n_1 = 4$

$$\therefore \frac{1}{\lambda_2} = R \left[\frac{1}{3^2} - \frac{1}{4^2} \right] = R \left[\frac{1}{9} - \frac{1}{16} \right] \dots (ii)$$

On dividing Eq. (ii) by Eq. (i), we get

$$\frac{\lambda_1}{\lambda_2} = \frac{\left[\frac{1}{9} - \frac{1}{16} \right]}{\left[\frac{1}{1} - \frac{1}{16} \right]} = \frac{7}{9 \times 15} = \frac{7}{135}$$

Question143

The wavelength of the photon emitted by a hydrogen atom when an electron makes a transition from $n = 2$ to $n = 1$ state is

[25 Feb 2021 Shift 2]

Options:

A. 121.8nm

B. 194.8nm

C. 490.7nm

D. 913.3nm

Answer: A

Solution:

Given, electron is moving from $n = 2$ to $n = 1$.

From Bohr's hydrogen spectrum (Rydberg formula)

$$\frac{1}{\lambda} = R \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

where, λ = wavelength,

$R = 1.097 \times 10^7 \text{ m}^{-1}$ (Rydberg's constant)

$$\Rightarrow \frac{1}{\lambda} = 1.097 \times 10^7 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$$= 1.097 \times 10^7 \left[\frac{1}{1} - \frac{1}{4} \right] = \frac{3}{4} \times 1.097 \times 10^7$$

$$\lambda = \frac{4}{3 \times 1.097 \times 10^7} = 1.215 \times 10^{-7}$$

$$= 121.5 \times 10^{-9} \sim \text{eq} 121.8 \text{ nm}$$

Question 144

According to Bohr atom model, in which of the following transitions will the frequency be maximum ?

[24 Feb 2021 Shift 2]

Options:

A. $n = 4$ to $n = 3$

B. $n = 2$ to $n = 1$

C. $n = 5$ to $n = 4$

D. $n = 3$ to $n = 2$

Answer: B

Solution:

Let n_f, n_i be the final and initial orbit.

As we know that,

$$\frac{1}{\lambda} = 109 \times 10^7 \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

Now, checking for each option, we get

$$(a) \frac{1}{\lambda} \propto \left[\frac{1}{3^2} - \frac{1}{4^2} \right] = \left[\frac{1}{9} - \frac{1}{16} \right] = 0.05 \dots (i)$$

$$(b) \frac{1}{\lambda} \propto \left[\frac{1}{1} - \frac{1}{4} \right] = 0.75 \dots (ii)$$

$$(c) \frac{1}{\lambda} \propto \left[\frac{1}{16} - \frac{1}{25} \right] = 0.0225 \dots (iii)$$

$$(d) \frac{1}{\lambda} \propto \left[\frac{1}{4} - \frac{1}{9} \right] = 0.14 \dots (iv)$$

The option (b) has highest value.

$$\text{Since, frequency, } f = \frac{C}{\lambda} \Rightarrow f \propto \frac{1}{\lambda}$$

$\therefore$ Frequency will be maximum for transition $n = 2$ to $n = 1$.

Question145

A radioactive sample is undergoing α -decay. At any time t_1 , its activity is A and another time t_2 , the activity is $\frac{A}{5}$. What is the average life time for the sample?

[26 Feb 2021 Shift 2]

Options:

A. $\frac{\ln 5}{t_2 - t_1}$

B. $\frac{t_1 - t_2}{\ln 5}$

C. $\frac{t_2 - t_1}{\ln 5}$

D. $\frac{\ln(t_2 + t_1)}{2}$

Answer: C

Solution:

Given, activity of radioactive element at time t_1 ,

$$A_1 = A$$

and at time t_2 , $A_2 = A/5$

As we know that,

$$\text{activity at any time (A)} = A_0 e^{-\lambda t}$$

where A_0 is activity at time $t = 0$.

$$\therefore A = A_0 e^{-\lambda t_1} \dots (i)$$

$$\text{and } A/5 = A_0 e^{-\lambda t_2} \dots (ii)$$

$$\Rightarrow 5 = e^{-\lambda t_1 + \lambda t_2}$$

Taking log on both sides, we get

$$\ln 5 = \lambda(t_2 - t_1) \ln e$$

$$\Rightarrow \ln 5 = \lambda(t_2 - t_1) [\because \ln e = 1]$$

$$\Rightarrow \lambda = \ln 5 / (t_2 - t_1)$$

$$\Rightarrow \text{Also, average life, } \tau = \frac{1}{\lambda}$$

$$\Rightarrow \tau = \frac{t_2 - t_1}{\ln 5}$$

Question 146

Two radioactive substances X and Y originally have N_1 and N_2 nuclei, respectively. Half-life of X is half of the half-life of Y. After three half-lives of Y, number of nuclei of both are equal. The ratio $\frac{N_1}{N_2}$ will be equal to

[25 Feb 2021 Shift 1]

Options:

A. $\frac{1}{8}$

B. $\frac{3}{1}$

C. $\frac{8}{1}$

D. $\frac{1}{3}$

Answer: C

Solution:

Given, initial amount of X and Y be N_1 and N_2 .

Let half-life of X be t_x and y be t_y .

According to question, $t_x = \frac{t_y}{2} = t$

$\Rightarrow t_x = t$ and $t_y = 2t$

After 3 half-lives of Y ,

$$3t_y = 6t$$

As we know that,

$$N = N_0 e^{-\lambda t}$$

where, N is the number of nuclei left undecayed.

$$\text{and } t_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{t_{1/2}}$$

$$\lambda_1 = \frac{0.693}{t} \text{ and } \lambda_2 = \frac{0.693}{2t}$$

Since, after 3 half-lives of Y number of nuclei of both become equal.

$$\therefore N_1 e^{-\lambda_1 6t} = N_2 e^{-\lambda_2 6t}$$

$$\Rightarrow N_1 / N_2 = e^{6t(-\lambda_2 + \lambda_1)}$$

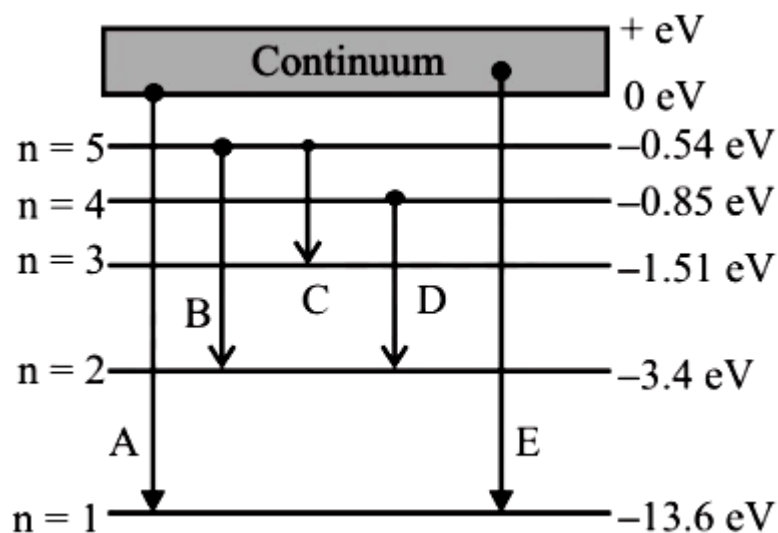
$$\Rightarrow N_1 / N_2 = e^{6t \left(\frac{0.693}{t} - \frac{0.693}{2t} \right)}$$

$$= e^{0.693 \left(\frac{6t}{t} - \frac{6t}{2t} \right)} = e^{0.693 \times 3} = 7.9 \sim \text{eq8}$$

$$\frac{N_1}{N_2} = \frac{8}{1}$$

Question147

In the given figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E. The transitions A, B and C respectively represent :



[24feb2021shift1]

Options:

- A. The ionization potential of hydrogen, second member of Balmer series and third member of Paschen series.
- B. The first member of the Lyman series, third member of Balmer series and second member of Paschen series.
- C. The series limit of Lyman series, third member of Balmer series and second member of Paschen series.
- D. The series limit of Lyman series, second member of Balmer series and second member of Paschen series.

Answer: C

Solution:

A corresponds to transition from $n = \infty$ to $n = 1$.

It is a series limit of Lyman series.

B corresponds to transition from $n = 5$ to $n = 2$.

It is a third member of Balmer series.

C corresponds to transition from $n = 5$ to $n = 3$, thus, it is second member of Paschen series.

Question148

Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionisation potential

**of this hydrogen atom will be
[18 Mar 2021 Shift 1]**

Options:

- A. 13.6eV
- B. 2815.2eV
- C. 331.2eV
- D. 27.2eV

Answer: B

Solution:

As we know that, the ionisation potential of the hydrogen,

$$IP_H = 13.6\text{eV}$$

We know that,

$$IP = \frac{Z^2 m q^4}{8n^2 h^2 \epsilon_0^2}$$

$$\Rightarrow IP \propto m q^4 \Rightarrow \frac{IP_\mu}{IP_H} = \frac{m_\mu q_\mu^4}{m_e q_e^4}$$

$$\Rightarrow \frac{IP}{IP_\mu} = \frac{m_\mu}{m_e} \quad [\because q_\mu = q_e]$$

$$IP_\mu = \frac{m_\mu}{m_e} \times IP_H = \frac{207m_e}{m_e} \times 13.6 \\ = 2815.2\text{eV}$$

Question149

The atomic hydrogen emits a line spectrum consisting of various series. Which series of hydrogen atomic spectra is lying in the visible region?

[17 Mar 2021 Shift 2]

Options:

- A. Brackett series
- B. Paschen series

C. Lyman series

D. Balmer series

Answer: D

Solution:

When an electron jumps from the higher energy level to $n = 2$ orbit, Balmer series of the line spectrum is obtained.

The Balmer series of the hydrogen atom lies in the visible region. However, Brackett and Paschen series of hydrogen atom lies in the infrared region and Lyman series of hydrogen atom lies in the ultraviolet region.

Question 150

A particle of mass m moves in a circular orbit in a central potential field $U(r) = U_0 r^4$. If Bohr's quantisation conditions are applied, radii of possible orbitals r_n vary with $n^{1/\alpha}$, where α is
[17 Mar 2021 Shift 2]

Answer: 3

Solution:

As we know, the force on the particle is given as

$$F = - \frac{dU(r)}{dr}$$

$$\text{Given, } U(r) = U_0 r^4$$

$$\Rightarrow F = - \frac{d(U_0 r^4)}{dr} \Rightarrow F = -4U_0 r^3$$

As we know, the force on the particle moving in a circular orbit of radius r will be centripetal force

$$\left| \frac{mv^2}{r} \right| = 4U_0 r^3$$

$$\Rightarrow v^2 \propto r^4$$

$$v \propto r^2 \dots (i)$$

Hence, the velocity of the particle is square of the radius of the orbit. Using the Bohr's quantisation condition, which states that the electron revolves around the nucleus only in those orbits for which

the angular momentum is some integral multiple of $\frac{h}{2\pi}$, here, h is

Planck's constant.

i.e,

$$n \frac{h}{2\pi} = mvr \quad [\text{From Eq. (i)}]$$

$$\Rightarrow r^3 \propto n \Rightarrow r \propto n^{1/3}$$

Comparing the above relation with $n^{1/\alpha}$, we get

$$\alpha = 3$$

Question151

Which level of the single ionized carbon has the same energy as the ground state energy of hydrogen atom?

[17 Mar 2021 Shift 1]

Options:

A. 1

B. 6

C. 4

D. 8

Answer: B

Solution:

Since we know that, energy of hydrogen atom is

$$E = -13.6 \frac{Z^2}{n^2}$$

where, Z is the atomic number and n is the energy state.

$\therefore$ For hydrogen atom, $Z = 1$ and $n = 1$

$$\Rightarrow E = \frac{-13.6(1)^2}{1^2}$$

$$\Rightarrow E = -13.6\text{eV} \dots (i)$$

For carbon atom, $Z = 6$, $n = ?$

$$\Rightarrow E = \frac{-13.6(6)^2}{n^2} \dots (ii)$$

According to question, carbon has same energy as the ground state energy of hydrogen atom.

$\therefore$ From Eqs. (i) and (ii), we get

$$-13.6 = \frac{-13.6(6)^2}{n^2} \Rightarrow n^2 = 6^2$$

$$\Rightarrow n = 6$$

Question152

If an electron is moving in the n th orbit of the hydrogen atom, then its velocity v_n for the n th orbit is given as

[17 Mar 2021 Shift 1]

Options:

A. $v_n \propto n$

B. $v_n \propto \frac{1}{n}$ C.

C. $v_n \propto n^2$

D. $v_n \propto \frac{1}{n^2}$

Answer: B

Solution:

As we know that, the velocity of electron in the n th shell of hydrogen atom is given by

$$v_n = \frac{Z e^2}{2 \epsilon_0 n h}$$

where, Z = atomic number,

n = order of orbit,

e = charge on an electron,

h = Planck's constant

and ϵ_0 = absolute permittivity of free space

$$\Rightarrow v_n \propto \frac{1}{n}$$

Question153

The first three spectral lines of H-atom in the Balmer series are given $\lambda_1, \lambda_2, \lambda_3$ considering the Bohr atomic model, the wavelengths

of first and third spectral lines $\left(\frac{\lambda_1}{\lambda_3} \right)$ are related by a factor of

approximately $x \times 10^{-1}$. The value of x to the nearest integer, is

.....

[16 Mar 2021 Shift 1]

Answer: 15

Solution:

Since, we know that,

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \dots\dots(i)$$

where, λ = wavelength of light emitted.

R = Rydberg's constant,

Z = atomic number,

n_1 = principal quantum number of lower energy level

and n_2 = principal quantum number of higher energy level.

Therefore, for 1st spectral line of Balmer series, $n_1 = 2$ and $n_2 = 3$

$$\frac{1}{\lambda_1} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$\Rightarrow \frac{1}{\lambda_1} = RZ^2 \left(\frac{5}{36} \right) \dots(ii)$$

Similarly, for 3rd spectral line, $n_1 = 2$ and $n_2 = 5$

$$\frac{1}{\lambda_3} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{5^2} \right)$$

$$\Rightarrow \frac{1}{\lambda_3} = RZ^2 \left(\frac{21}{100} \right) \dots\dots(iii)$$

Now, dividing Eq. (iii) by Eq. (ii), we get

$$\frac{\frac{1}{\lambda_3}}{\frac{1}{\lambda_1}} = \frac{RZ^2 \left(\frac{21}{100} \right)}{RZ^2 \left(\frac{5}{36} \right)} \Rightarrow \frac{\lambda_1}{\lambda_3} = \frac{21}{100} \times \frac{36}{5}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_3} = 1.512 = 15.12 \times 10^{-1}$$

Comparing with the given value in the question i.e., $x \times 10^{-1}$, the value of $x = 15$.

Question154

The decay of a proton to neutron is

[18 Mar 2021 Shift 2]

Options:

- A. not possible as proton mass is less than the neutron mass
- B. possible only inside the nucleus
- C. not possible but neutron to proton conversion is possible
- D. always possible as it is associated only with β^+ decay

Answer: B

Solution:

The decay of a proton to neutron is possible only inside the nucleus because the mass of proton is less than the mass of neutron. This mass difference is compensated with the binding energy of nucleus. This decay is not possible outside the nucleus because particle cannot be decayed into the greater mass.

Question155

A radioactive sample disintegrates via two independent decay processes having half-lives $T_{1/2}^{(1)}$ and $T_{1/2}^{(2)}$, respectively. The effective half-life $T_{1/2}$ of the nuclei is

[18 Mar 2021 Shift 1]

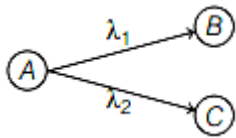
Options:

- A. $T_{V2} = \frac{T_{1/2}^{(1)} + T_{1/2}^{(2)}}{T_{1/2}^{(1)} - T_{1/2}^{(2)}}$
- B. $T_{1/2} = T_{1/2}^{(0)} + T_{1/2}^{(2)}$
- C. $T_{1/2} = \frac{T_{V2}^{(1)} T_{V2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$
- D. None of these

Answer: C

Solution:

The radioactive sample disintegrates via two independent decays as shown in the figure,



Here, λ is the net disintegrate constant.

λ_1 is the first decay constant,

and λ_2 is the second decay constant.

From the above figure,

$$\lambda = \lambda_1 + \lambda_2 \dots (i)$$

As, we know that,

$$\text{Half-life, } T_{1/2} = \frac{\ln(2)}{\lambda} \Rightarrow \lambda = \frac{\ln(2)}{T_{1/2}}$$

Substituting in Eq. (i), we get

$$\frac{\ln(2)}{T_{1/2}} = \frac{\ln(2)}{T_{1/2}^{(1)}} + \frac{\ln(2)}{T_{1/2}^{(2)}}$$

$$\Rightarrow T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$$

Question156

Amplitude of a mass-spring system, which is executing simple harmonic motion decreases with time. If mass = 500g, decay constant = 20g/s, then how much time is required for the amplitude of the system to drop to half of its initial value? ($\ln 2 = 0.693$)
[16 Mar 2021 Shift 2]

Options:

A. 34.65s

B. 17.32s

C. 0.034s

D. 15.01s

Answer: A

Solution:

Given,

Mass, $m = 500\text{g}$

Decay constant, $\lambda = 20\text{g/s}$

For damped oscillation,

$$A = A_0 e^{-bvt/2m} \dots (i)$$

Here, b = damped constant = decay constant

As per question, the amplitude of the system is dropped to half of its initial value.

So, Eq. (i) becomes

$$\frac{A_0}{2} = A_0 e^{-bt/2m} \Rightarrow \frac{1}{2} = e^{-bt/2m}$$

$$\Rightarrow 2 = e^{bt/2m}$$

$$\Rightarrow 2 = e^{bt/2m}$$

Taking log on both sides of above equation, we get

$$\log 2 = \log(e^{bt/2m})$$

$$\Rightarrow \log 2 = \frac{bt}{2m} \Rightarrow t = \frac{2m}{b} \log 2$$

$$= \frac{2 \times 500 \times 0.693}{20}$$

$$[\because \log 2 = 0.693]$$

$$\Rightarrow t = 34.65\text{s}$$

Question157

Calculate the time interval between 33% decay and 67% decay if half-life of a substance is 20min.

[16 Mar 2021 Shift 2]

Options:

A. 60min

B. 20min

C. 40min

D. 13min

Answer: B

Solution:

As we know that,

$$N = N_0 e^{-\lambda t}$$

where,

N = number of radioactive atoms left undecayed,

N_0 = original number of radioactive atoms,

t = elapsed time

and λ = decay constant.

$$\Rightarrow N_1 = N_0 e^{-\lambda \cdot t_1} \dots (i)$$

$$\text{and } N_2 = N_0 e^{-\lambda t_2} \dots (ii)$$

Consider Eq. (i), $N_1 = N_0 e^{-\lambda t_1}$

$$\Rightarrow \frac{N_1}{N_0} = e^{-\lambda t_1} \Rightarrow 0.67 = e^{-\lambda t_1}$$

$$\Rightarrow \log(0.67) = \log(e^{-\lambda t_1})$$

$$\Rightarrow \log 0.67 = -\lambda t_1 \dots (iii)$$

Consider Eq. (ii), $N_2 = N_0 e^{-\lambda t_2}$

$$\Rightarrow \frac{N_2}{N_0} = e^{-\lambda t_2} \Rightarrow 0.33 = e^{-\lambda t_2}$$

$$\Rightarrow \log(0.33) = \log(e^{-\lambda t_2})$$

$$\Rightarrow \log(0.33) = -\lambda t_2 \dots (iv)$$

Subtracting Eq. (iv) from Eq. (iii), we get

$$\log 0.67 - \log 0.33 = -\lambda t_1 + \lambda t_2$$

$$\Rightarrow \log\left(\frac{0.67}{0.33}\right) = \lambda(t_2 - t_1) \Rightarrow \lambda(t_2 - t_1) = \log 2$$

$$\Rightarrow t_2 - t_1 = \frac{\log 2}{\lambda} = t_{1/2}$$

$$\therefore \text{Half-life} = t_{1/2} = 20 \text{ min}$$

Question 158

The half-life of Au^{198} is 2.7 days. The activity of 1.50 mg of Au^{198} if its atomic weight is 198 g mol^{-1} is ($N_A = 6 \times 10^{23} / \text{mol}$)

[16 Mar 2021 Shift 2]

Options:

A. 240 Ci

B. 357 Ci

C. 535 Ci

D. 252 Ci

Answer: B

Solution:

Given,

Half-life of Au^{198} , $T_{1/2} = 2.7$ days

Amount of Au^{198} , $m = 1.5\text{mg} = 1.5 \times 10^{-3}\text{g}$

Atomic weight of Au^{198} , $A = 198\text{gmol}^{-1}$

We know that activity, $A_0 = \lambda N_0$

where, $\lambda = \text{decay constant} = \frac{\ln 2}{T_{1/2}}$

$N_0 = \text{initial number of atoms} = \frac{m}{A} N_A$

$$\Rightarrow A_0 = \lambda \cdot \frac{m}{A} N_A = \frac{\ln 2}{T_{1/2}} \times \frac{1.5 \times 10^{-3}}{198} \times 6.023 \times 10^{23}$$

$$\Rightarrow A_0 = \frac{\ln 2 \times 1.5 \times 10^{-3} \times 6 \times 10^{23}}{2.7 \times 3600 \times 24 \times 198 \times 3.7 \times 10^{10}} \text{Ci}$$

[$\because 1 \text{ curie (Ci)} = 3.7 \times 10^{10} \text{Bq}$]

$$\Rightarrow A_0 = 3.65 \times 10^2 \text{Ci} = 365 \text{Ci}$$

The answer is close to 357Ci.

Question159

The radiation corresponding to $3 \rightarrow 2$ transition of a hydrogen atom falls on a gold surface to generate photoelectrons. These electrons are passed through a magnetic field of $5 \times 10^{-4} \text{T}$. Assume that the radius of the largest circular path followed by these electrons is 7mm, the work function of the metal is:

(Mass of electron = $9.1 \times 10^{-31} \text{kg}$)

[20 Jul 2021 Shift 1]

Options:

A. 1.36 eV

B. 1.88 eV

C. 0.16 eV

D. 0.82 eV

Answer: D

Solution:

$$1.51 \text{ eV} \rightarrow n = 3$$

$$3.4 \text{ eV} \rightarrow n = 3$$

$$13.6 \text{ eV} \rightarrow n = 1$$

$$3 \rightarrow 2 \Rightarrow 1.89 \text{ eV}$$

$$5 \times 10^{-4} \text{ T} \quad r = 7 \text{ mm}$$

$$r = \frac{mv}{qB} \Rightarrow mv = qrB$$

$$\Rightarrow E = \frac{p^2}{2m} = \frac{(qRB)^2}{2m}$$

$$= \frac{(1.6 \times 10^{-19} \times 7 \times 10^{-3} \times 5 \times 10^{-4})^2}{2 \times 9.1 \times 10^{-31} \text{ Joule}}$$

$$= \frac{3136 \times 10^{-52}}{18.2 \times 10^{-31} \times 1.6 \times 10^{-19}} \times \text{eV}$$

$$= 1.077 \text{ eV}$$

We know work function = energy incident $(K E)_{\text{electron}}$

$$\phi = 1.89 - 1.077 = 0.813 \text{ eV}$$

Question 160

The K_{α} X-ray of molybdenum has wavelength 0.071 nm . If the energy of a molybdenum atom with a K electron knocked out is 27.5 keV , the energy of this atom when an L electron is knocked out will be _____ keV. (Round off to the nearest integer)

$$[h = 4.14 \times 10^{-15} \text{ eV s}, c = 3 \times 10^8 \text{ ms}^{-1}]$$

[27 Jul 2021 Shift 2]

Answer: 10

Solution:

$$E_{K_{\alpha}} = E_K - E_L$$

$$\frac{hc}{\lambda_{K_{\alpha}}} = E_K - E_L$$

$$E_L = E_K - \frac{hc}{\lambda_{K_{\alpha}}}$$

$$= 27.5 \text{ keV} - \frac{12.42 \times 10^{-7} \text{ eV m}}{0.071 \times 10^{-9} \text{ m}}$$

$$E_L = (27.5 - 17.5)\text{keV} \\ = 10\text{keV}$$

Question161

An electron and proton are separated by a large distance. The electron starts approaching the proton with energy 3 eV. The proton captures the electrons and forms a hydrogen atom in second excited state. The resulting photon is incident on a photosensitive metal of threshold wavelength 4000 Å. What is the maximum kinetic energy of the emitted photoelectron?

[27 Jul 2021 Shift 2]

Options:

- A. 7.61 eV
- B. 1.41 eV
- C. 3.3 eV
- D. No photoelectron would be emitted

Answer: B

Solution:

Initially, energy of electron = +3eV

finally, in 2nd excited state,

$$\text{energy of electron} = -\frac{(13.6\text{eV})}{3^2}$$

$$= -1.51\text{eV}$$

Loss in energy is emitted as photon,

$$\text{So, photon energy } \frac{hc}{\lambda} = 4.51\text{eV}$$

Now, photoelectric effect equation

$$K E_{\max} = \frac{hc}{\lambda} - \phi = 4.51 - \left(\frac{hc}{\lambda_{\text{th}}} \right)$$

$$= 4.51\text{eV} - \frac{12400\text{eV Å}}{4000\text{Å}}$$

$$= 1.41\text{eV}$$

Question162

Consider the following statements :

- A. Atoms of each element emit characteristics spectrum.
- B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.
- C. The density of nuclear matter depends on the size of the nucleus.
- D. A free neutron is stable but a free proton decay is possible.
- E. Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below :

[27 Jul 2021 Shift 2]

Options:

- A. A, B, C, D and E
- B. A, B and E only
- C. B and D only
- D. A, C and E only

Answer: B

Solution:

- (A) True, atom of each element emits characteristic spectrum.
 - (B) True, according to Bohr's postulates $mvr = \frac{nh}{2\pi}$ and hence electron resides into orbits of specific radius called stationary orbits.
 - (C) False, density of nucleus is constant
 - (D) False, A free neutron is unstable decays into proton and electron and antineutrino.
 - (E) True unstable nucleus show radioactivity.
-

Question163

A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance 200 μF is first charged and later connected with resistor 'R'. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R'

should be _____ Ω .
[27 Jul 2021 Shift 1]

Answer: 150

Solution:

$$T_m = 30\text{ms}$$

$$C = 200\mu\text{F}$$

$$\frac{q}{N} = \frac{Q_0 e^{-t/RC}}{N_0 e^{-\lambda t}} = \frac{Q_0}{N_0} e^{t\left(\lambda - \frac{1}{RC}\right)}$$

Since q/N is constant hence

$$\lambda = \frac{1}{RC}$$

$$R = \frac{1}{\lambda C} = \frac{T_m}{C} = \frac{30 \times 10^{-3}}{200 \times 10^{-6}} = 150\Omega$$

Question164

If 'f' denotes the ratio of the number of nuclei decayed (N_d) to the number of nuclei at $t = 0$ (N_0) then for a collection of radioactive nuclei, the rate of change of f' with respect to time is given as :
[λ is the radioactive decay constant]

[27 Jul 2021 Shift 1]

Options:

A. $-\lambda(1 - e^{-\lambda t})$

B. $\lambda(1 - e^{-\lambda t})$

C. $\lambda e^{-\lambda t}$

D. $-\lambda e^{-\lambda t}$

Answer: C

Solution:

$$N = N_0 e^{-\lambda t}$$

$$N_d = N_0 - N$$

$$N_d = N_0(1 - e^{-\lambda t})$$

$$\frac{N_d}{N_0} = f = 1 - e^{-\lambda t}$$

$$\frac{df}{dt} = \lambda e^{-\lambda t}$$

Question 165

The nuclear activity of a radioactive element becomes $\left(\frac{1}{8}\right)^{\text{th}}$ of its initial value in 30 years. The half-life of radioactive element is _____ years.

[25 Jul 2021 Shift 2]

Answer: 10

Solution:

$$A = A_0 e^{-\lambda t}$$

$$\frac{A_0}{8} = A_0 e^{-\lambda t} \Rightarrow \lambda t = \ln 8$$

$$\lambda t = 3 \ln 2$$

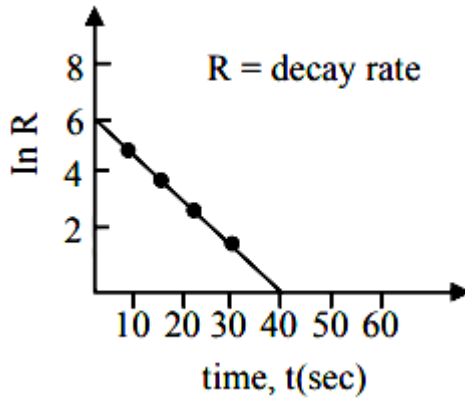
$$\frac{\ln 2}{\lambda} = \frac{t}{3} = \frac{30}{3} = 10 \text{ years}$$

Question 166

For a certain radioactive process the graph between $\ln R$ and $t(\text{sec})$ is obtained as shown in the figure.

Then the value of half life for the unknown radioactive material is

approximately :



[20 Jul 2021 Shift 2]

Options:

A. 9.15 sec

B. 6.93 sec

C. 2.62 sec

D. 4.62 sec

Answer: D

Solution:

$$R = R_0 e^{-\lambda t}$$

$$\ln R = \ln R_0 - \lambda t$$

$-\lambda$ is slope of straight line

$$\lambda = \frac{3}{20}$$

$$t_{1/2} = \frac{\ln 2}{\lambda} = 4.62$$

Question167

A radioactive substance decays to $\left(\frac{1}{16}\right)^{\text{th}}$ of its initial activity in 80 days. The half life of the radioactive substance expressed in days is _____.

[20 Jul 2021 Shift 2]

Answer: 20

Solution:

$$N_0 \xrightarrow{t_{1/2}} \frac{N_0}{2} \xrightarrow{t_{1/2}} \frac{N_0}{4} \xrightarrow{t_{1/2}} \frac{N_0}{8} \xrightarrow{t_{1/2}} \frac{N_0}{16}$$

$$4 \times t_{1/2} = 80$$

$$t_{1/2} = 20 \text{ days}$$

Question 168

A radioactive material decays by simultaneous emissions of two particles with half lives of 1400 years and 700 years respectively. What will be the time after the which one third of the material remains ? (Take $\ln 3 = 1.1$)

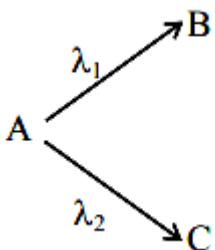
[20 Jul 2021 Shift 1]

Options:

- A. 1110 years
- B. 700 years
- C. 340 years
- D. 740 years

Answer: D

Solution:



Given $\lambda_1 = \frac{\ln 2}{700} / \text{year}, \lambda_2 = \frac{\ln 2}{1400} / \text{year}$

$$\therefore \lambda_{\text{net}} = \lambda_1 + \lambda_2 = \ln 2 \left[\frac{1}{700} + \frac{1}{1400} \right]$$

$$= \frac{3 \ln 2}{1400} / \text{year}$$

Now, Let initial no. of radioactive nuclei be N_0 .

$$\therefore \frac{N}{N_0} = e^{-\lambda_{\text{net}} t}$$

$$\Rightarrow \ln \frac{1}{3} = -\lambda_{\text{net}} t$$

$$\Rightarrow 1.1 = \frac{3 \times 0.693}{1400} t \Rightarrow t \approx 740 \text{ years}$$

Hence option 4

Question 169

From the given data, the amount of energy required to break the nucleus of aluminium ${}_{13}^{27}\text{Al}$ is _____ $\times 10^{-3} \text{ J}$

Mass of neutron = 1.00866u

Mass of proton = 1.00726u

Mass of Aluminium nucleus = 27.18846u

(Assume 1 u corresponds to x J of energy)

(Round off to the nearest integer)

[25 Jul 2021 Shift 2]

Answer: 27

Solution:

$$\Delta m = (Z m_p + (A - Z) m_n) - M_{\text{Al}}$$

$$= (13 \times 1.00726 + 14 \times 1.00866) - 27.18846$$

$$= 27.21562 - 27.18846$$

$$= 0.02716 \text{ u}$$

$$E = 27.16 \times 10^{-3} \text{ J}$$

Question 170

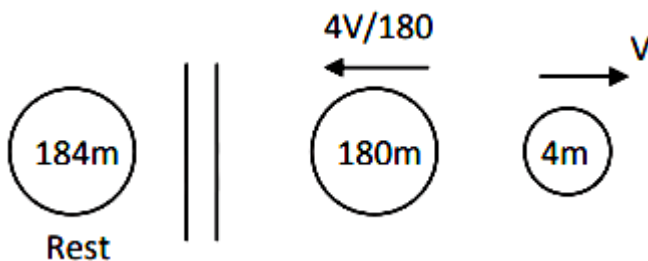
A nucleus with mass number 184 initially at rest emits an α -particle. If the Q value of the reaction is 5.5 MeV, calculate the kinetic energy of the α -particle.
[22 Jul 2021 Shift 2]

Options:

- A. 5.0 MeV
- B. 5.5 MeV
- C. 0.12 MeV
- D. 5.38 MeV

Answer: D

Solution:



$$\frac{1}{2}(4m)v^2 + \frac{1}{2}(180m)\left(\frac{4v}{180}\right)^2 = 5.5 \text{ MeV}$$

$$\Rightarrow \frac{1}{2}4mv^2 \left[1 + 45 \left(\frac{4}{180} \right)^2 \right] = 5.5 \text{ MeV}$$

$$\Rightarrow K.E._\alpha = \frac{5.5}{1 + 45 \cdot \left(\frac{4}{180} \right)^2} \text{ MeV}$$

$$K.E._\alpha = 5.38 \text{ MeV}$$

Question 171

A nucleus of mass M emits γ -ray photon of frequency ' ν '. The loss of internal energy by the nucleus is :

[Take 'c' as the speed of electromagnetic wave]

[20 Jul 2021 Shift 1]

Options:

A. $h\nu$

B. 0

C. $h\nu \left[1 - \frac{h\nu}{2Mc^2} \right]$

D. $h\nu \left[1 + \frac{h\nu}{2Mc^2} \right]$

Answer: D

Solution:

Energy of γ ray $[E_\gamma] = h\nu$

Momentum of γ ray

$$[P_\gamma] = \frac{h}{\lambda} = \frac{h\nu}{C}$$

Total momentum is conserved.

$$\vec{P}_\gamma + \vec{P}_{Nu} = 0$$

Where $\vec{P}_{Nu}$ = Momentum of decayed nuclei

$$\Rightarrow P_\gamma = P_{Nu}$$

$$\Rightarrow \frac{h\nu}{C} = P_{Nu}$$

$\Rightarrow$ K.E. of nuclei

$$= \frac{1}{2} M v^2 = \frac{(P_{Nu})^2}{2M} = \frac{1}{2M} \left[\frac{h\nu}{C} \right]^2$$

Loss in internal energy $= E_\gamma + K.E._{Nu}$

$$= h\nu + \frac{1}{2M} \left[\frac{h\nu}{C} \right]^2$$

$$= h\nu \left[1 + \frac{h\nu}{2MC^2} \right]$$

Question172

x different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are excited to states with principal quantum number $n = 6$? The value of x is

[27 Aug 2021 Shift 2]

Answer: 15

Solution:

Given, principal quantum number, $n = 6$ Number of wavelength, $N = \frac{n(n-1)}{2}$
 $= \frac{6(6-1)}{2} = 3(5) = 15$
 $x = 15$
 $\therefore x = 15$

Question 173

A free electron of 2.6 eV energy collides with a H^+ ion. This results in the formation of a hydrogen atom in the first excited state and a photon is released. Find the frequency of the emitted photon.

($h = 6.6 \times 10^{-34} \text{ Js}$)

[31 Aug 2021 Shift 2]

Options:

A. $1.45 \times 10^{16} \text{ MHz}$

B. $0.19 \times 10^{15} \text{ MHz}$

C. 1.4510^9 MHz

D. $9.0 \times 10^{27} \text{ MHz}$

Answer: C

Solution:

Given that, energy of free electron,

$$E_1 = 2.6 \text{ eV}$$

We know that, the energy of H-atom in its first excited state ($n = 2$),

$$E_2 = \frac{-13.6}{2^2} = -\frac{13.6}{4} \text{ eV}$$

Now, the energy of emitted photons will be the difference of these two energies E_1 and E_2 .

$$\Delta E = E_1 - E_2$$

$$\therefore h\nu = \left[2.6 - \left(-\frac{13.6}{4} \right) \right] \text{ eV}$$

$$\Rightarrow \nu = \frac{(10.4 + 13.6) \times 1.6 \times 10^{-19}}{4 \times 6.6 \times 10^{-34}} = 1.45 \times 10^9 \text{ MHz}$$

Question 174

A particular hydrogen like ion emits radiation of frequency $2.92 \times 10^{15} \text{ Hz}$ when it makes transition from $n = 3$ to $n = 1$. The frequency in Hz of radiation emitted in transition from $n = 2$ to $n = 1$ will be

[26 Aug 2021 Shift 1]

Options:

A. 0.44×10^{15}

B. 6.57×10^{15}

C. 4.38×10^{15}

D. 246×10^{15}

Answer: D

Solution:

The frequency of the emitted radiation by a particular hydrogen like ion when it makes transition from $n_1 = n_{\text{initial}}$ to $n_2 = n_{\text{final}}$ is given by

$$f = RZ^2C \left(\frac{1}{n_2} - \frac{1}{n_1} \right)$$

where, c = velocity of light
and R = Rydberg's constant.

For H-atom, $Z = 1$

So, for transition from $n = 3$ to $n = 1$

$$f_1 = R_C \left(1 - \frac{1}{3^2} \right) \dots (i)$$

Similarly for transition from $n = 2$ to $n = 1$,

$$f_2 = R_C \left(1 - \frac{1}{2^2} \right) \dots (ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{f_1}{f_2} = \frac{Rc\left(1 - \frac{1}{9}\right)}{Rc\left(1 - \frac{1}{4}\right)} = \frac{\frac{8}{9}}{\frac{3}{4}} = \frac{32}{27}$$

Here, $f_1 = 2.92 \times 10^{15} \text{ Hz}$

$$\Rightarrow \frac{2.92 \times 10^{15}}{f_2} = \frac{32}{27}$$

$$\Rightarrow f_2 = \frac{2.92 \times 10^{15} \times 27}{32}$$

$$= 2.46 \times 10^{15} \text{ Hz}$$

Question 175

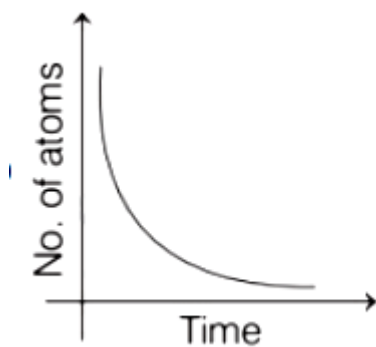
A sample of a radioactive nucleus A disintegrates to another radioactive nucleus B, which in turn disintegrates to some other stable nucleus C. Plot of a graph showing the variation of number of atoms of nucleus B versus time is

(Assume that at $t = 0$, there are no B atoms in the sample)

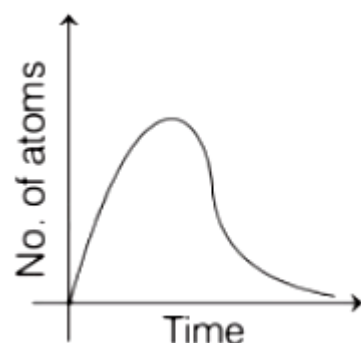
[31 Aug 2021 Shift 1]

Options:

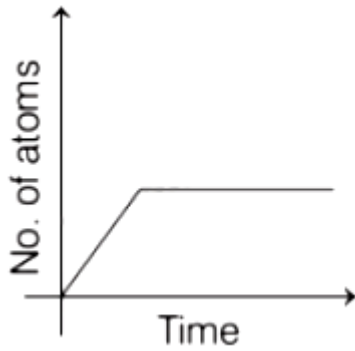
A.



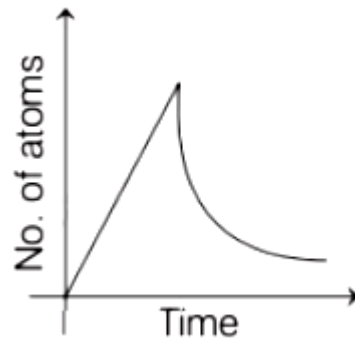
B.



C.



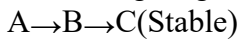
D.



Answer: B

Solution:

According to given question, the decay is as shown below



Let N_0 be the original number of atoms in the radioactive nucleus A.

Initially at $t = 0$,

Number of atoms of B in the sample is zero.

According to radioactive decay law,

$$\text{i.e. } N = N_0 e^{-\lambda t} \dots (i)$$

With the increase in time t , the number of atoms of B will start increasing and reaches a maximum value.

After reaching the maximum value, B will start decaying into C.

Since, number of atoms varies with time following an exponential law i.e. Eq. (i).

So, the graph between number of atoms and t will be exponential in nature. Thus, the correct option is (b).

Question176

There are 10^{10} radioactive nuclei in a given radioactive element. Its half-life time is 1 min. How many nuclei will remain after 30 s?
 $(\sqrt{2} = 1.414)$
[27 Aug 2021 Shift 1]

Options:

A. 2×10^{10}

B. 7×10^9

C. 10^5

D. 4×10^{10}

Answer: B

Solution:

Initial number of radioactive nuclei, $N_0 = 10^{10}$

Half-life, $t_{\frac{1}{2}} = 1 \text{ min} = 60\text{s}$

Time, $t = 30\text{s}$

We know that, number of nuclei remaining in radioactive decay after time t is given by

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

Substituting the values in above relation, we get

$$\frac{N}{10^{10}} = \left(\frac{1}{2}\right)^{\frac{30}{60}} = \left(\frac{1}{2}\right)^{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow N = \frac{1}{\sqrt{2}} \times 10^{10} = 0.707 \times 10^{10}$$

$$\Rightarrow N \approx 7 \times 10^9$$

Thus, the radioactive nuclei remaining after 30s is 7×10^9 .

Question177

At time $t = 0$, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is λ . However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the

evolution of $\frac{N_B(t)}{N_B(0)}$ with respect to time t ?

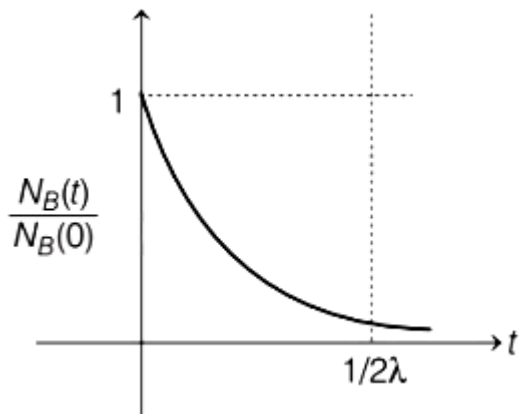
$[N_A(0) = \text{Number of A atoms at } t = 0]$

$[N_B(0) = \text{Number of B atoms at } t = 0]$

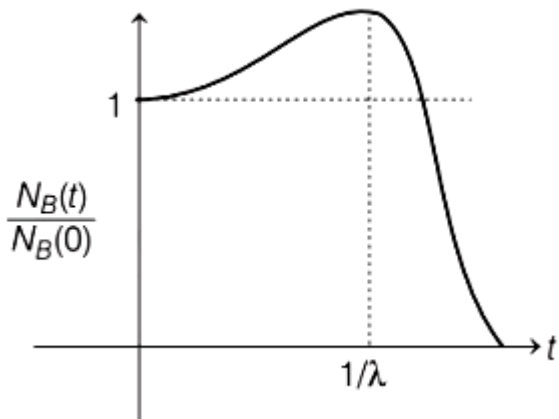
[26 Aug 2021 Shift 2]

Options:

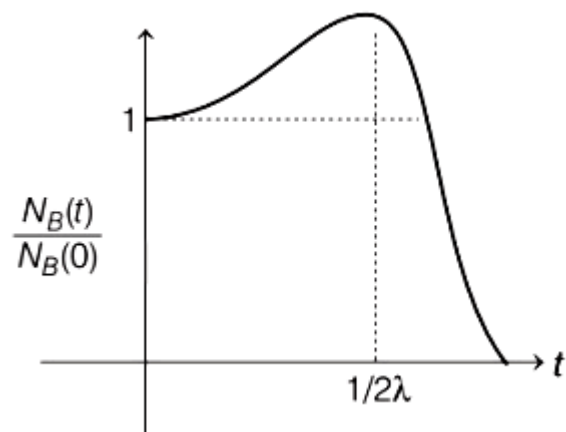
A.



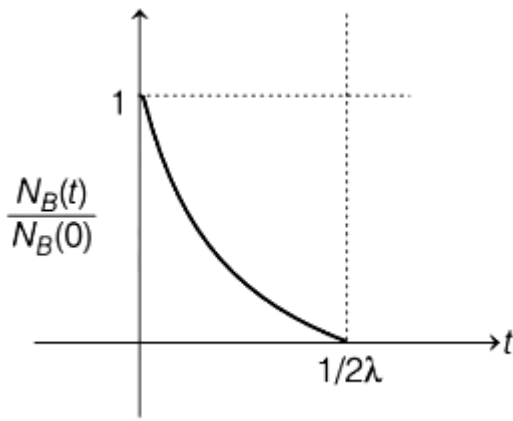
B.



C.



D.



Answer: C

Solution:

Given, at time $t = 0$, $N_A(0) = 2N_B(0)$

Decay constant is same for both radioactive atoms as λ .

For $A \rightarrow B$,

$$\frac{dN_B(t)}{dt} = \lambda N_A(t) - \lambda N_B(t)$$

Substituting $N_A(0)e^{-\lambda t}$ for $N_A(t)$ in above expression, we get

$$\frac{dN_B(t)}{dt} = \lambda N_A(0)e^{-\lambda t} - \lambda N_B(t)$$

$$= 2\lambda N_B(0)e^{-\lambda t} - \lambda N_B(t)$$

$$\Rightarrow \frac{dN_B(t)}{dt} + \lambda N_B(t) = 2\lambda N_B(0)e^{-\lambda t}$$

Multiplying both sides by $e^{\lambda t}$, we get

$$e^{\lambda t} \left[\frac{dN_B(t)}{dt} + \lambda N_B(t) \right] = 2\lambda N_B(0)e^{-\lambda t} \times e^{\lambda t}$$

$$\Rightarrow \frac{d}{dt}[N_B(t)e^{\lambda t}] = 2\lambda N_B(0)$$

Integrating both sides,

$$[N_B(t)e^{\lambda t}] = 2\lambda N_B(0)t + C \dots (i)$$

Putting $t = 0$ in above expression,

$$[N_B(0)e^{\lambda \times 0}] = 2\lambda N_B(0) \times 0 + C$$

$$\Rightarrow C = N_B(0)$$

Putting value of C in Eq. (i)

$$[N_B(t)e^{\lambda t}] = 2\lambda N_B(0)t + N_B(0) \dots (ii)$$

$$\Rightarrow N_B(t) = N_B(0)[1 + 2\lambda t]e^{-\lambda t}$$

$$\Rightarrow \frac{N_B(t)}{N_B(0)} = [1 + 2\lambda t]e^{-\lambda t}$$

$$\text{As, } N_B(t) = N_B(0)[1 + 2\lambda t]e^{-\lambda t}$$

$$\Rightarrow N_B(t) = C[1 + 2\lambda t]e^{-\lambda t} \dots (iii)$$

The maximum value of $N_B(t)$ is obtained at

$$\frac{dN_B(t)}{dt} = 0$$

From Eq. (ii),

$$\frac{dN_B(t)}{dt} = -\lambda C[1 + 2\lambda t]e^{-\lambda t} + 2C\lambda e^{-\lambda t} = 0$$

Solving the above expression,

$$t = \frac{1}{2\lambda} s$$

Thus, maximum value of function will be at $t = \frac{1}{2\lambda} s$.

Hence, graph (c) is correct option.

Question178

The graph which depicts the results of Rutherford gold foil experiment with α -particles is:

θ : Scattering angle

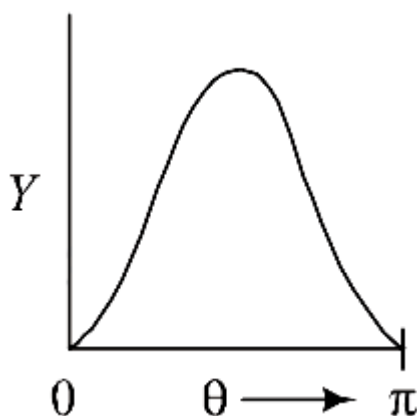
α : Number of scattered α -particles detected

(Plots are schematic and not to scale)

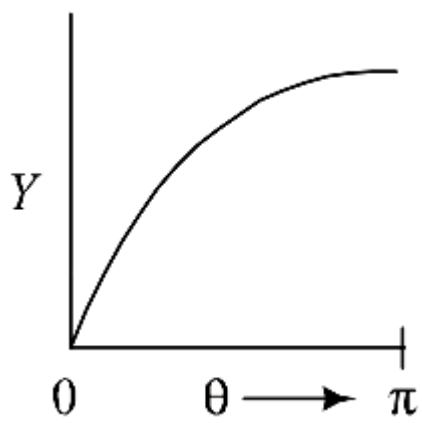
[8 Jan. 2020 I]

Options:

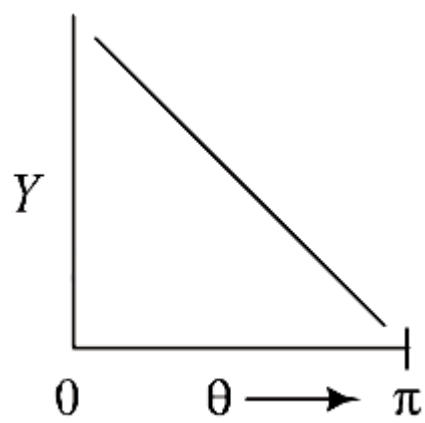
A.



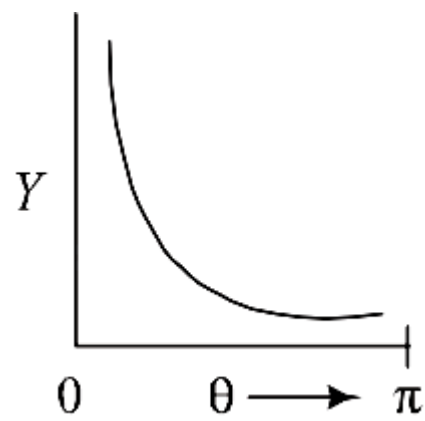
B.



C.



D.



Answer: C

The energy required to ionise a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state?

[9 Jan. 2020 II]

Options:

A. 24.2 nm

B. 11.4 nm

C. 35.8 nm

D. 8.6 nm

Answer: B

Solution:

According to Bohr's Theory the wavelength of the radiation emitted from hydrogen atom is given by

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\therefore Z = 3$$

$$\therefore \frac{1}{\lambda} = 9R \left(1 - \frac{1}{9} \right)$$

$$\Rightarrow \lambda = \frac{1}{8R} = \frac{1}{8 \times 10973731.6} \text{ m} \quad (R = 10973731.6 \text{ m}^{-1})$$

$$\Rightarrow \lambda = 11.39 \text{ nm}$$

Question180

The first member of the Balmer series of hydrogen atom has a wavelength of 6561 Å. The wavelength of the second member of the Balmer series (in nm) is _____.

[NA 8 Jan. 2020 II]

Answer: 486

Solution:

The wavelength of the spectral line of hydrogen spectrum is given by formula

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Where, R = Rydberg constant

For the first member of Balmer series $n_f = 2$, $n_i = 3$

$$\therefore \frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \dots (i)$$

For last member of Balmer series, $n_f = 2$, $n_i = 4$

$$\text{So, } \frac{1}{\lambda'} = R \left[\frac{1}{4} - \frac{1}{16} \right] \dots (ii)$$

Dividing (i) by (ii), we get

$$\Rightarrow \frac{\lambda'}{\lambda} = \frac{5 \times 16}{9 \times 4 \times 3}$$

$$\Rightarrow \lambda' = \frac{5 \times 4 \times 656.1}{9 \times 3} (\text{nm}) = 486 \text{ nm}$$

Question181

The time period of revolution of electron in its ground state orbit in a hydrogen atom is $1.6 \times 10^{-16} \text{ s}$. The frequency of revolution of the electron in its first excited state (in s^{-1}) is:
[7 Jan. 2020 I]

Options:

A. 1.6×10^{14}

B. 7.8×10^{14}

C. 6.2×10^{15}

D. 5.6×10^{12}

Answer: B

Solution:

For first excited state $n' = 3$

$$\text{Time period } T \propto \frac{n^3}{Z^2}$$

$$\Rightarrow \frac{T_2}{T_1} = \frac{n^3}{n'^3}$$

$$\therefore T_2 = 8T_1 = 8 \times 1.6 \times 10^{-16} \text{ s}$$

$$\therefore \text{Frequency, } \nu = \frac{1}{T_2} = \frac{1}{8 \times 1.6 \times 10^{-16}}$$

$$\approx 7.8 \times 10^{14} \text{ Hz}$$

Question 182

The activity of a radioactive sample falls from 700 s^{-1} to 500 s^{-1} in 30 minutes. Its half life is close to:
[7 Jan. 2020, II]

Options:

A. 72 min

B. 62 min

C. 66 min

D. 52 min

Answer: B

Solution:

We know that

$$\text{Activity, } A = A_0 e^{-\lambda t}$$

$$A = A_0 e^{-t \ln 2 / T_{1/2}} \left(\because \lambda = \frac{\ln 2}{T_{1/2}} \right)$$

$$\Rightarrow 500 = 700 e^{-t \ln 2 / T_{1/2}}$$

$$\Rightarrow \ln \frac{5}{7} = \frac{30 \ln 2}{T_{1/2}} \quad (\because t = 30 \text{ minute})$$

$$\Rightarrow T_{1/2} = 30 \frac{\ln 2}{\ln 1.4} = 61.8 \text{ minute}$$

$$(\because \ln 2 = 0.693 \text{ and } \ln 1.4 = 0.336)$$

$$\Rightarrow T_{1/2} \approx 62 \text{ minute}$$

Question 183

A particle of mass $200M \text{ eV} / c^2$ collides with a hydrogen atom at rest. Soon after the collision the particle comes to rest, and the atom recoils and goes to its first excited state. The initial kinetic energy of the particle (in eV) is $\frac{N}{4}$. The value of N is :

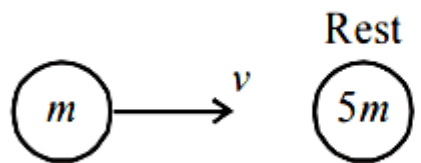
(Given the mass of the hydrogen atom to be $1 \text{ GeV}/c^2$)

[NA Sep. 05, 2020 (I)]

Answer: 51

Solution:

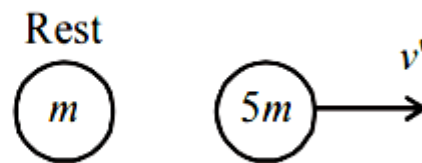
Before collision



Particle

Hydrogen

After collision



Particle

Hydrogen

From linear momentum conservation, $L_i = L_f$

$$mV + 0 = 0 + 5mV' \Rightarrow V' = \frac{V}{5}$$

$$\text{Loss of K E} = K E_i - K E_f = \frac{1}{2}mv^2 - \frac{1}{2}(5m)\left(\frac{v}{5}\right)^2$$

$$= \frac{1}{2}mv^2 \left(1 - \frac{1}{5}\right) = \frac{4}{5} \left(\frac{mv^2}{2}\right)$$

$$= \frac{4}{5} K E_i = 10.2 \text{ eV}$$

[$\because$ Energy in first excited state of atom = 10.2 eV]

$$K E_i = 12.75 \text{ eV} = \frac{N}{4} \Rightarrow N = 51$$

The value of N = 51

Question 184

In the line spectra of hydrogen atom, difference between the largest and the shortest wavelengths of the Lyman series is 304 Å. The corresponding difference for the Paschen series in Å is : _____.
[NA Sep. 04, 2020 (I)]

Answer: 10553.14

Solution:

From Bohr's formula for hydrogen atom,

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$R = 1.097 \times 10^7 \text{ m}^{-1}$$

$$\frac{1}{\lambda_{\min}} = R(1) = R \because n_2 = \infty \text{ and } n_1 = 1$$

$$1/\lambda_{\max} = R \left\{ 1 - \frac{1}{4} \right\} = \frac{3R}{4} \because n_2 = 2, n_1 = 1$$

$$\therefore \lambda_{\max} - \lambda_{\min} = \frac{4}{3R} - \frac{1}{R} = \frac{1}{3R} = 304 \text{ (Given)}$$

For Paschen series :

$$\lambda'_{\min} = R \left(\frac{1}{9} \right) \text{ and } \lambda'_{\max} = R \left(\frac{1}{9} - \frac{1}{16} \right) = \frac{7R}{16 \times 9}$$

$$\lambda'_{\max} - \lambda'_{\min} = \frac{16 \times 9}{7R} - \frac{9}{R} = \frac{81}{7R}$$

$$\text{or, } \lambda'_{\max} - \lambda'_{\min} = \frac{81}{7R} = \frac{81 \times 3}{7 \times 3R} = \frac{81 \times 3}{7} \times 304 \left(\because \frac{1}{3R} = 304 \text{ Å} \right)$$

$$\therefore \text{For Paschen series, } \lambda'_{\max} - \lambda'_{\min} = 10553.14$$

Question 185

In a hydrogen atom the electron makes a transition from $(n+1)^{\text{th}}$ level to the n^{th} level. If $n \gg 1$, the frequency of radiation emitted is proportional to :
[Sep. 02, 2020 (II)]

Options:

A. $\frac{1}{n}$

B. $\frac{1}{n^3}$

C. $\frac{1}{n^2}$

D. $\frac{1}{n^4}$

Answer: B

Solution:

Total energy of electron in n^{th} orbit of hydrogen atom

$$E_n = -\frac{Rhc}{n^2}$$

Total energy of electron in $(n+1)^{\text{th}}$ level of hydrogen atom

$$E_{n+1} = -\frac{Rhc}{(n+1)^2}$$

When electron makes a transition from $(n+1)^{\text{th}}$ level to n^{th} level

Change in energy,

$$\Delta E = E_{n+1} - E_n$$

$$h\nu = Rhc \cdot \left[\frac{1}{n^2} - \frac{1}{(n+1)^2} \right] (\because E = h\nu)$$

$$\nu = R \cdot c \left[\frac{(n+1)^2 - n^2}{n^2(n+1)^2} \right]$$

$$\nu = R \cdot c \left[\frac{1 + 2n}{n^2(n+1)^2} \right]$$

For $n \gg 1$

$$\Rightarrow \nu = R \cdot c \left[\frac{2n}{n^2 \times n^2} \right] = \frac{2RC}{n^3}$$

$$\Rightarrow \nu \propto \frac{1}{n^3}$$

Question 186

The radius R of a nucleus of mass number A can be estimated by the formula $R = (1.3 \times 10^{-15})A^{1/3}$ m. It follows that the mass density of a nucleus is of the order of :

$$(M_{\text{prot.}} \cong M_{\text{neut.}} \simeq 1.67 \times 10^{-27} \text{ kg})$$

[Sep. 03, 2020 (II)]

Options:

A. 10^3kgm^{-3}

B. 10^{10}kgm^{-3}

C. 10^{24}kgm^{-3}

D. 10^{17}kgm^{-3}

Answer: D

Solution:

$$\text{Density of nucleus, } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{mA}{\frac{4}{3}\pi R^3}$$

$$\Rightarrow \rho = \frac{mA}{\frac{4}{3}\pi (R_0 A^{1/3})^3} \quad (\because R = R_0 A^{1/3})$$

Here m = mass of a nucleon

$$\therefore \rho = \frac{3 \times 1.67 \times 10^{-27}}{4 \times 3.14 \times (1.3 \times 10^{-15})^3} \quad (\text{Given, } R_0 = 1.3 \times 10^{-15})$$

$$\Rightarrow \rho = 2.38 \times 10^{17} \text{kg/m}^3$$

Question187

You are given that mass of ${}_3^7\text{Li} = 7.0160\text{u}$

Mass of ${}_2^4\text{He} = 4.0026\text{u}$

and Mass of ${}_1^1\text{H} = 1.0079\text{u}$

When 20g of ${}_3^7\text{Li}$ is converted into ${}_2^4\text{He}$ by proton capture, the energy liberated, (in kW h), is :

[Mass of nucleon left. = $1\text{GeV} / c^2$]

[Sep. 06,2020 (I)]

Options:

A. 4.5×10^5

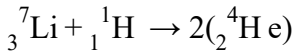
B. 8×10^6

C. 6.82×10^5

D. 1.33×10^6

Answer: D

Solution:



$$\Delta m \rightarrow [m_{\text{Li}} + m_{\text{H}}] - 2[M_{\text{He}}]$$

$$\text{Energy released} = \Delta mc^2$$

$$\text{In use of 1g Li energy released} = \frac{\Delta mc^2}{m_{\text{Li}}}$$

$$\begin{aligned} \text{In use of 20g energy released} &= \frac{\Delta mc^2}{m_{\text{Li}}} \times 20\text{g} \\ &= \frac{[(7.016 + 1.0079) - 2 \times 4.0026]\text{u} \times c^2}{7.016 \times 1.6 \times 10^{-24}} \times 20\text{g} \\ &= \left(\frac{0.0187 \times 1.6 \times 10^{-19} \times 10^9}{7.016 \times 1.6 \times 10^{-24}} \times 20 \right) = 480 \times 10^{10}\text{J} \end{aligned}$$

$$\because 1\text{J} = 2.778 \times 10^{-7}\text{kWh}$$

$$\begin{aligned} \therefore \text{Energy released} &= 480 \times 10^{10} \times 2.778 \times 10^{-7} \\ &= 1.33 \times 10^6\text{kWh} \end{aligned}$$

Question188

Given the masses of various atomic particles $m_p = 1.0072\text{u}$

$m_n = 1.0087\text{u}$, $m_e = 0.000548\text{u}$, $m_{\bar{\nu}} = 0$, $m_d = 2.0141\text{u}$, where p equiv proton, $n \equiv$ neutron, $e \equiv$ electron, $\bar{\nu} \equiv$ antineutrino and $d \equiv$ deuteron. Which of the following process is allowed by momentum and energy conservation?

[Sep. 06,2020 (II)]

Options:

A. $n + n \rightarrow$ deuterium atom (electron bound to the nucleus)

B. $p \rightarrow n + e^+ + \bar{\nu}$

C. $n + p \rightarrow d + \gamma$

D. $e^+ + e^- \rightarrow \gamma$

Answer: C

Solution:

For the momentum and energy conservation, mass defect (Δm) should be positive. Since some energy is lost in every process.

$$(m_p + m_n) > m_d$$

Question189

Find the Binding energy per nucleon for $_{50}^{120}\text{Sn}$. Mass of proton $m_p = 1.00783\text{U}$, mass of neutron $m_n = 1.00867\text{U}$ and mass of tin nucleus $m_{\text{Sn}} = 119.902199\text{U}$.

(take $1\text{ U} = 931\text{MeV}$)

[Sep. 04,2020 (II)]

Options:

A. 7.5 MeV

B. 9.0 MeV

C. 8.0 MeV

D. 8.5 MeV

Answer: D

Solution:

Mass defect,

$$\begin{aligned}\Delta m &= (50m_p + 70m_n) - (m_{\text{Sn}}) \\ &= (50 \times 1.00783 + 70 \times 1.008) - (119.902199) \\ &= 1.096\end{aligned}$$

$$\text{Binding energy} = (\Delta m)C^2 = (\Delta m) \times 931 = 1020.56$$

$$\frac{\text{Binding energy}}{\text{Nucleon}} = \frac{1020.5631}{120} = 8.5\text{MeV}$$

Question190

In a reactor, 2kg of ${}_{92}\text{U}^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200MeV. Given that the Avogadro number, $N = 6.023 \times 10^{26}$ per kilo mole and $1\text{eV} = 1.6 \times 10^{-19}\text{J}$. The power output of the reactor is close to:
[Sep. 02, 2020 (I)]

Options:

- A. 35 MW
- B. 60 MW
- C. 125 MW
- D. 54 MW

Answer: B

Solution:

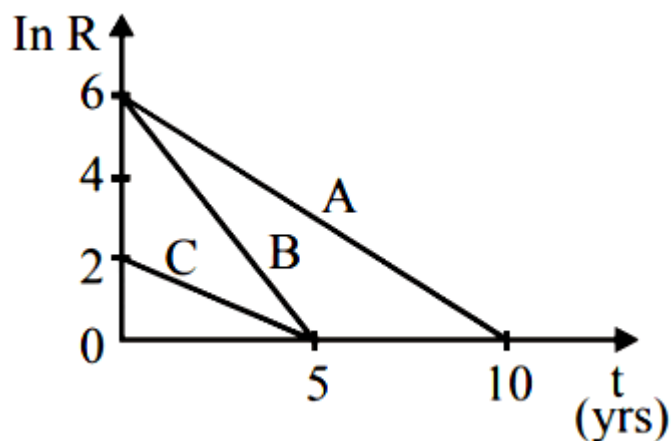
Power output of the reactor,

$$P = \frac{\text{energy}}{\text{time}}$$
$$= \frac{2}{235} \times \frac{6.023 \times 10^{26} \times 200 \times 1.6 \times 10^{-19}}{30 \times 24 \times 60 \times 60} \simeq 60\text{MW}$$

Question191

Activities of three radioactive substances A, B and C are represented by the curves A, B and C, in the figure. Then their half-lives

$T_{\frac{1}{2}}(A) : T_{\frac{1}{2}}(B) : T_{\frac{1}{2}}(C)$ are in the ratio:



[Sep. 05, 2020 (I)]

Options:

A. 2 : 1 : 1

B. 3 : 2 : 1

C. 2 : 1 : 3

D. 4 : 3 : 1

Answer: C

Solution:

Since, $R = R_0 e^{-\lambda t}$

$$\ln R = \ln R_0 + (-\lambda \ln t)$$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \text{Slope}$$

$$\lambda_A = \frac{6}{10} \Rightarrow T_A = \frac{10}{6} \ln 2$$

$$\lambda_B = \frac{6}{5} \Rightarrow T_B = \frac{5 \ln 2}{6}$$

$$\lambda_C = \frac{2}{5} \Rightarrow T_C = \frac{5 \ln 2}{6}$$

$$\therefore T_{\frac{1}{2}A} : T_{\frac{1}{2}B} : T_{\frac{1}{2}C} = \frac{10}{6} : \frac{5}{6} : \frac{15}{6} = 2 : 1 : 3$$

Question 192

A radioactive nucleus decays by two different processes. The half life for the first process is 10 s and that for the second is 100 s. The effective half life of the nucleus is close to :
[Sep. 05, 2020 (II)]

Options:

- A. 9 sec.
- B. 6 sec.
- C. 55 sec.
- D. 12 sec.

Answer: A

Solution:

Let λ_1 and λ_2 be the decay constants of two process.

N be the number of nuclei left undecayed after two process.

From the law of radioactive decay we have $-\frac{dN}{dt} = \lambda_1 N + \lambda_2 N$ $\left[\because -\frac{dN}{dt} = \lambda N \right]$

$$\Rightarrow -\frac{dN}{dt} = (\lambda_1 + \lambda_2)N$$

$$\Rightarrow \lambda_{eq.} = (\lambda_1 + \lambda_2)$$

$$\Rightarrow \frac{\ln 2}{T} = \frac{\ln 2}{T_1} + \frac{\ln 2}{T_2} \left(\because \lambda = \frac{\ln 2}{T} \right)$$

$$\Rightarrow \frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$$

$$\Rightarrow \frac{1}{T} = \frac{1}{10} + \frac{1}{100} = \frac{11}{100} \text{ [Given: } T_1 = 10\text{s \& } T_2 = 100\text{s]}$$

$$\Rightarrow T = \frac{100}{11} = 9 \text{ sec}$$

Question193

In a radioactive material, fraction of active material remaining after time t is 9/16. The fraction that was remaining after t/2 is :
[Sep. 03, 2020 (I)]

Options:

A. $\frac{4}{5}$

B. $\frac{3}{5}$

C. $\frac{3}{4}$

D. $\frac{7}{8}$

Answer: C

Solution:

As we know, for first order decay, $N(t) = N_0 e^{-\lambda t}$

According to question,

$$\frac{N(t)}{N_0} = \frac{9}{16} = e^{-\lambda t}$$

After time, $t/2$

$$N(t/2) = N_0 e^{-\lambda(t/2)}$$

$$\frac{N(t/2)}{N_0} = \sqrt{e^{-\lambda t}} = \sqrt{\frac{9}{16}}$$

$$\therefore N(t/2) = \frac{3}{4} N_0$$

Question194

In a hydrogen like atom, when an electron jumps from the M-shell to the L-shell, the wavelength of emitted radiation is λ . If an electron jumps from N-shell to the L-shell, the wavelength of emitted radiation will be:

[11 Jan 2019 II]

Options:

A. $\frac{27}{20}\lambda$

B. $\frac{16}{25}\lambda$

C. $\frac{25}{16}\lambda$

D. $\frac{20}{27}\lambda$

Answer: D

Solution:

When electron jumps from M $\rightarrow$ L shell

$$\frac{1}{\lambda} = K \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{K \times 5}{36} \dots\dots (i)$$

When electron jumps from N $\rightarrow$ L shell

$$\frac{1}{\lambda'} = K \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{K \times 3}{16} \dots\dots(ii)$$

solving equation (i) and (ii) we get

$$\lambda' = \frac{20}{27}\lambda$$

Question195

Consider the nuclear fission



Given that the binding energy/nucleon of N e²⁰, H e⁴ and C¹² are, respectively, 8.03M eV , 7.07M eV and 7.86M eV identify the correct statement:

[10 Jan. 2019 II]

Options:

- A. energy of 12.4 MeV will be supplied
- B. 8.3 MeV energy will be released
- C. energy of 3.6 MeV will be released
- D. energy of 11.9 MeV has to be supplied

Answer: D

Question196

In a radioactive decay chain, the initial nucleus is ${}_{90}^{232}\text{Th}$. At the end there are 6α -particles and 4β -particles which are emitted. If the end nucleus is ${}_Z^AX$, A and Z are given by :

[12 Jan. 2019, II]

Options:

A. $A = 208$; $Z = 80$

B. $A = 202$; $Z = 80$

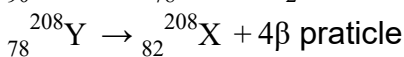
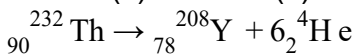
C. $A = 208$; $Z = 82$

D. $A = 200$; $Z = 81$

Answer: C

Solution:

When one α - particle emitted then daughter nuclei has 4 unit less mass number (A) and 2 unit less atomic (z) number (z).



Question197

Using a nuclear counter the count rate of emitted particles from a radioactive source is measured. At $t = 0$ it was 1600 counts per second and $t = 8$ seconds it was 100 counts per second. The count rate observed, as counts per second, at $t = 6$ seconds is close to:

[10 Jan. 2019 I]

Options:

A. 200

B. 150

C. 400

D. 360

Answer: A

Solution:

According to question, at $t = 0$, $A_0 = \frac{dN}{dt} = 1600 \text{ C/s}$

and at $t = 8 \text{ s}$, $A = 100 \text{ C/s}$

$$\therefore \frac{A}{A_0} = \frac{1}{16} \text{ in } 8 \text{ s}$$

Therefore half life period, $t_{1/2} = 2 \text{ s}$

$$\therefore \text{Activity at } t = 6 \text{ s} = 1600 \left(\frac{1}{2} \right)^3 = 200 \text{ C/s}$$

Question 198

A sample of radioactive material A, that has an activity of 10 mCi ($1 \text{ Ci} = 3.7 \times 10^{10} \text{ decays/s}$), has twice the number of nuclei as another sample of a different radioactive material B which has an activity of 20 mCi. The correct choices for half-lives of A and B would then be respectively:

[9 Jan. 2019 I]

Options:

A. 5 days and 10 days

B. 10 days and 40 days

C. 20 days and 5 days

D. 20 days and 10 days

Answer: C

Solution:

Activity $A = \lambda N$

For material, $A \rightarrow 10 = (\lambda N_0)1A$

For material, $B \rightarrow 20 = N_{01}B$

$$\Rightarrow \lambda_B = 4\lambda_A \therefore T_{1/A} = 4T_{1/2B} \left[\because T_{1/2} = \frac{0.693}{\lambda} \right]$$

i.e. 20 days half-lives for A and 5 days $(T_{1/2})_B$ For material B.

Question199

At a given instant, say $t = 0$, two radioactive substances A and B have equal activities. The ratio $\frac{R_B}{R_A}$ of their activities after time t itself decays with time t as e^{-3t} . If the half-life of A is $\ln 2$, the half-life of B is:

[9 Jan. 2019 II]

Options:

A. $4 \ln 2$

B. $\frac{\ln 2}{2}$

C. $\frac{\ln 2}{4}$

D. $2 \ln 2$

Answer: C

Solution:

Halflife of A = $\ln 2$

$$\left(\frac{t_1}{2}\right)^A = \frac{\ln 2}{\lambda}$$

$$\therefore \lambda_A = 1$$

at $t = 0$ $R_A = R_B$

$$N_A e^{-\lambda_A t} = N_B e^{-\lambda_B t}$$

$$N_A = N_B \text{ at } t = 0$$

$$\text{At } t = t \quad \frac{R_B}{R_A} = \frac{N_0 e^{-\lambda_B t}}{N_0 e^{-\lambda_A t}}$$

$$e^{-(\lambda_B - \lambda_A)t} = e^{-3t}$$

$$\Rightarrow \lambda_B - \lambda_A = 3$$

$$\lambda_B = 3 + \lambda_A = 4$$

$$(t_{1/2})_B = \frac{\ln 2}{\lambda_B} = \frac{\ln 2}{4}$$

Question200

An excited He^+ ion emits two photons in succession, with wavelengths 108.5nm and 30.4nm, in making a transition to ground state. The quantum number n, corresponding to its initial excited state is (for photon of wavelength », energy $E = \frac{1240\text{eV}}{\lambda(\text{ in nm })}$)
[12 April 2019 II]

Options:

A. $n = 4$

B. $n = 5$

C. $n = 7$

D. $n = 6$

Answer: B

Solution:

$$E = E_1 + E_2$$

$$13.6 \frac{Z^2}{n^2} = \frac{1240}{\lambda_1} + \frac{1240}{\lambda_2}$$

$$\text{or } \frac{13.6(2)^2}{n^2} = 1240 \left(\frac{1}{108.5} + \frac{1}{30.4} \right) \times \frac{1}{10^{-9}}$$

On solving, $n = 5$

Question201

The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths, $\frac{\lambda_1}{\lambda_2}$, of the photons

**emitted in this process is :
[12 April 2019 II]**

Options:

A. 20/7

B. 27/5

C. 7/5

D. 9/7

Answer: A

Solution:

$$\frac{1}{\lambda_1} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = \frac{7R}{16 \times 9}$$

$$\text{And } \frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36}$$

$$\text{Now } \frac{\lambda_1}{\lambda_2} = \frac{(5R/36)}{7R/(16 \times 9)} = \frac{20}{7}$$

Question202

**Consider an electron in a hydrogen atom, revolving in its second excited state (having radius 4.65 Å). The de- Broglie wavelength of this electron is :
[12 April 2019 II]**

Options:

A. 3.5 Å

B. 6.6 Å

C. 12.9 Å

D. 9.7 Å

Answer: D

Solution:

$$v = \frac{c}{137n} = \frac{c}{137 \times 3}$$

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\left(\frac{m \times c}{3 \times 137}\right)} = \frac{h}{mc} \times (3 \times 137) = 9.7 \text{ \AA}$$

Question 203

In Li^{++} , electron in first Bohr orbit is excited to a level by a radiation of wavelength λ . When the ion gets deexcited to the ground state in all possible ways (including intermediate emissions), a total of six spectral lines are observed. What is the value of λ ?

(Given : $h = 6.63 \times 10^{-34} \text{ J s}$; $c = 3 \times 10^8 \text{ ms}^{-1}$)

[10 April 2019 II]

Options:

A. 11.4 nm

B. 9.4 nm

C. 12.3 nm

D. 10.8 nm

Answer: D

Solution:

Spectral lines obtained on account of transition from n th orbit to various lower orbits is $\frac{n(n-1)}{2}$

$$\Rightarrow 6 = \frac{n(n-1)}{2}$$

$$\Rightarrow n = 4$$

$$\Delta E = \frac{hc}{\lambda} = \frac{-Z^2}{n^2} (13.6 \text{ eV})$$

$$\Rightarrow \frac{1}{\lambda} = Z^2 \left(\frac{13.6 \text{ eV}}{hc} \right) \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$= (13.4)(3)^2 \left[1 - \frac{1}{16} \right] \text{eV}$$

$$\Rightarrow \lambda = \frac{1242 \times 16}{(13.4) \times (9)(15)} \text{nm} \simeq 10.8 \text{nm}$$

Question204

Taking the wavelength of first Balmer line in hydrogen spectrum ($n = 3$ to $n = 2$) as 660 nm, the wavelength of the 2nd Balmer line ($n = 4$ to $n = 2$) will be;
[9 April 2019 I]

Options:

A. 889.2 nm

B. 488.9 nm

C. 642.7 nm

D. 388.9 nm

Answer: B

Solution:

$$\frac{1}{\lambda_1} = -R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = \frac{5R}{36}$$

$$\frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3R}{16}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{80}{108}$$

$$\lambda_2 = \frac{80}{108} \lambda_1 = \frac{80}{108} \times 660 = 488.9 \text{nm}$$

Question205

A He^+ ion is in its first excited state. Its ionization energy is:
[9 April 2019 II]

Options:

- A. 48.36 eV
- B. 54.40 eV
- C. 13.60 eV
- D. 6.04 eV

Answer: C

Solution:

$$E_n = -13.6 \frac{Z^2}{n^2}$$

$$\text{For } H e^+, E_2 = \frac{-13.6(2)^2}{2^2} = -13.60 \text{ eV}$$

$$\text{Ionization energy} = 0 - E_2 = 13.60 \text{ eV}$$

Question 206

**Radiation coming from transitions $n = 2$ to $n = 1$ of hydrogen atoms fall on He^+ ions in $n = 1$ and $n = 2$ states. The possible transition of helium ions as they absorb energy from the radiation is :
[8 April 2019 I]**

Options:

- A. $n = 2 \rightarrow n = 3$
- B. $n = 1 \rightarrow n = 4$
- C. $n = 2 \rightarrow n = 5$
- D. $n = 2 \rightarrow n = 4$

Answer: D

Solution:

Energy released by hydrogen atom for transition $n = 2$ to $n = 1$

$$\therefore \Delta E_1 = 13.6 \times \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3}{4} \times 13.6 \text{ eV}$$

$$= 10.2\text{eV}$$

This energy is absorbed by He^+ ion in transition from $n = 2$ to $n = n_1$ (say)

$$\therefore \Delta E_2 = 13.6 \times 4 \times \left(\frac{1}{4} - \frac{1}{n_1^2} \right) = 10.2\text{eV}$$

$$\Rightarrow n_1 = 4$$

So, possible transition is $n = 2 \rightarrow n = 4$

Question 207

A hydrogen atom, initially in the ground state is excited by absorbing a photon of wavelength $980\text{\AA}$. The radius of the atom in the excited state, in terms of Bohr radius a_0 , will be:

[11 April 2019 I]

Options:

A. $25a_0$

B. $9a_0$

C. $16a_0$

D. $4a_0$

Answer: C

Solution:

$$\text{Energy of photon} = \frac{hc}{\lambda} = \frac{12500}{980} = 12.75\text{eV}$$

Energy of electron in n^{th} orbit is given by

$$E_n = \frac{-13.6}{n^2} \Rightarrow E_n - E_1 = -13.6 \left[\frac{1}{n^2} - \frac{1}{1^2} \right]$$

$$\Rightarrow 12.75 = 13.6 \left[\frac{1}{1^2} - \frac{1}{n^2} \right] \Rightarrow n = 4$$

$\therefore$ Electron will excite to $n = 4$

We know that ' R ' $\propto n^2$

$\therefore$ Radius of atom will be $16a_0$

Question208

The ratio of the mass densities of nuclei of ^{40}Ca and ^{16}O is close to :
[8 April 2019 II]

Options:

A. 1

B. 0.1

C. 5

D. 2

Answer: A

Solution:

Nuclear density is independent of atomic number.

Question209

Two radioactive materials A and B have decay constants 10λ and λ , respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of A to that of B will be $1/e$ after a time :

[10 April 2019, I]

Options:

A. $\frac{1}{9\lambda}$

B. $\frac{1}{11\lambda}$

C. $\frac{11}{10\lambda}$

D. $\frac{1}{10\lambda}$

Answer: A

Solution:

$$\text{As, } N = N_0 e^{-\lambda t}$$

$$\text{so, } \frac{N_A}{N_B} = e^{(\lambda_B - \lambda_A)t} = \frac{1}{e} \Rightarrow (\lambda_B - \lambda_A)t = -1$$

$$\Rightarrow (\lambda_A - \lambda_B) \cdot t = 1$$

$$\Rightarrow t = \frac{1}{(\lambda_B - \lambda_A)} = \frac{1}{10\lambda - \lambda} = \frac{1}{9\lambda}$$

Question 210

Two radioactive substances A and B have decay constants 5λ and λ respectively. At $t = 0$, a sample has the same number of the two nuclei. The time taken for the ratio of the number of nuclei to

become $\left(\frac{1}{e}\right)^2$ will be

[10 April 2019, II]

Options:

A. $1/2\lambda$

B. $1/4\lambda$

C. $1/\lambda$

D. $2/\lambda$

Answer: A

Solution:

Let N_1 and N_2 be the number of radioactive nuclei of substance at anytime t .

$$N_1(\text{ at } t) = N_0 e^{-5\lambda t} \dots\dots(i)$$

$$N_2(\text{ at } t) = N_0 e^{-\lambda t} \dots\dots(ii)$$

Dividing equation (i) by (ii), we get

$$\frac{N_1}{N_2} = \frac{1}{e^2} = e^{-4\lambda t} \Rightarrow 4\lambda t = 2$$

$$\Rightarrow t = \frac{2}{4\lambda} = \left(\frac{1}{2\lambda} \right)$$

Question 211

An electron from various excited states of hydrogen atom emit radiation to come to the ground state. Let λ_n, λ_g be the de Broglie wavelength of the electron in the n^{th} state and the ground state respectively. Let Λ_n be the wavelength of the emitted photon in the transition from the n^{th} state to the ground state. For large n , (A, B are constants)

[2018]

Options:

A. $\Lambda_n \approx A + \frac{B}{\lambda_n^2}$

B. $\Lambda_n \approx A + B\lambda_n$

C. $\Lambda_n^2 \approx A + B\lambda_n^2$

D. $\Lambda_n^2 \approx \lambda$

Answer: A

Solution:

Wavelength of emitted photon from n^{th} state to the ground state,

$$\frac{1}{\Lambda_n} = RZ^2 \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$$

$$\Lambda_n = \frac{1}{RZ^2} \left(1 - \frac{1}{n^2} \right)^{-1}$$

Since n is very large, using binomial theorem

$$\Lambda_n = \frac{1}{RZ^2} \left(1 + \frac{1}{n^2} \right)$$

$$\Lambda_n = \frac{1}{RZ^2} + \frac{1}{RZ^2} \left(\frac{1}{n^2} \right)$$

As we know, $\lambda_n = \frac{2\pi r}{n} = 2\pi \left(\frac{n^2 h^2}{4\pi^2 m Z e^2} \right) \frac{1}{n} \propto n$

$$\Lambda_n \approx A + \frac{B}{\lambda_n^2}$$

Question212

If the series limit frequency of the Lyman series is ν_1 , then the series limit frequency of the P-fund series is
[2018]

Options:

A. $25\nu_L$

B. $16\nu_L$

C. $\nu_L/16$

D. $\nu_L/25$

Answer: D

Solution:

$$h\nu_L = E_\infty - E_1 \dots (i)$$

$$h\nu_f = E_\infty - E_5 \dots (ii)$$

$$E \propto \frac{Z^2}{n^2} \Rightarrow \frac{E_5}{E_1} = \left(\frac{1}{5} \right)^2 = \frac{1}{25}$$

$$\text{Eqn(i)}/(ii) \Rightarrow \frac{h\nu_L}{h\nu_f} = \frac{E_1}{E_5}$$

$$\Rightarrow \frac{\nu_L}{\nu_f} = \frac{25}{1} \Rightarrow \nu_f = \frac{\nu_L}{25}$$

Question213

The de-Broglie wavelength (λ_B) associated with the electron orbiting in the second excited state of hydrogen atom is related to that in the

ground state (λ_G) by
[Online April 16, 2018]

Options:

A. $\lambda_B = \lambda_G/3$

B. $\lambda_B = \lambda_G/2$

C. $\lambda_B = 2\lambda_G$

D. $\lambda_B = 3\lambda_G$

Answer: D

Solution:

de-Broglie wavelength, $\lambda = \frac{h}{p}$

$$\frac{\lambda_B}{\lambda_G} = \frac{p_a}{p_B} = \frac{mv_G}{mv_B}$$

Speed of electron $V \propto \frac{Z}{n}$

$$\text{so } \frac{\lambda_B}{\lambda_G} = \frac{n_B}{n_G} = \frac{3}{1} \Rightarrow \lambda_B = 3\lambda_G$$

Question214

The energy required to remove the electron from a singly ionized Helium atom is 2.2 times the energy required to remove an electron from Helium atom. The total energy required to ionize the Helium atom completely is:

[Online April 15, 2018]

Options:

A. 20 eV

B. 79 eV

C. 109 eV

D. 34 eV

Answer: B

Solution:

Energy required to remove e^- from singly ionized helium atom $= \frac{(13.6)Z^2}{1^2} = 54.4\text{eV}$ ($\because Z = 2$)

Energy required to remove e^- from helium atom $= x\text{eV}$

According to question, $54.4\text{eV} = 2.2x \Rightarrow x = 24.73\text{eV}$

Therefore, energy required to ionize helium atom $= (54.4 + 24.73)\text{eV} = 79.12\text{eV}$

Question215

Muon (μ^{-1}) is negatively charged ($|q| = |e|$) with a mass $m_\mu = 200m_e$, where m_e is the mass of the electron and e is the electronic charge. If μ^{-1} is bound to a proton to form a hydrogen like atom, identify the correct statements

(A) Radius of the muonic orbit is 200 times smaller than that of the electron

(B) the speed of the μ^{-1} in the n th orbit is $\frac{1}{200}$ times that of the electron in the n th orbit

(C) The ionization energy of muonic atom is 200 times more than that of an hydrogen atom

(D) The momentum of the muon in the n th orbit is 200 times more than that of the electron

[Online April 15, 2018]

Options:

A. (A), (B), (D)

B. (B), (D)

C. (C), (D)

D. (A), (C), (D)

Answer: D

Solution:

$$(A) \text{ Radius of muon} = \frac{\text{Radius of hydrogen}}{200}$$

$$\text{Radius of H atom} = r = \frac{\epsilon_0 n^2 h^2}{\pi m e^2}$$

$$\text{Radius of muon} = r_\mu = \frac{\epsilon_0 n^2 h^2}{\pi \times 200 m e^2}$$

$$r_\mu = \frac{r}{200}$$

(B) Velocity relation given is wrong

(C) Ionization energy in e^-H atom

$$E = \frac{+m e^4}{8 \epsilon_0^2 n^2 h^2}$$

$$E_\mu = \frac{200 m e^4}{8 \epsilon_0^2 n^2 h^2} = 200E$$

(D) Momentum of H-atom

$$mvr = \frac{nh}{2\pi}$$

Hence (A), (C), (D) are correct.

Question216

**An unstable heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 8:27. The ratio of the radii of the nuclei (assumed to be spherical) is:
[Online April 15, 2018]**

Options:

A. 8 : 27

B. 2 : 3

C. 3 : 2

D. 4 : 9

Answer: C

Solution:

Let heavy nucleus breaks into two nuclei of mass m_1 and m_2 and move away with velocities V_1 and V_2 ? respectively.

According to question,

$$\frac{V_1}{V_2} = \frac{8}{27}$$

$$m_1 V_1 = m_2 V_2 \text{ (Law of momentum conservation)}$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{V_2}{V_1} = \frac{27}{8}$$

$$\frac{\rho \times \frac{4}{3}\pi R_1^3}{\rho \times \frac{4}{3}\pi R_2^3} \left(\because \text{density } \rho = \frac{\text{mass}}{\text{volume}} \right)$$

$$\Rightarrow \left(\frac{R_1}{R_2} \right) = \left(\frac{27}{8} \right)^{\frac{1}{3}} = \left(\frac{3}{2} \right)^{3 \times \frac{1}{3}} \therefore \frac{R_1}{R_2} = \frac{3}{2}$$

Question217

At some instant, a radioactive sample S_1 having an activity $5\mu\text{Ci}$ has twice the number of nuclei as another sample S_2 which has an activity of $10\mu\text{Ci}$. The half lives of S_1 and S_2 are [Online April 16, 2018]

Options:

- A. 10 years and 20 years, respectively
- B. 5 years and 20 years, respectively
- C. 20 years and 10 years, respectively
- D. 20 years and 5 years, respectively

Answer: D

Solution:

$$S_1 = \frac{dN}{dt} = \lambda_1 N_1 = 5;$$

$$S_2 = \frac{dN}{dt} = \lambda_2 N_2 = 10$$

$$\text{(Given } N_1 = 2N_2 \text{)}$$

$$\lambda_1(2N_2) = 5 \rightarrow (1);$$

$$\lambda_2 N_2 = 10 \rightarrow (2)$$

$\Rightarrow$ dividing (1) & (2)

$$\frac{2\lambda_1}{\lambda_2} = \frac{1}{2} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{1}{4}$$

We know that $\lambda = \frac{\ln 2}{t_{1/2}}$

$$\therefore \lambda \propto \frac{1}{t_{1/2}}$$

$$\Rightarrow \frac{t_{1/2}^1}{t_{1/2}^2} = \frac{4}{1}$$

$$t_{1/2}^1 = 4t_{1/2}^2$$

$$\Rightarrow t_{1/2}^1 = 20 \text{ [half life of } s_1 \text{]}$$

$$t_{1/2}^2 = 5 \text{ [half life of } s_2 \text{]}$$

Question 218

A solution containing active cobalt ${}_{27}^{60}\text{Co}$ having activity of $0.8\mu\text{Ci}$ and decay constant λ is injected in an animal's body. If 1cm^3 of blood is drawn from the animal's body after 10 hrs of injection, the activity found was 300 decays per minute. What is the volume of blood that is flowing in the body? ($1\text{Ci} = 3.7 \times 10^{10}$ decay per second and at $t = 10\text{hrs}$ $e^{-\lambda t} = 0.84$)
[Online April 15, 2018]

Options:

- A. 6 litres
- B. 7 litres
- C. 4 litres
- D. 5 litres

Answer: D

Solution:

Let initial activity = $N_0 = 0.8\mu\text{Ci} = 0.8 \times 3.7 \times 10^4 \text{ dps}$
Activity in 1cm^3 of blood at $t = 10\text{hr}$

$$n = \frac{300}{60} \text{ d ps} = 5 \text{ d ps}$$

N = Activity of whole blood at time $t = 10 \text{ hr}$

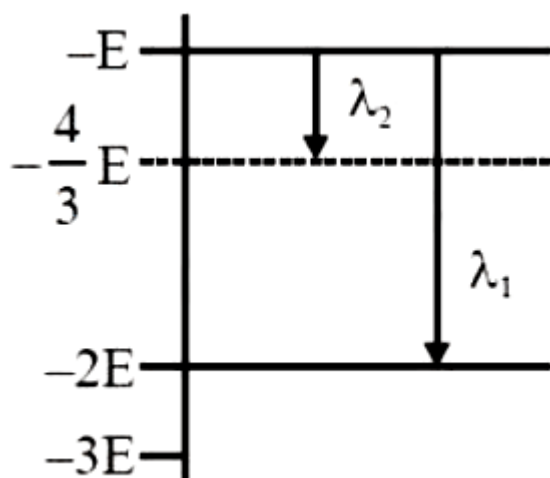
Total volume of the blood in the person, $V = \frac{N}{n}$

$$= \frac{N_0 e^{-\lambda t}}{n} = \frac{0.8 \times 3.7 \times 10^4 \times 0.7927}{5} \cong 5 \text{ litres}$$

Question 219

Some energy levels of a molecule are shown in the figure.

The ratio of the wavelengths $r = \lambda_1 / \lambda_2$, is given by



[2017]

Options:

A. $r = \frac{3}{4}$

B. $r = \frac{1}{3}$

C. $r = \frac{4}{3}$

D. $r = \frac{2}{3}$

Answer: B

Solution:

From energy level diagram, using $\Delta E = \frac{hc}{\lambda}$

For wavelength $\lambda_1 \Delta E = -E - (-2E) = \frac{hc}{\lambda_1}$

$$\therefore \lambda_1 = \frac{hc}{E}$$

For wavelength $\lambda_2 \Delta E = -E - \left(-\frac{4E}{3}\right) = \frac{hc}{\lambda_2}$

$$\therefore \lambda_2 = \frac{hc}{\left(\frac{E}{3}\right)}$$

$$\therefore r = \frac{\lambda_1}{\lambda_2} = \frac{1}{3}$$

Question220

The acceleration of an electron in the first orbit of the hydrogen atom (z = 1) is :

[Online April 9, 2017]

Options:

A. $\frac{h^2}{\pi^2 m^2 r^3}$

B. $\frac{h^2}{8\pi^2 m^2 r^3}$

C. $\frac{h^2}{4\pi^2 m^2 r^3}$

D. $\frac{h^2}{4\pi m^2 r^3}$

Answer: C

Solution:

Speed of electron in first orbit (n = 1) of hydrogen atom (z = 1),

$$v = \frac{e^2}{2\epsilon_0 h}$$

radius of Bohr's first orbit,

$$r = \frac{h^2 \epsilon_0}{\pi m e^2} \Rightarrow \epsilon_0 = \frac{r \pi m e^2}{h^2} \dots\dots (i)$$

Acceleration of electron,

$$\frac{v^2}{r} = \frac{e^4}{4\epsilon_0^2 h^2} \times \frac{\pi m e^2}{h^2 \epsilon_0}$$

$$= \frac{e^4 \times \pi m e^2}{4 h^4 \epsilon_0^3} \dots\dots (ii)$$

eliminating ϵ_0 from eq (ii),

$$= \frac{e^4 \pi m e^2 h^6}{4 h^4 r^3 \pi^3 m^3 e^6} \text{ from eq}^n(i)$$

$$= \frac{h^2}{4 \pi^2 m^2 r^3}$$

Question221

According to Bohr's theory, the time averaged magnetic field at the centre (i.e. nucleus) of a hydrogen atom due to the motion of electrons in the n^{th} orbit is proportional to :

(n = principal quantum number)

[Online April 8, 2017]

Options:

A. n^{-4}

B. n^{-5}

C. n^{-3}

D. n^{-2}

Answer: D

Solution:

Magnetic field at the centre of nucleus of H-atom,

$$B = \frac{\mu_0 I}{2r} \dots\dots (i)$$

According to Bohr's model, radius of orbit $r \propto n^2$ from eq. (i) we can also write as $B \propto n^{-2}$

Question222

Imagine that a reactor converts all given mass into energy and that it operates at a power level of 10^9 watt. The mass of the fuel consumed per hour in the reactor will be :

(velocity of light, c is 3×10^8 m/s)

[Online April 9, 2017]

Options:

A. 0.96gm

B. 0.8gm

C. 4×10^{-2} gm

D. 6.6×10^{-5} gm

Answer: C

Solution:

Power level of reactor, $P = \frac{E}{\Delta t} = \Delta mc^2 \Delta t$

mass of the fuel consumed per hour in the reactor,

$$\frac{\Delta m}{\Delta t} = \frac{P}{c^2} = \frac{10^9}{(3 \times 10^8)^2} = 4 \times 10^{-2} \text{ gm}$$

Question223

Two deuterons undergo nuclear fusion to form a Helium nucleus.

Energy released in this process is : (given binding energy per nucleon for deuteron=1.1 MeV and for helium=7.0 MeV)

[Online April 8, 2017]

Options:

A. 30.2 MeV

B. 32.4 MeV

C. 23.6 MeV

D. 25.8 MeV

Answer: C

Solution:

${}_1\text{H}^2 + {}_1\text{H}^2 \rightarrow {}_2\text{He}^4$ Total binding energy of two deuterium nuclei = $1.1 \times 4 = 4.4 \text{ MeV}$

Binding energy of a $({}_2\text{He}^4)$ nuclei = $4 \times 7 = 28 \text{ MeV}$

Energy released in this process = $28 - 4.4 = 23.6 \text{ MeV}$

Question 224

A radioactive nucleus A with a half life T , decays into a nucleus B. At t = 0, there is no nucleus B. At sometime, the ratio of the number of B to that of A is 0.3. Then, t is given by [2017]

Options:

A. $t = T \log(1.3)$

B. $t = \frac{T}{\log(1.3)}$

C. $t = T \frac{\log 2}{\log 1.3}$

D. $t = \frac{\log 1.3}{\log 2}$

Answer: D

Solution:

Let initially there are total N_0 number of nuclei At time t

$$\frac{N_B}{N_A} = 0.3 \text{ (given)}$$

$$\Rightarrow N_B = 0.3 N_A$$

$$N_0 = N_A + N_B = N_A + 0.3 N_A$$

$$\therefore N_A = \frac{N_0}{1.3}$$

$$\text{As we know } N_t = N_0 e^{-\lambda t}$$

$$\text{or, } \frac{N_0}{1.3} = N_0 e^{-\lambda t}$$

$$\frac{1}{1.3} = e^{-\lambda t} \Rightarrow \ln(1.3) = \lambda t$$

$$\text{or, } t = \frac{\ln(1.3)}{\lambda} \Rightarrow t = \frac{\ln(1.3)}{\frac{\ln(2)}{T}} = \frac{\ln(1.3)}{\ln(2)} T$$

Question225

**A hydrogen atom makes a transition from $n = 2$ to $n = 1$ and emits a photon. This photon strikes a doubly ionized lithium atom ($z = 3$) in excited state and completely removes the orbiting electron. The least quantum number for the excited state of the ion for the process is :
[Online April 9, 2016]**

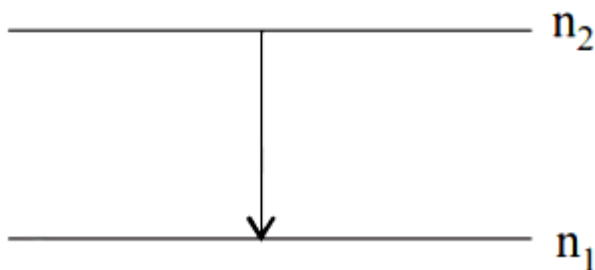
Options:

- A. 2
- B. 4
- C. 5
- D. 3

Answer: B

Solution:

A hydrogen atom makes a transition from $n = 2$ to $n = 1$



$$\text{Then wavelength} = Rcz^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = Rc(1)^2 \left[1 - \frac{1}{4} \right]$$

$$\lambda = Rc \left[\frac{3}{4} \right] \dots\dots(1)$$

For ionized lithium

$$\lambda = Rc(3)^2 \left[\frac{1}{n^2} \right] = Rc9 \left[\frac{1}{n^2} \right] \dots\dots(2)$$

$$R_c \left[\frac{3}{4} \right] = R_c 9 \left[\frac{1}{n^2} \right]$$

$$\Rightarrow \frac{3}{4} = \frac{9}{n^2} \Rightarrow n = \sqrt{12} = 2\sqrt{3}$$

$\therefore$ The least quantum number must be 4.

Question226

Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed number of A and B nuclei will be :

[2016]

Options:

A. 1 : 4

B. 5 : 4

C. 1 : 16

D. 4 : 1

Answer: B

Solution:

For $A_{t/2} = 20$ min, $t = 80$ min, number of half lifes $n = 4$

$\therefore$ Nuclei remaining $= \frac{N_0}{2^4}$. Therefore nuclei decayed $= N_0 - \frac{N_0}{2^4}$

For $B_{t/2} = 40$ min., $t = 80$ min, number of half lifes $n = 2$

$\therefore$ Nuclei remaining $= \frac{N_0}{2^2}$. Therefore nuclei decayed $= N_0 - \frac{N_0}{2^2}$

$$\therefore \text{Required ratio} = \frac{N_0 - \frac{N_0}{2^4}}{N_0 - \frac{N_0}{2^2}} = \frac{1 - \frac{1}{16}}{1 - \frac{1}{4}} = \frac{15}{16} \times \frac{4}{3} = \frac{5}{4}$$

Question227

As an electron makes a transition from an excited state to the ground state of a hydrogen - like atom/ion :
[2015]

Options:

- A. kinetic energy decreases, potential energy increases but total energy remains same
- B. kinetic energy and total energy decrease but potential energy increases
- C. its kinetic energy increases but potential energy and total energy decrease
- D. kinetic energy, potential energy and total energy decrease

Answer: C

Solution:

Kinetic energy of electron is

$$\text{K.E.} \propto \left(\frac{Z}{N}\right)^2$$

When the electron makes transition from excited state to ground state, then n increases and kinetic energy increases.

Total energy = $-KE$

$\therefore$ Total energy also decreases.

Potential energy is lowest for ground state.

Question228

The de–Broglie wavelength associated with the electron in the $n = 4$ level is :
[Online April 11, 2015]

Options:

- A. $\frac{1}{4}$ th of the de–Broglie wavelength of the electron in the ground state.
- B. four times the de–Broglie wavelength of the electron in the ground state
- C. two times the de–Broglie wavelength of the electron in the ground state
- D. half of the de–Broglie wavelength of the electron in the ground state

Answer: B

Solution:

De-Broglie wavelength of electron $\lambda = \frac{h}{mV}$

As we know, $V \propto \frac{1}{n}$

So, $\lambda \propto n$

$$\lambda_4 = 4\lambda_1$$

λ_1 is the de-Broglie wavelength of the electron in the ground state.

Question229

If one were to apply Bohr model to a particle of mass ‘m’ and charge ‘q’ moving in a plane under the influence of a magnetic field ‘B’, the energy of the charged particle in the nth level will be :
[Online April 10, 2015]

Options:

A. $n \left(\frac{hqB}{2\pi m} \right)$

B. $n \left(\frac{hqB}{8\pi m} \right)$

C. $n \left(\frac{hqB}{4\pi m} \right)$

D. $n \left(\frac{hqB}{\pi m} \right)$

Answer: C

Solution:

$$qVB = \frac{mv^2}{r} \dots\dots(i)$$

$$\frac{nh}{2\pi} = mvr \dots\dots(ii)$$

Multiplying equation (i) and (ii),

$$\frac{qBnh}{2\pi} = m^2v^2$$

Now multiplying both sides by $\frac{1}{2m}$,

$$n \frac{qBh}{4\pi m} = \frac{1}{2}mv^2$$

$$\text{i.e. K E} = n \left[\frac{qBh}{4\pi m} \right]$$

Question230

Let N_β be the number of β particles emitted by 1 gram of Na^{24} radioactive nuclei (half life = 15hrs) in 7.5 hours, N_β is close to (Avogadro number = 6.023×10^{23} /g. mole)
[Online April 11, 2015]

Options:

A. 6.2×10^{21}

B. 7.5×10^{21}

C. 1.25×10^{22}

D. 1.75×10^{22}

Answer: B

Solution:

We know that $N_\beta = N_0(1 - e^{-\lambda t})$

$$N_\beta = \frac{6.023 \times 10^{23}}{24} \left[1 - e^{-\frac{\ln 2}{15} \times 7.5} \right]$$

on solving we get,

$$N_\beta = 7.4 \times 10^{21}$$

Question231

The radiation corresponding to $3 \rightarrow 2$ transition of hydrogen atom falls on a metal surface to produce photoelectrons. These electrons

are made to enter a magnetic field of $3 \times 10^{-4} \text{ T}$. If the radius of the largest circular path followed by these electrons is 10.0mm, the work function of the metal is close to:
[2014]

Options:

A. 1.8 eV

B. 1.1 eV

C. 0.8 eV

D. 1.6 eV

Answer: B

Solution:

Radius of circular path followed by electron is given by,

$$r = \frac{mv}{qB} = \frac{\sqrt{2meV}}{eB} = \frac{1}{B} \sqrt{\frac{2m}{e} V}$$

$$\Rightarrow V = \frac{B^2 r^2 e}{2m} = 0.8 \text{ V}$$

For transition between 3 to 2.

$$E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{13.6 \times 5}{36} = 1.88 \text{ eV}$$

$$\text{Work function} = 1.88 \text{ eV} - 0.8 \text{ eV} = 1.08 \text{ eV} \approx 1.1 \text{ eV}$$

Question232

Hydrogen (${}_1\text{H}^1$), Deuterium (${}_1\text{H}^2$), singly ionised Helium(${}_2\text{He}^4$)⁺ and doubly ionised lithium (${}_3\text{Li}^6$)⁺⁺ all have one electron around the nucleus. Consider an electron transition from $n = 2$ to $n = 1$. If the wavelengths of emitted radiation are λ_1 , λ_2 , λ_3 and λ_4 respectively then approximately which one of the following is correct?
[2014]

Options:

A. $4\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

B. $\lambda_1 = 2\lambda_2 = 2\lambda_3 = \lambda_4$

C. $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

D. $\lambda_1 = 2\lambda_2 = 3\lambda_3 = 4\lambda_4$

Answer: C

Solution:

Wave number $\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n^2} \right]$

$\Rightarrow \lambda \propto \frac{1}{Z^2}$

$\therefore \lambda Z^2 = \text{constant}$

By question $n = 1$ and $n_1 = 2$

Then, $\lambda_1 = \lambda_2 = 4\lambda_3 = 9\lambda_4$

Question233

Match List - I (Experiment performed) with List-II (Phenomena discovered/associated) and select the correct option from the options given below the lists:

	List - I		List - II
(1)	Davisson and Germer experiment	(i)	Wave nature of electrons
(2)	Millikan's oil drop experiment	(ii)	Charge of an electron
(3)	Rutherford experiment	(iii)	Quantisation of energy levels
(4)	Franck-Hertz experiment	(iv)	Existence of nucleus

[Online April 19, 2014]

Options:

A. (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)

B. (1)-(i), (2)-(ii), (3)-(iv), (4)-(iii)

C. (1)-(iii), (2)-(iv), (3)-(i), (4)-(ii)

D. (1)-(iv), (2)-(iii), (3)-(ii), (4)-(i)

Answer: B

Solution:

(1) Davisson and Gemner experiment-wave nature of electrons.

(2) Millikan's oil drop experiment - charge of an electron.

(3) Rutherford experiment - Existence of nucleus.

(4) Frank-Hertz experiment - Quantisation of energy levels.

Question234

The binding energy of the electron in a hydrogen atom is 13.6 eV, the energy required to remove the electron from the first excited state of Li^{++} is:

[Online April 9, 2014]

Options:

A. 122.4 eV

B. 30.6 eV

C. 13.6 eV

D. 3.4 eV

Answer: B

Solution:

For first excited state, $n = 2$ and for $\text{Li}^{++} Z = 3$

$$E_n = \frac{13.6}{n^2} \times Z^2 = \frac{13.6}{4} \times 9 = 30.6\text{eV}$$

Question235

A piece of wood from a recently cut tree shows 20 decays per minute. A wooden piece of same size placed in a museum (obtained from a tree cut many years back) shows 2 decays per minute. If half life of C^{14} is 5730 years, then age of the wooden piece placed in the museum is approximately:
[Online April 19, 2014]

Options:

- A. 10439 years
- B. 13094 years
- C. 19039 years
- D. 39049 years

Answer: C

Solution:

$$\text{Given: } \frac{dN_0}{dt} = 20 \text{ decays /min}$$

$$\frac{dN}{dt} = 2 \text{ decays/min}$$

$$T_{1/2} = 5730 \text{ years}$$

As we know,

$$N = N_0 e^{-\lambda t}$$

$$\log \frac{N_0}{N} = \lambda t$$

$$\begin{aligned} \therefore t &= \frac{1}{\lambda} \log \frac{N_0}{N} \\ &= \frac{2.303 \times T_{1/2}}{0.693} \times \log_{10} \frac{N_0}{N} \end{aligned}$$

$$\text{But } \frac{\frac{dN_0}{dt}}{\frac{dN}{dt}} = \frac{N_0}{N} = \frac{20}{2} = 10$$

$$\begin{aligned} \therefore t &= \frac{2.303 \times 5730}{0.693} \times 1 \\ &= 19039 \text{ years} \end{aligned}$$

Question236

A piece of bone of an animal from a ruin is found to have ^{14}C activity of 12 disintegrations per minute per gm of its carbon content. The ^{14}C activity of a living animal is 16 disintegrations per minute per gm. How long ago nearly did the animal die? (Given half life of ^{14}C is $t_{1/2} = 5760$ years)

[Online April 12, 2014]

Options:

- A. 1672 years
- B. 2391 years
- C. 3291 years
- D. 4453 years

Answer: B

Solution:

Given, for ^{14}C

$$A_0 = 16 \text{ dis min}^{-1} \text{ g}^{-1}$$

$$A = 12 \text{ dis min}^{-1} \text{ g}^{-1}$$

$$t_{1/2} = 5760 \text{ years}$$

$$\text{Now, } \lambda = \frac{0.693}{t_{1/2}}$$

$$\lambda = \frac{0.693}{5760} \text{ per year}$$

$$\text{Then, from, } t = \frac{2.303}{\lambda} \log_{10} \frac{A_0}{A}$$

$$= \frac{2.303 \times 5760}{0.693} \log_{10} \frac{16}{12}$$

$$= \frac{2.303 \times 5760}{0.693} \log_{10} 1.333$$

$$= \frac{2.303 \times 5760 \times 0.1249}{0.693} = 2390.81 \approx 2391 \text{ years.}$$

Question237

A radioactive nuclei with decay constant $\frac{0.5}{s}$ is being produced at a constant rate of 100 nuclei/s. If at $t = 0$ there were no nuclei, the time when there are 50 nuclei is:
[Online April 11, 2014]

Options:

A. 1 s

B. $2 \ln\left(\frac{4}{3}\right)$ s

C. $\ln 2$ s

D. $\ln\left(\frac{4}{3}\right)$ s

Answer: B

Solution:

Let N be the number of nuclei at any time t then,

$$\frac{dN}{dt} = 100 - \lambda N \quad \text{or} \quad \int_0^N \frac{dN}{(100 - \lambda N)} = \int_0^t dt$$

$$-\frac{1}{\lambda} [\log(100 - \lambda N)]_0^N = t$$

$$\log(100 - \lambda N) - \log 100 = -\lambda t$$

$$\log \frac{100 - \lambda N}{100} = -\lambda t$$

$$\frac{100 - \lambda N}{100} = e^{-\lambda t} \quad 1 - \lambda N / 100 = e^{-\lambda t}$$

$$N = \frac{100}{\lambda} (1 - e^{-\lambda t})$$

As, $N = 50$ and $\lambda = 0.5/\text{sec}$

$$\therefore 50 = \frac{100}{0.5} (1 - e^{-0.5t})$$

Solving we get,

$$t = 2 \ln\left(\frac{4}{3}\right) \text{ sec}$$

Question 238

In a hydrogen like atom electron make transition from an energy level with quantum number n to another with quantum number

$(n - 1)$. If $n \gg 1$, the frequency of radiation emitted is proportional to :
[2013]

Options:

A. $\frac{1}{n}$

B. $\frac{1}{n^2}$

C. $\frac{1}{n^3/2}$

D. $\frac{1}{n^3}$

Answer: D

Solution:

$$\Delta E = h\nu$$

$$\nu = \frac{\Delta E}{h} = k \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right] = \frac{k(2n-1)}{n^2(n-1)^2}$$

$$\approx \frac{2k}{n^3} \text{ or } \nu \propto \frac{1}{n^3}$$

Question239

A 12.5 eV electron beam is used to bombard gaseous hydrogen at room temperature. It will emit :
[Online April 25, 2013]

Options:

A. 2 lines in the Lyman series and 1 line in the Balmer series

B. 3 lines in the Lyman series

C. 1 line in the Lyman series and 2 lines in the Balmer series

D. 3 lines in the Balmer series

Answer: A

Solution:

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{12.5 \times 1.6 \times 10^{-19}} = 993 \text{Å}$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

(where Rydberg constant, $R = 1.097 \times 10^7$)

$$\text{or, } \frac{1}{993 \times 10^{-10}} = 1.097 \times 10^7 \left(\frac{1}{1^2} - \frac{1}{n_2^2} \right)$$

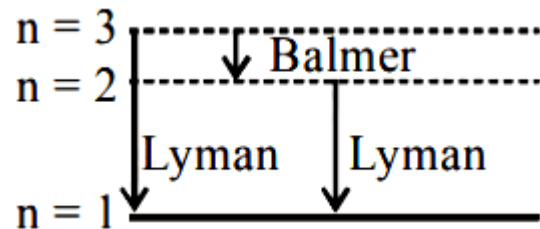
Solving we get $n_2 = 3$

Spectral lines

Total number of spectral lines = 3

Two lines in Lyman series for $n_1 = 1, n_2 = 2$ and $n_1 = 1, n_2 = 3$ and one in Balmer series for

$n_1 = 2, n_2 = 3$



Question240

In the Bohr's model of hydrogen-like atom the force between the nucleus and the electron is modified as $F = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right)$, where β is a constant. For this atom, the radius of the n^{th} orbit in terms of the

Bohr radius $\left(a_0 = \frac{\epsilon_0 h^2}{m\pi e^2} \right)$ is:

[Online April 23, 2013]

Options:

A. $r_n = a_0 n - \beta$

B. $r_n = a_0 n^2 + \beta$

$$C. r_n = a_0 n^2 - \beta$$

$$D. r_n = a_0 n + \beta$$

Answer: C

Solution:

$$\text{As } F = \frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right)$$

$$\text{and } mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr}$$

$$\therefore m \left(\frac{nh}{2\pi mr} \right)^2 \times \frac{1}{r} = \frac{e^2}{4\pi\epsilon_0} \left(\frac{1}{r^2} + \frac{\beta}{r^3} \right)$$

$$\text{or, } \frac{1}{r^2} + \frac{\beta}{r^3} = \frac{mn^2 h^2 4\pi\epsilon_0}{4\pi^2 m^2 e^2 r^3}$$

$$\text{or, } \frac{a_0 n^2}{r^3} = \frac{1}{r^2} + \frac{\beta}{r^3} \left(\because a_0 = \frac{\epsilon_0 h^2}{m\pi e^2} \text{ Given} \right)$$

For n^{th} atom

$$\therefore r_n = a_0 n^2 - \beta$$

Question241

Orbits of a particle moving in a circle are such that the perimeter of the orbit equals an integer number of deBroglie wavelengths of the particle. For a charged particle moving in a plane perpendicular to a magnetic field, the radius of the n^{th} orbital will therefore be proportional to :

[Online April 22, 2013]

Options:

A. n^2

B. n

C. $n^{1/2}$

D. $n^{1/4}$

Answer: C

Solution:

According to the question,

$$2\pi r = n\lambda = \frac{nh}{p} = \frac{nh}{mv}$$

$$\text{or } mvr = \frac{nh}{2\pi} \text{ or } mv = \frac{nh}{2\pi r}$$

$$F = qv_B = \frac{mv^2}{r} \text{ or, } q_B = \frac{mv}{r} = \frac{nh}{2\pi r \cdot r}$$

$$\text{or, } r^2 = \frac{nh}{2\pi q_B} \text{ or, } r = \sqrt{\frac{nh}{2\pi q_B}}$$

$$\text{i.e., } r \propto n^{1/2}$$

Question242

In the Bohr model an electron moves in a circular orbit around the proton. Considering the orbiting electron to be a circular current loop, the magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, is :
[Online April 9, 2013]

Options:

A. $\left(\frac{e}{2m} \frac{n^2 h}{2\pi} \right)$

B. $\left(\frac{e}{m} \right) \frac{nh}{2\pi}$

C. $\left(\frac{e}{2m} \right) \frac{nh}{2\pi}$

D. $\left(\frac{e}{m} \right) \frac{n^2 h}{2\pi}$

Answer: C

Solution:

Magnetic moment of the hydrogen atom, when the electron is in n^{th} excited state, i.e., $n' = (n + 1)$

As magnetic moment $M_n = I_n A = i_n (\pi r_n^2)$

$$i_n = eV_n = \frac{mz^2 e^5}{4\epsilon_0^2 n^3 h^3}$$

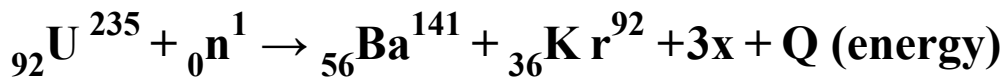
$$r_n = \frac{n^2 h^2}{4\pi^2 k m e^2} \left(k = \frac{1}{4\pi\epsilon_0} \right)$$

Solving we get magnetic moment of the hydrogen atom for n^{th} excited state

$$M_{n'} = \left(\frac{e}{2m} \right) \frac{n\hbar}{2\pi}$$

Question243

When Uranium is bombarded with neutrons, it undergoes fission. The fission reaction can be written as :



where three particles named x are produced and energy Q is released. What is the name of the particle x ?

[Online April 9, 2013]

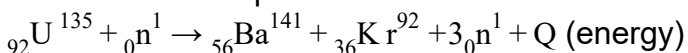
Options:

- A. electron
- B. α -particle
- C. neutron
- D. neutrino

Answer: C

Solution:

Nuclear fission equation



Hence particle x is neutron.

Question244

**The half-life of a radioactive element A is the same as the mean-life of another radioactive element B. Initially both substances have the same number of atoms, then :
[Online April 22, 2013]**

Options:

- A. A and B decay at the same rate always.
- B. A and B decay at the same rate initially.
- C. A will decay at a faster rate than B.
- D. B will decay at a faster rate than A.

Answer: D

Solution:

$$(T_{1/2})_A = (t_{\text{mean}})_B$$

$$\Rightarrow \frac{0.693}{\lambda_A} = \frac{1}{\lambda_B} \Rightarrow \lambda_A = 0.693\lambda_B$$

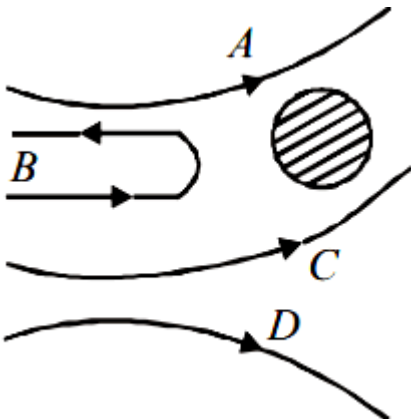
or $\lambda_A < \lambda_B$

Also rate of decay $= \lambda N$

Initially number of atoms (N) of both are equal but since $\lambda_B > \lambda_A$, therefore B will decay at a faster rate than A.

Question245

In the Rutherford experiment, a-particles are scattered from a nucleus as shown. Out of the four paths, which path is not possible?



[Online May 7, 2012]

Options:

A. D

B. B

C. C

D. A

Answer: C

Solution:

As α -particles are doubly ionised helium He^{++} i.e. Positively charged and nucleus is also positively charged and we know that like charges repel each other.

Question246

**Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be :
[2012]**

Options:

A. 2

B. 3

C. 5

D. 6

Answer: D

Solution:

For ground state, the principal quantum no. $(n) = 1$.

Principal quantum number 4 belongs to 3rd excited state.

The possible number of the spectral lines from a state n to ground state is

$$= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

Question247

A diatomic molecule is made of two masses m_1 and m_2 which are separated by a distance r . If we calculate its rotational energy by applying Bohr's rule of angular momentum quantization, its energy will be given by: (n is an integer)
[2012]

Options:

A. $\frac{(m_1 + m_2)^2 n^2 h^2}{2m_1^2 m_2^2 r^2}$

B. $\frac{n^2 h^2}{2(m_1 + m_2)r^2}$

C. $\frac{2n^2 h^2}{(m_1 + m_2)r^2}$

D. $\frac{(m_1 + m_2)n^2 h^2}{2m_1 m_2 r^2}$

Answer: D

Solution:

The energy of the system of two atoms of diatomic molecule $E = \frac{1}{2} I \omega^2$

where I = moment of inertia

ω = Angular velocity = $\frac{L}{I}$,

L = Angular momentum

$$I = \frac{1}{2}(m_1 r_1^2 + m_2 r_2^2)$$

$$\text{Thus, } E = \frac{1}{2}(m_1 r_1^2 + m_2 r_2^2) \omega^2 \dots\dots (i)$$

$$E = \frac{1}{2}(m_1 r_1^2 + m_2 r_2^2) \frac{L^2}{I^2}$$

$$L = nh$$

(According to Bohr's Hypothesis)

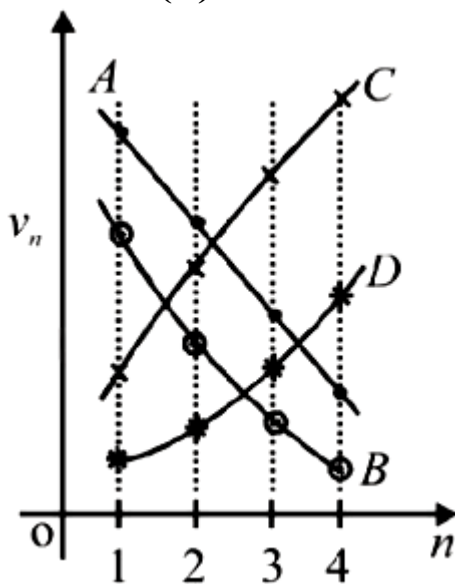
$$E = \frac{1}{2}(m_1 r_1^2 + m_2 r_2^2) \frac{L^2}{(m_1 r_1^2 + m_2 r_2^2)^2}$$

$$E = \frac{1}{2} \frac{L^2}{(m_1 r_1^2 + m_2 r_2^2)} = \frac{n^2 h^2}{8\pi^2 (m_1 r_1^2 + m_2 r_2^2)}$$

$$E = \frac{(m_1 + m_2) n^2 h^2}{8\pi^2 r^2 m_1 m_2}$$

Question248

Which of the plots shown in the figure represents speed (v_n) of the electron in a hydrogen atom as a function of the principal quantum number (n)?



[Online May 26, 2012]

Options:

- A. B
- B. D
- C. C
- D. A

Answer: A

Solution:

Velocity of electron in n^{th} orbit of hydrogen atom is given by :

$$V_n = \frac{2\pi K Z e^2}{nh}$$

Substituting the values we get,

$$V_n = \frac{2.2 \times 10^6}{n} \text{ m/s or } V_n \propto \frac{1}{n}$$

As principal quantum number increases, velocity decreases.

Question249

A doubly ionised Li atom is excited from its ground state ($n = 1$) to $n = 3$ state. The wavelengths of the spectral lines are given by λ_{32} , λ_{31} and λ_{21} . The ratio $\lambda_{32}/\lambda_{31}$ and $\lambda_{21}/\lambda_{31}$ are, respectively
[Online May 12,2012]

Options:

A. 8.1, 0.67

B. 8.1, 1.2

C. 6.4, 1.2

D. 6.4, 0.67

Answer: C

Solution:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ where } R = \text{Rydberg constant}$$

$$\frac{1}{\lambda_{32}} = \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5}{36} \Rightarrow \lambda_{32} = \frac{36}{5}$$

Similarly solving for λ_{31} and λ_{21}

$$\lambda_{31} = \frac{9}{8} \text{ and } \lambda_{21} = \frac{4}{3}$$

$$\therefore \frac{\lambda_{32}}{\lambda_{31}} = 6.4 \text{ and } \frac{\lambda_{21}}{\lambda_{31}} \simeq 1.2$$

Question250

A hypothetical atom has only three energy levels. The ground level has energy, $E_1 = -8\text{eV}$. The two excited states have energies,

$E_2 = -6\text{eV}$ and $E_3 = -2\text{eV}$. Then which of the following wavelengths will not be present in the emission spectrum of this atom?

[Online May 12,2012]

Options:

A. 207 nm

B. 465 nm

C. 310 nm

D. 620 nm

Answer: B

Solution:

$$E = \frac{hc}{\lambda}$$

Question251

The electron of a hydrogen atom makes a transition from the $(n + 1)^{\text{th}}$ orbit to the n^{th} orbit. For large n the wavelength of the emitted radiation is proportional to

[Online May 7,2012]

Options:

A. n

B. n^3

C. n^4

D. n^2

Answer: B

Solution:

If $n_1 = n$ and $n_2 = n + 1$

$$\text{Maximum wavelength } \lambda_{\max} = \frac{n^2(n+1)^2}{(2n+1)R}$$

Therefore, for large n , $\lambda_{\max} \propto n^3$

Question 252

Assume that a neutron breaks into a proton and an electron. The energy released during this process is : (mass of neutron

$= 1.6725 \times 10^{-27} \text{ kg}$, mass of proton $= 1.6725 \times 10^{-27} \text{ kg}$, mass of electron $= 9 \times 10^{-31} \text{ kg}$)

[2012]

Options:

A. 0.51 MeV

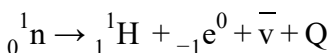
B. 7.10 MeV

C. 6.30 MeV

D. 5.4 MeV

Answer: A

Solution:



The mass defect during the process

$$\begin{aligned}\Delta m &= m_n - m_H - m_e = 1.6725 \times 10^{-27} - (1.6725 \times 10^{-27} + 9 \times 10^{-31} \text{ kg}) \\ &= -9 \times 10^{-31}\end{aligned}$$

The energy released during the process

$$E = \Delta mc^2$$

$$E = 9 \times 10^{-31} \times 9 \times 10^{16} = 81 \times 10^{-15} \text{ Joules}$$

$$E = \frac{81 \times 10^{-15}}{1.6 \times 10^{-19}} = 0.511 \text{ MeV}$$

Question253

Ionisation energy of Li (Lithium) atom in ground state is 5.4 eV.

Binding energy of an electron in Li^+ ion in ground state is 75.6 eV.

Energy required to remove all three electrons of Lithium (Li) atom is

[Online May 19, 2012]

Options:

A. 81.0 eV

B. 135.4 eV

C. 203.4 eV

D. 156.6 eV

Answer: D

Question254

The counting rate observed from a radioactive source at $t = 0$ was 1600 counts s^{-1} , and $t = 8$ s, it was 100 counts s^{-1} . The counting rate observed as counts s^{-1} at $t = 6$ s will be

[Online May 26, 2012]

Options:

A. 250

B. 400

C. 300

D. 200

Answer: D

Solution:

As we know,

$$\left[\frac{N}{N_0} \right] = \left[\frac{1}{2} \right]^n \dots\dots(i)$$

n = no. of half life

N - no. of atoms left

N_0 - initial no. of atoms

By radioactive decay law,

$$\frac{dN}{dt} = -kN$$

k - disintegration constant

$$\therefore \frac{\frac{dN}{dt}}{N_0} = \frac{N}{N_0} \dots\dots(ii)$$

From (i) and (ii) we get

$$\frac{\frac{dN}{dt}}{N_0} = \left[\frac{1}{2} \right]^n$$

$$\text{or, } \left[\frac{100}{1600} \right] = \left[\frac{1}{2} \right]^n \Rightarrow \left[\frac{1}{2} \right]^4 = \left[\frac{1}{2} \right]^n$$

$\therefore n = 4$, Therefore, in 8 seconds 4 half life had occurred in which counting rate reduces to 100 counts s^{-1} .

$$\therefore \text{Half life, } \frac{T_{1/2}}{2} = 2 \text{ sec}$$

In 6 sec, 3 half life will occur

$$\therefore \left[\frac{\frac{dN}{dt}}{1600} \right] = \left[\frac{1}{2} \right]^3 \Rightarrow \frac{dN}{dt} = 200 \text{ counts } s^{-1}.$$

Question255

The decay constants of a radioactive substance for α and β emission are λ_α and λ_β respectively. If the substance emits α and β simultaneously, then the average half life of the material will be [Online May 19, 2012]

Options:

A. $\frac{2T_\alpha T_\beta}{T_\alpha + T_\beta}$

B. $T_\alpha + T_\beta$

C. $\frac{T_{\alpha}T_{\beta}}{T_{\alpha}+T_{\beta}}$

D. $\frac{1}{2}(T_{\alpha}+T_{\beta})$

Answer: C

Solution:

$$T_{av} = \frac{T_{\alpha}T_{\beta}}{T_{\alpha}+T_{\beta}}$$

If α and β are emitted simultaneously.

Question 256

Which of the following Statements is correct?
[Online May 12, 2012]

Options:

- A. The rate of radioactive decay cannot be controlled but that of nuclear fission can be controlled.
- B. Nuclear forces are short range, attractive and charge dependent.
- C. Nuclei of atoms having same number of neutrons are known as isobars.
- D. Wavelength of matter waves is given by de Broglie formula but that of photons is not given by the same formula

Answer: A

Solution:

Radioactive decay is a continuous process. Rate of radioactive decay cannot be controlled. Nuclear fission can be controlled but not of nuclear fusion.

Question 257

A sample originally contained 10^{20} radioactive atoms, which emit α - particles. The ratio of α -particles emitted in the third year to that emitted during the second year is 0.3. How many α -particles were emitted in the first year?
[Online May 7, 2012]

Options:

A. 3×10^{18}

B. 3×10^{19}

C. 5×10^{18}

D. 7×10^{19}

Answer: B

Question258

Energy required for the electron excitation in Li^{++} from the first to the third Bohr orbit is :
[2011]

Options:

A. 36.3 eV

B. 108.8 eV

C. 122.4 eV

D. 12.1 eV

Answer: B

Solution:

Energy of excitation (ΔE) is

$$\Delta E = 13.6Z^2 \left(\frac{1}{n_1} - \frac{1}{n_2} \right) \text{eV}$$

$$\Rightarrow \Delta E = 13.6(3)^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = 108.8 \text{eV}$$

Question 259

After absorbing a slowly moving neutron of mass m_N (momentum ≈ 0) a nucleus of mass M breaks into two nuclei of masses m_1 and $5m_1$ ($6m_1 = M + m_N$) respectively. If the de Broglie wavelength of the nucleus with mass m_1 is λ , the de Broglie wavelength of the nucleus will be
[2011]

Options:

A. 5λ

B. $\lambda/5$

C. λ

D. 25λ

Answer: C

Solution:

Initial momentum of system, $p_i = 0$

Let p_1 and p_2 be the momentum of broken nuclei of masses m_1 and $5m_1$ respectively.

$$p_f = p_1 + p_2$$

From the conservation of momentum

$$p_i = p_f$$

$$0 = p_1 + p_2$$

$$p_1 = -p_2$$

From de Broglie relation, wavelength

$$\lambda_1 = \frac{h}{p_1} \text{ and } \lambda_2 = \frac{h}{p_2}$$

$$|\lambda_1| = |\lambda_2|$$

$$\lambda_1 = \lambda_2 = \lambda.$$

Question260

The half life of a radioactive substance is 20 minutes. The approximate time interval ($t_2 - t_1$) between the time t_2 when $\frac{2}{3}$ of it had decayed and time t_1 when $\frac{1}{3}$ of it had decayed is :

[2011]

Options:

A. 14 min

B. 20 min

C. 28 min

D. 7 min

Answer: B

Solution:

Number of undecayed atom after time t_2 ;

$$\frac{N_0}{3} = N_0 e^{-\lambda t_2} \dots\dots(i)$$

Number of undecayed atom after time t_1 ;

$$\frac{2N_0}{3} = N_0 e^{-\lambda t_1} \dots\dots(ii)$$

Dividing (ii) by (i), we get

$$2 = e^{\lambda(t_2 - t_1)}$$

$$\Rightarrow \ln 2 = \lambda(t_2 - t_1)$$

$$\Rightarrow t_2 - t_1 = \ln 2 / \lambda$$

Question261

Statement - 1 : A nucleus having energy E_1 decays by β^- emission to daughter nucleus having energy E_2 , but the β^- rays are emitted with a continuous energy spectrum having end point energy $E_1 - E_2$.

Statement - 2 : To conserve energy and momentum in β^- decay at least three particles must take part in the transformation.

[2011 RS]

Options:

- A. Statement-1 is correct but statement-2 is not correct.
- B. Statement-1 and statement-2 both are correct and statement-2 is the correct explanation of statement-1.
- C. Statement-1 is correct, statement-2 is correct and statement-2 is not the correct explanation of Statement-1
- D. Statement-1 is incorrect, statement-2 is correct.

Answer: B

Solution:

Statement-1: A nucleus having energy E_1 decays by β^- - emission to daughter nucleus having energy E_2 then β^- - rays are emitted with continuous energy spectrum with energy $E_1 - E_2$

Statement-2: For energy conservation and momentum conservation at least three particles, daughter nucleus, β^- particle and antineutrino are required.

Question 262

The binding energy per nucleon for the parent nucleus is E_1 and that for the daughter nuclei is E_2 . Then

[2010]

Options:

- A. $E_2 = 2E_1$
- B. $E_1 > E_2$

C. $E_2 > E_1$

D. $E_1 = 2E_2$

Answer: C

Solution:

In nuclear fission, the binding energy per nucleon of daughter nuclei is always greater than the parent nucleus.

Question 263

**The speed of daughter nuclei is
[2010]**

Options:

A. $c \frac{\Delta m}{M + \Delta m}$

B. $c \sqrt{\frac{2\Delta m}{M}}$

C. $c \sqrt{\frac{\Delta m}{M}}$

D. $c \sqrt{\frac{\Delta m}{M + \Delta m}}$

Answer: B

Solution:

$$\text{Mass defect, } \Delta M = \left[(M + \Delta m) - \left(\frac{M}{2} + \frac{M}{2} \right) \right]$$

$$= [M + \Delta m - M] = \Delta m$$

$$\text{Energy released, } Q = \Delta M c^2 = \Delta m c^2 \dots (i)$$

From the law of conservation of momentum

$$(M + \Delta m) \times 0 = M \cdot 2v_1 - \frac{M}{2} \times v_2$$

$$\Rightarrow v_1 = v_2$$

$$\text{Now, } Q = \frac{1}{2} \left(\frac{M}{2} \right) v_1^2 + \frac{1}{2} \left(\frac{M}{2} \right) v_2^2 - \frac{1}{2}$$

$$(M + \Delta m) \times (0)^2 \\ = \frac{M}{2} v_1^2 (\because v_1 = v_2) \dots\dots(ii)$$

From equation (i) and (ii), we get

$$\left(\frac{M}{2} \right) v_1^2 = \Delta m c^2 \\ \Rightarrow v_1^2 = \frac{2\Delta m c^2}{M} \Rightarrow v_1 = c \sqrt{\frac{2\Delta m}{M}}$$

Question 264

A radioactive nucleus (initial mass number A and atomic number Z emits 3 α - particles and 2 positrons. The ratio of number of neutrons to that of protons in the final nucleus will be [2010]

Options:

A. $\frac{A - Z - 8}{Z - 4}$

B. $\frac{A - Z - 4}{Z - 8}$

C. $\frac{A - Z - 12}{Z - 4}$

D. $\frac{A - Z - 4}{Z - 2}$

Answer: B

Solution:

When a radioactive nucleus emits 1 α -particle, the mass number decreases by 4 units and atomic number decreases by 2 units. When a radioactive nucleus emits 1 positron the atomic number decreases by 1 unit but mass number remains constant.

$$\therefore \text{Mass number of final nucleus} = A - 12$$

$$\text{Atomic number of final nucleus} = Z - 8$$

$$\therefore \text{Number of neutrons, } N_n = (A - 12) - (Z - 8) = A - Z - 4$$

$$\text{Number of protons, } N_p = Z - 8$$

$$\therefore \text{Required ratio} = \frac{N_n}{N_p} = \frac{A - Z - 4}{Z - 8}$$

Question265

The transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition from :

[2009]

Options:

A. $3 \rightarrow 2$

B. $4 \rightarrow 2$

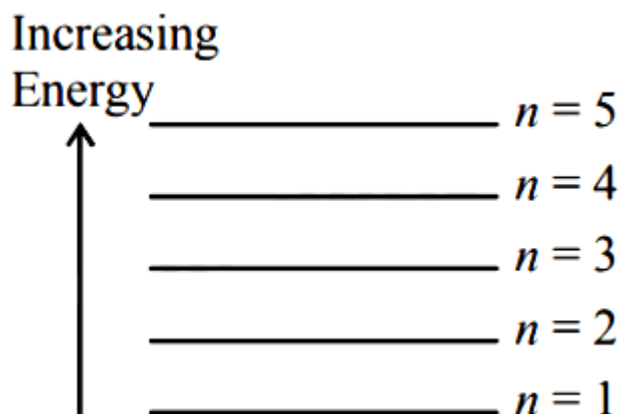
C. $5 \rightarrow 4$

D. $2 \rightarrow 1$

Answer: C

Solution:

It is given that transition from the state $n = 4$ to $n = 3$ in a hydrogen like atom result in ultraviolet radiation. For infrared radiation $\left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ should be less. The only option is $5 \rightarrow 4$.



Question266

Suppose an electron is attracted towards the origin by a force $\frac{k}{r}$ where 'k' is a constant and 'r' is the distance of the electron from the origin. By applying Bohr model to this system, the radius of the

**n^{th} orbital of the electron is found to be ' r_n ' and the kinetic energy of the electron to be ' T_n '. Then which of the following is true?
[2008]**

Options:

A. $T_n \propto \frac{1}{n^2}$, $r_n \propto n^2$

B. T_n independent of n , $r_n \propto n$

C. $T_n \propto \frac{1}{n}$, $r_n \propto n$

D. $T_n \propto \frac{1}{n^3}$, $r_n \propto n^2$

Answer: B

Solution:

Given,

$$\text{Centripetal force} = \frac{k}{r}$$

Then

$$\frac{k}{r} = \frac{mv^2}{r}$$

$$\Rightarrow k = mv^2 \Rightarrow T_n = \frac{1}{2}mv^2 = \frac{1}{2}k$$

T_n is independent of n

Also,

$$\text{Angular momentum, } L = \frac{nh}{2\pi}$$

$$\Rightarrow mvr_n = \frac{nh}{2\pi} (\because L = mvr)$$

$$\Rightarrow r_n = \frac{nh}{2\pi\sqrt{km}} (\because m^2v^2 = km \text{ or } mv = \sqrt{km})$$

Clearly, $r_n \propto n$

Question 267

Statement-1: Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion and

Statement-2 : For heavy nuclei, binding energy per nucleon

increases with increasing Z while for light nuclei it decreases with increasing Z .

[2008]

Options:

A. Statement-1 is false, Statement-2 is true

B. Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1

C. Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1

D. Statement-1 is true, Statement-2 is false

Answer: D

Solution:

We know that energy is released when heavy nuclei undergo fission or light nuclei undergo fusion. Therefore statement (1) is correct.

The second statement is false because for heavy nuclei the binding energy per nucleon decreases with increasing Z and for light nuclei, B.E/nucleon increases with increasing Z .

Question268

Which of the following transitions in hydrogen atoms emit photons of highest frequency?

[2007]

Options:

A. $n = 1$ to $n = 2$

B. $n = 2$ to $n = 6$

C. $n = 6$ to $n = 2$

D. $n = 2$ to $n = 1$

Answer: D

Solution:

We have to find the frequency of emitted photons.

For emission of photons electron should make a transition from higher energy level to lower energy level. so, option (a) and (b) are incorrect.

Frequency of emitted photon is given by

$$h\nu = -13.6 \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

For transition from $n = 6$ to $n = 2$

$$\nu_1 = \frac{-13.6}{h} \left(\frac{1}{6^2} - \frac{1}{2^2} \right) = \frac{2}{9} \times \left(\frac{13.6}{h} \right)$$

For transition from $n = 2$ to $n = 1$,

$$\nu_2 = \frac{-13.6}{h} \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = \frac{3}{4} \times \left(\frac{13.6}{h} \right)$$

$$\therefore \nu_1$$

Question 269

Which of the following are the constituents of the nucleus?
[2007]

Options:

- A. Electrons and protons
- B. Neutrons and protons
- C. Electrons and neutrons
- D. Neutrons and positrons

Answer: B

Question 270

If M_O is the mass of an oxygen isotope ${}_8\text{O}^{17}$, M_P and M_N are the masses of a proton and a neutron respectively, the nuclear binding

**energy of the isotope is
[2007]**

Options:

A. $(M_{\text{O}} - 17M_{\text{N}})c^2$

B. $(M_{\text{O}} - 8M_{\text{P}})c^2$

C. $(M_{\text{O}} - 8M_{\text{P}} - 9M_{\text{N}})c^2$

D. $M_{\text{O}}c^2$

Answer: C

Solution:

Number of protons in oxygen isotope, $Z = 8$

Number of neutrons $= 17 - 8 = 9$

Binding energy

$$= [Z M_{\text{p}} + (A - Z)M_{\text{N}} - M]c^2$$

$$= [8M_{\text{p}} + (17 - 8)M_{\text{N}} - M]c^2$$

$$= [8M_{\text{p}} + 9M_{\text{N}} - M]c^2$$

$$= [8M_{\text{p}} + 9M_{\text{N}} - M_{\text{O}}]c^2$$

Question271

**The half-life period of a radio-active element X is same as the mean life time of another radio-active element Y. Initially they have the same number of atoms. Then
[2007]**

Options:

A. X and Y decay at same rate always

B. X will decay faster than Y

C. Y will decay faster than X

D. X and Y have same decay rate initially

Answer: C

Solution:

Let λ_X and λ_Y be the decay constant of X and Y .

Half life of X , = average life of Y

$$T_{1/2} = T_{av}$$

$$\Rightarrow \frac{0.693}{\lambda_X} = \frac{1}{\lambda_Y}$$

$$\Rightarrow \lambda_X = (0.693) \cdot \lambda_Y$$

$$\therefore \lambda_X < \lambda_Y$$

Now, the rate of decay is given by

$$-\left(\frac{dN}{dt}\right)_x = \lambda_X N_0$$

$$-\left(\frac{dN}{dt}\right)_y = \lambda_Y N_0$$

As the rate of decay is directly proportional to decay constant, Y will decay faster than X.

Question272

An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to [2006]

Options:

A. v^2

B. $\frac{1}{m}$

C. $\frac{1}{v^2}$

D. $\frac{1}{Ze}$

Answer: B

Solution:

Work done to stop the α particle is equal to K.E.

$$\therefore qV = \frac{1}{2}mv^2 \Rightarrow q \times \frac{K(Ze)}{r} = \frac{1}{2}mv^2$$

$$\Rightarrow r = \frac{2(2e)K(Ze)}{mv^2} = \frac{4KZe^2}{mv^2}$$

$$\Rightarrow r \propto \frac{1}{v^2} \text{ and } r \propto \frac{1}{m}$$

Question273

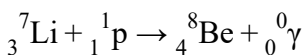
When ${}_3\text{Li}^7$ nuclei are bombarded by protons, and the resultant nuclei are ${}_4\text{Be}^8$, the emitted particles will be
[2006]

Options:

- A. alpha particles
- B. beta particles
- C. gamma photons
- D. neutrons

Answer: C

Solution:



We see that both proton number and mass number are equal in both sides, so emitted particle should be massless gamma photons.

Question274

If the binding energy per nucleon in ${}_3^7\text{Li}$ and ${}_2^4\text{He}$ nuclei are 5.60 MeV and 7.06 MeV respectively, then in the reaction
 $\text{p} + {}_3^7\text{Li} \rightarrow 2{}_2^4\text{He}$

**energy of proton must be
[2006]**

Options:

A. 28.24 MeV

B. 17.28 MeV

C. 1.46 MeV

D. 39.2 MeV

Answer: B

Solution:

Given,

Binding energy per nucleon of ${}^7_3\text{Li} = 5.60\text{MeV}$

Binding energy per nucleon of ${}^4_2\text{He} = 7.06\text{MeV}$

Let E be the energy of proton, then

$$E + 7 \times 5.6 = 2 \times [4 \times 7.06]$$

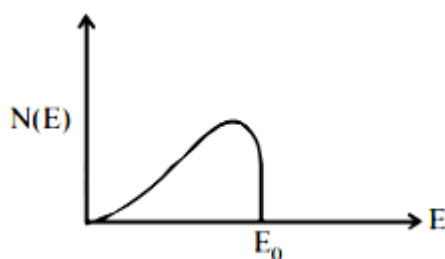
$$\Rightarrow E = 56.48 - 39.2 = 17.28\text{MeV}$$

Question275

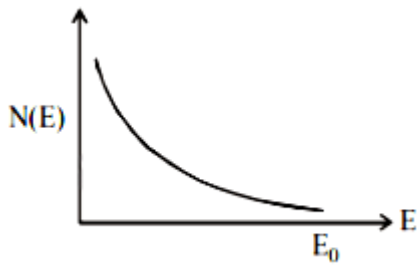
**The energy spectrum of β -particles [number $N(E)$ as a function of β -energy E] emitted from a radioactive source is
[2006]**

Options:

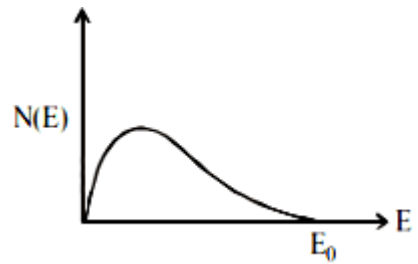
A.



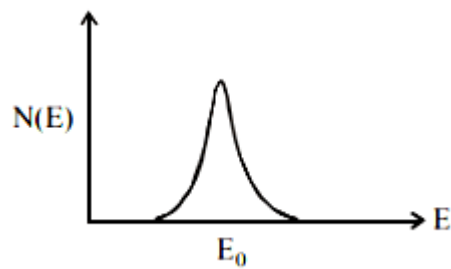
B.



C.



D.



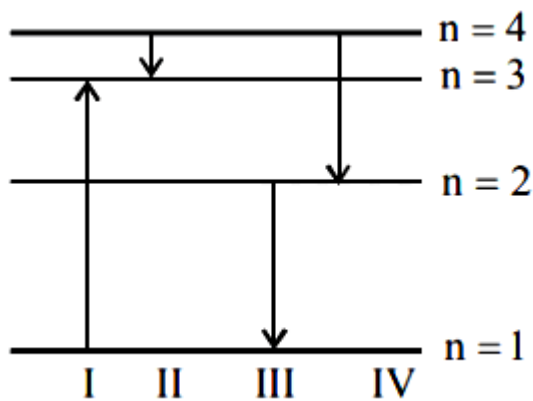
Answer: C

Solution:

The range of energy of β -particles is from zero to some maximum value.

Question276

The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



[2005]

Options:

- A. IV
- B. III
- C. II
- D. I

Answer: B

Solution:

Energy of radiation that corresponds to energy difference between two energy levels n_1 and n_2 is given as

$$E = Rhc \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \therefore E \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

E will be maximum for the transition for which $\left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ is maximum. Here n_2 is the higher energy level.

Clearly, $\left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ is maximum for the third transition, i.e., $2 \rightarrow 1$. I transition is showing the absorption of energy.

Question 277

If radius of the $_{13}^{27}\text{Al}$ nucleus is estimated to be 3.6 fermi then the radius of $_{52}^{125}\text{Te}$ nucleus be nearly

[2005]

Options:

- A. 8 fermi
- B. 6 fermi
- C. 5 fermi
- D. 4 fermi

Answer: B

Solution:

Radius of a nucleus,

$$R = R_0(A)^{1/3}$$

Here, R_0 is a constant

A = atomic mass number

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2} \right)^{1/3} = \left(\frac{27}{125} \right)^{1/3} = \frac{3}{5}$$

$$\Rightarrow R_2 = \frac{5}{3} \times 3.6 = 6 \text{ fermi}$$

Question278

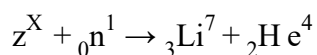
A nuclear transformation is denoted by $X (n, \alpha)_3^7\text{Li}$ Which of the following is the nucleus of element X ?
[2005]

Options:

- A. ${}_5^{10}\text{B}$
- B. ${}_{12}^{12}\text{C}_6$
- C. ${}_4^{11}\text{Be}$
- D. ${}_5^9\text{B}$

Answer: A

Solution:



Using conservation of mass number

$$A + 1 = 4 + 7$$

$$\Rightarrow A = 10$$

Using conservation of charge number

$$Z + 0 = 2 + 3 \Rightarrow Z = 5$$

It is boron ${}_5^{10}\text{B}$

Question 279

Starting with a sample of pure ${}^{66}\text{Cu}$, $\frac{7}{8}$ of it decays into Zn in 15 minutes. The corresponding half life is [2005]

Options:

A. 15 minutes

B. 10 minutes

C. $7\frac{1}{2}$ minutes

D. 5 minutes

Answer: D

Solution:

It is given that

$\frac{7}{8}$ of Cu decays in 15 minutes

$\therefore$ Cu left undecayed is

$$N = 1 - \frac{7}{8} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$$

$\therefore$ No. of half lifes = 3

$$n = \frac{t}{T} \Rightarrow 3 = \frac{15}{T}$$

$$\Rightarrow T = \text{half life period} = \frac{15}{3} = 5 \text{ minutes}$$

Question280

The intensity of gamma radiation from a given source is I . On passing through 36mm of lead, it is reduced to $\frac{I}{8}$. The thickness of lead which will reduce the intensity to $\frac{I}{2}$ will be

[2005]

Options:

- A. 9 mm
- B. 6 mm
- C. 12 mm
- D. 18 mm

Answer: C

Solution:

Let intensity of gamma radiation from source be I_0 .

$$\text{Intensity } I = I_0 \cdot e^{-\mu d}$$

Where d is the thickness of lead.

Applying logarithm on both sides,

$$-\mu d = \log\left(\frac{I}{I_0}\right)$$

$$\text{For } d = 36\text{mm, intensity} = \frac{I}{8}$$

$$-\mu \times 36 = \log\left(\frac{I/8}{I}\right) \dots\dots\dots(i)$$

For intensity $I/2$, thickness = d

$$-\mu \times d = \log\left(\frac{I/2}{I}\right) \dots\dots\dots(ii)$$

Dividing (i) by (ii),

$$\frac{36}{d} = \frac{\log\left(\frac{1}{8}\right)}{\log\left(\frac{1}{2}\right)} = \frac{3 \log\left(\frac{1}{2}\right)}{\log\left(\frac{1}{2}\right)} = 3 \text{ or } d = \frac{36}{3} = 12\text{mm}$$

Question281

**An alpha -particle of energy 5M eV is scattered through 180° by a fixed uranium nucleus. The distance of closest approach is of the order of
[2004]**

Options:

A. 10^{-12}cm

B. 10^{-10}cm

C. 10^{-14}cm

D. 10^{-15}cm

Answer: A

Solution:

Distance of closest approach

$$r_0 = \frac{Z e(2e)}{4\pi\epsilon_0 \left(\frac{1}{2}mv^2\right)}$$

Energy, $E = 5 \times 10^6 \times 1.6 \times 10^{-19}\text{J}$

$$\therefore r_0 = \frac{9 \times 10^9 \times (92 \times 1.6 \times 10^{-19})(2 \times 1.6 \times 10^{-19})}{5 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$\Rightarrow r_h = 5.2 \times 10^{-14}\text{m} = 5.3 \times 10^{-12}\text{cm}$$

Question282

**A nucleus disintegrated into two nuclear parts which have their velocities in the ratio of 2 : 1. The ratio of their nuclear sizes will be
[2004]**

Options:

A. $3^{1/2} : 1$

B. $1 : 2^{1/3}$

C. $2^{1/3} : 1$

D. $1 : 3^{1/2}$

Answer: B

Solution:

Given :

$$\frac{v_1}{v_2} = \frac{2}{1}$$

From conservation of momentum $m_1 v_1 = m_2 v_2$

$$\Rightarrow \left(\frac{m_1}{m_2} \right) = \left(\frac{v_2}{v_1} \right) = \frac{1}{2}$$

We know that mass of nucleus, $m \propto A$

Nuclear size $R \propto A^{1/3} \propto m^{1/3}$

$$\frac{R_1}{R_2} = \left(\frac{m_1}{m_2} \right)^{1/3} \Rightarrow \frac{R_1^3}{R_2^3} = \frac{1}{2} \Rightarrow \left(\frac{R_1}{R_2} \right) = \left(\frac{1}{2} \right)^{1/3}$$

Question283

The binding energy per nucleon of deuteron (${}_1^2\text{H}$) and helium nucleus (${}_2^4\text{He}$) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is
[2004]

Options:

A. 23.6 MeV

B. 26.9 MeV

C. 13.9 MeV

D. 19.2 MeV

Answer: A

Solution:

The chemical reaction of process is $2 {}_1^2\text{H} \rightarrow {}_2^4\text{He}$
 Binding energy of two deuterons, $4 \times 1.1 = 4.4 \text{ MeV}$
 Binding energy of helium nucleus $= 4 \times 7 = 28 \text{ MeV}$
 Energy released $= 28 - 4.4 = 23.6 \text{ MeV}$

Question284

The wavelengths involved in the spectrum of deuterium (${}_1^2\text{D}$) are slightly different from that of hydrogen spectrum, because [2003]

Options:

- A. the size of the two nuclei are different
- B. the nuclear forces are different in the two cases
- C. the masses of the two nuclei are different
- D. the attraction between the electron and the nucleus is different in the two cases

Answer: C

Solution:

The wavelength of spectrum is given by

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\text{where } R = \frac{1.097 \times 10^7}{1 + \frac{m}{M}}$$

where m = mass of electron

M = mass of nucleus.

Thus, wavelength involved in the spectrum of hydrogen like atom depends upon masses of nucleus. The mass number of hydrogen and deuterium is 1 and 2 respectively, so spectrum of deuterium will be different from hydrogen.

Question285

When a U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed ' u ', the recoil speed of the residual nucleus

is
[2003]

Options:

A. $\frac{4u}{238}$

B. $-\frac{4u}{234}$

C. $\frac{4u}{234}$

D. $-\frac{4u}{238}$

Answer: C

Solution:

Mass of α particle, $m_\alpha = 4u$

Mass of nucleus after fission, $m_n = 234u$ From conservation of linear momentum we have

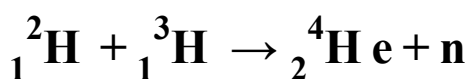
$$238 \times 0 = 4u + 234v$$

$$\therefore v = -\frac{4}{234}u$$

$$\therefore \text{Speed} = \left| \vec{v} \right| = \frac{4}{234}u$$

Question 286

In the nuclear fusion reaction



given that the repulsive potential energy between the two nuclei is $\sim 7.7 \times 10^{-14} \text{ J}$, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's Constant

$$k = 1.38 \times 10^{-23} \text{ J / K }]$$

[2003]

Options:

A. 10^7 K

B. 10^5K

C. 10^3K

D. 10^9K

Answer: D

Solution:

The average kinetic energy per molecule at temperature T is

$$= \frac{3}{2}kT$$

Where k = Boltzmann's constant

This kinetic energy should be able to provide the repulsive potential energy

$$\therefore \frac{3}{2}kT = 7.7 \times 10^{-14}$$

$$\Rightarrow T = \frac{2 \times 7.7 \times 10^{-14}}{3 \times 1.38 \times 10^{-23}} = 3.7 \times 10^9\text{K}$$

Question287

Which of the following cannot be emitted by radioactive substances during their decay?

[2003]

Options:

A. Protons

B. Neutrinoes

C. Helium nuclei

D. Electrons

Answer: A

Solution:

The radioactive substances emit α -particles (Helium nucleus), β -particles (electrons) and neutrinoes. Protons cannot be emitted by radioactive substances during their decay.

Question288

A nucleus with $Z = 92$ emits the following in a sequence:

$\alpha, \beta^-, \beta^-, \alpha, \alpha, \alpha, \alpha, \beta^-, \beta^-, \alpha, \beta^+, \beta^+, \alpha$

Then Z of the resulting nucleus is

[2003]

Options:

A. 76

B. 78

C. 82

D. 74

Answer: B

Solution:

The number of α -particles released = 8

Decrease in atomic number = $8 \times 2 = 16$

The number of μ^- -particles released = 4

Increase in atomic number = $4 \times 1 = 4$

Also the number of β^+ particles released is 2, which should decrease the atomic number by 2 .

Therefore the final atomic number of resulting nucleus

$$= Z - 16 + 4 - 2 = Z - 14$$

$$= 92 - 14 = 78$$

Question289

A radioactive sample at any instant has its disintegration rate 5000 disintegrations per minute. After 5 minutes, the rate is 1250 disintegrations per minute. Then, the decay constant (per minute) is [2003]

Options:

A. $0.4 \ln 2$

B. $0.2 \ln 2$

C. $0.1 \ln 2$

D. $0.8 \ln 2$

Answer: A

Solution:

Initial activity, $A_o = 5000$ disintegration per minute

Activity after 5 min, $A = 1250$ disintegration per minute

$$A = A_o e^{-\lambda t}$$

$$\Rightarrow e^{-\lambda t} = \frac{A_o}{A}$$

$$\Rightarrow \lambda = \frac{1}{t} \log_e \frac{A_o}{A} = \frac{1}{5} \log_e \frac{5000}{1250}$$

$$= \frac{2}{5} \log_e 2 = 0.4 \log_e 2$$

Question290

If 13.6 eV energy is required to ionize the hydrogen atom, then the energy required to remove an electron from $n = 2$ is [2002]

Options:

A. 10.2 eV

B. 0 eV

C. 3.4 eV

D. 6.8 eV

Answer: C

Solution:

The energy required to remove the electron from the n^{th} orbit of hydrogen is given by

$$E_n = \frac{13.6}{n^2} \text{eV /atom}$$

For $n = 2$, $E_n = \frac{13.6}{4} = 3.4\text{eV}$

Therefore the energy required to remove electron from $n = 2$ is $+3.4\text{eV}$

Question291

At a specific instant emission of radioactive compound is deflected in a magnetic field. The compound can emit

(i) electrons

(ii) protons

(iii) He^{2+}

(iv) neutrons

The emission at instant can be

[2002]

Options:

A. i, ii, iii

B. i, ii, iii, iv

C. iv

D. ii, iii

Answer: A

Solution:

Charged particles are deflected in magnetic field.

Electrons, protons and He^{2+} all are charged species. Hence, correct option is (a).

Question292

If N_0 is the original mass of the substance of half-life period $t_{1/2} = 5$ years, then the amount of substance left after 15 years is

[2002]

Options:

A. $N_0/8$

B. $N_0/16$

C. $N_0/2$

D. $N_0/4$

Answer: A

Solution:

After every half-life, the mass of the substance reduces to half its initial value.

$$N_0 \xrightarrow{5\text{years}} \frac{N_0}{2} \xrightarrow{5\text{years}} \frac{N_0/2}{2} = \frac{N_0}{4} \xrightarrow{5\text{years}} \frac{N_0/4}{2} = \frac{N_0}{8}$$
